Junior IOAA

Problems by Topic

2022–2025

Problem statements and official solutions

About this edition

This collection brings together the individual Junior IOAA papers for 2022–2025. Problems are grouped by subject. Observational tasks form a separate chapter, grouped by round. Each chapter includes its corresponding official solutions.

Problem statements and solution text are transcribed as editable LaTeX. Figures, charts, and marked answer overlays are vector crops from the supplied papers. The companion crop edition preserves the original page content. Numerical values and mathematical notation should be checked against those source papers when the books are used for teaching.

The original papers are the work of their authors and organising committees. Editorial titles support topic navigation. Notes identify missing solutions and differences between question and solution versions in the supplied archive.

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Chapter 1

The Celestial Sphere

Theory problems

Problem 1.1 The Sun on the Horizon

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q2 Solution p. 5

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

The astronomical refraction correction, ρ_r, has the minimum value, $\rho_{r,\min} = 0$, when the star is at the zenith of the Earth's surface observer ($z = 0$, $h = 90°$), and has the maximum value, $\rho_{r,\max}$, when the star, whose light passes through the Earth's atmosphere, in order to reach the Earth's surface observer, is at the horizon ($z = 90°$, $h = 0$), that is, when the star rises or sets.

It is known, that for an observer, atmospheric refraction raises the Sun in the moments of sunrise and sunset, that is, when it is below the horizon, approximately at its edge, by an apparent disk.

It is given: the radius of the Sun, $R_S = 6.96 \times 10^5$ km, the distance between the centre of the Earth and the centre of the Sun, $d = 15 \times 10^7$ km. The value of $\rho_{r,\max}$ is

(a) $\rho_{r,\max} \approx 23.7'$.

(b) $\rho_{r,\max} \approx 43.7'$.

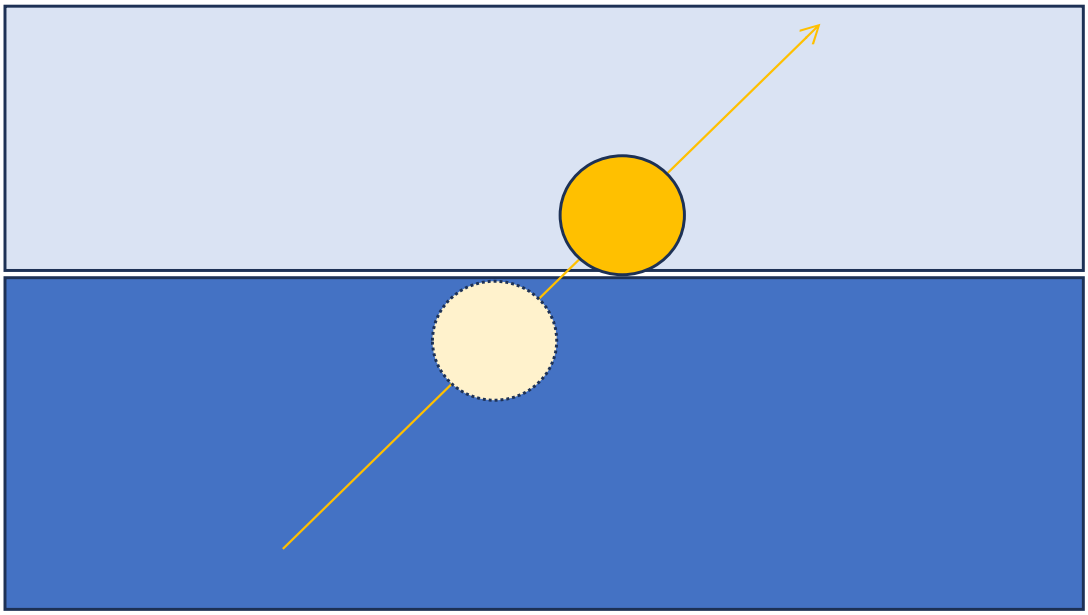
(c) $\rho_{r,\max} \approx 53.7'$.

(d) $\rho_{r,\max} \approx 33.7'$.

Problem 1.2 Horizon crossing

Junior IOAA 2023, Volos, Greece Theory problems Q1 10 points Solution p. 5

The Sun, at some location, crosses the horizon at sunrise on autumnal equinox day at an angle of 45°. The angular diameter of the Sun is 32 arcminutes. How long does it approximately take for the entire solar disk to cross the horizon? You may neglect atmospheric effects.



Problem 1.3 Rising Sagarmatha and angular separation

Junior IOAA 2024, Kathmandu, Nepal Theory problems T8 15 points Solution p. 6

As we saw in the group competition, Nepal got the chance to name a star in the constellation of Leo. The name given to that star is Sagarmatha. The coordinates of this star are $\alpha = 11^{\mathrm{h}}35^{\mathrm{m}}52^{\mathrm{s}}$, $\delta = -4°45'21''$. Given that the Autumnal equinox for the year 2024 happened on Sep 22 at 6:28 pm Nepal time, and neglecting the effects of atmospheric refraction,

1. Find the rise time of Sagarmatha as seen from Kathmandu on 6 October 2024. The rise time here refers to local time in Kathmandu.

2. We want to find Sagarmatha through a telescope. Since Sagarmatha is too faint, we first point to ϕ Leo ($\alpha = 11^{\mathrm{h}}16^{\mathrm{m}}38^{\mathrm{s}}$, $\delta = -3°38'58''$). Find the angular separation between ϕ Leo and Sagarmatha.

Supplied examination data. Kathmandu's latitude is 27.7172°N.

Problem 1.4 Where is Theodor?

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.3 10 points Solution p. 6

Theodor is in a random city somewhere in the world. Using his knowledge of geography, he narrows down the possible locations to the four cities listed below. Then, on October 20th, he measures the time between sunrise and sunset and finds it to be $t = 10^{\mathrm{h}}42^{\mathrm{m}}$. Can you help him figure out which city he is in? Ignore atmospheric refraction and Sun's apparent size.

(a) Compute the Sun's equatorial coordinates on that date. (Consider that the sun moves with constant velocity on the ecliptic from the autumn equinox onward.)

(b) Based on the measured daytime duration t, determine which of the four cities Theodor is in.

(c) On this day, exactly at the moment when the Sun sets for Theodor, determine the geographical coordinates of the point on Earth where the Sun is directly overhead.

No.	City	Latitude	Longitude
1	Kutaisi	42.27° N	42.66° E
2	Vassouras	22.41° S	43.66° W
3	Mumbai	18.96° N	72.83° E
4	Chorzów	50.31° N	18.97° E



1)Kutaisi (φ=42.27° N, L=42.66° E)



2)Vassouras (φ=22.41° S,L=43.66° W)



3)Mumbai (φ=18.96° N, L=72.83° E)



4)Chorzów (φ= 50.31° N, L=18.97° E)

Problem 1.5 Ceahlau Mountain

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.5 15 points Solution p. 7

Ceahlău Mountain is located in Neamţ County, not far from the city of Piatra Neamţ. An observer on the top of the mountain ($\varphi = 46°53'57''$ N, $L = 26°24'41''$ E, altitude $h = 1907\,\text{m}$ above sea level), looks at the starry sky, making observations of the star Almach in the constellation Andromeda.

(a) Determine whether, from the top of Mount Ceahlău, the star Almach (γ AND, $\alpha = 02^{\text{h}}03^{\text{m}}54^{\text{s}}$, $\delta = 42°19'47''$) is circumpolar. The radius of the Earth is known to be $R_\oplus = 6378\,\text{km}$.

(b) In 1778, it was discovered that γ AND is a double star. The two stars are separated by an angle of about $10''$. What should be the diameter of the objective lens of a telescope to see the two components distinctly (the wavelength is known to be $\lambda = 550\,\text{nm}$)?

(c) It was later discovered that the Almach star is a multiple star system consisting of the primary component γ_1 (A) with apparent magnitude $m_1 = 2.27$ and a subsystem γ_2 with apparent magnitude $m_2 = 4.84$. Calculate the apparent magnitude of the system consisting of the star γ_1 and the subsystem γ_2.

(d) What is the absolute magnitude of the star, knowing that it is approximately 355 light-years away?

Problem 1.6 The Flying Aquila

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.6 15 points Solution p. 7

Radu is observing the night sky from Aveiro ($\varphi = 40.67°$ N, $L = 8.65°$ W) in the year 2024. During his observations, he sees Altair ($\alpha_{\text{Altair}} = 19^{\text{h}}51^{\text{m}}$, $\delta_{\text{Altair}} = 8°55'50''$) rising at local solar time $T_{\text{Local,1}} = 22{:}39{:}43$. Later, at local solar time $T_{\text{Local,2}} = 02{:}42{:}58$, he observes that Altair and Arcturus ($\delta_{\text{Arcturus}} = 19°33'6''$) are at the same altitude. The observations are made from sea level. Ignore atmospheric effects.

(a) Determine the sidereal time at the moment Altair rises.

(b) Determine the current date (at the moment Altair rises). Ignore the equation of time, and consider that the right ascension of the Sun is increasing uniformly through the year, at a rate of $3^{\text{m}}56^{\text{s}}$ per day. The spring equinox occurs on March 21st.

(c) Determine the right ascension of Arcturus. Take into account that Arcturus is in the Western hemisphere of the sky at $T_{\text{Local,2}}$.

In your solution, use a for altitude, H for hour angle, δ for declination, and α for right ascension.

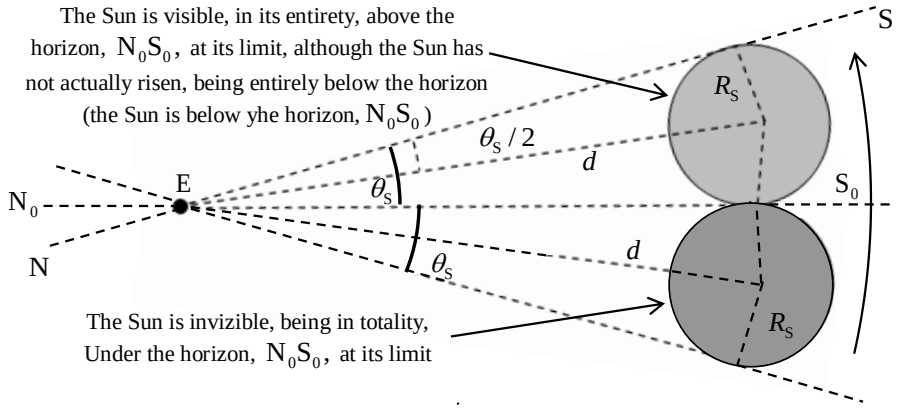
Solutions

Solution 1.1 The Sun on the Horizon

Problem p. 1

[The official key prints 0.0098 rad and 33.7 arcmin in the same calculation. Those numbers and the source wording about sunrise are inconsistent; they are reproduced with source comments.]

Atmospheric refraction raises the Sun above its horizon, at its limit, at sunrise and sunset, approximately by an apparent disk, as shown in the figure.



The upper arc of the real solar disk rises after the moment when the whole apparent disk is already visible above the horizon. The official explanation then states that sunrise and sunset are seen later than they actually occur, making the day a little longer. Under these conditions,

$$\begin{aligned}\rho_{r,\max} &= \theta_S,\\ \sin\frac{\theta_S}{2} &= \frac{R_S}{d} \approx \frac{\theta_S}{2},\\ \theta_S &= \frac{2R_S}{d} = \frac{2\times 6.96\times 10^5}{15\times 10^7} = 0.0098\,\text{rad}.\end{aligned}$$

Using $1\,\text{rad} = 180\times 60'/3.14$, the source gives

$$\theta_S = 0.0098\frac{180\times 60'}{3.14} \approx 33.7' = \rho_{r,\max}.$$

Thus the official answer is (d).

Solution 1.2 Horizon crossing

Problem p. 1

The Sun covers approximately 361° on the celestial sphere in 24 hours, thus the velocity will be

$$v = \frac{361}{24\times 3600}\ \text{degrees}\,\text{s}^{-1}.$$

(5 points)

The Sun will need to move $s = (32/60)/\cos(\pi/4)$ degrees to cross the horizon. Therefore, the time it will need is

$$t = \frac{s}{v} = \frac{(32/60)/\cos(\pi/4)}{361/(24\times 3600)} = 3\ \text{minutes}.$$

(5 points)

Solution 1.3 Rising Sagarmatha and angular separation

Problem p. 2

[The official rise-time calculation assumes local-noon culmination and does not use the supplied equinox epoch, date or right ascension. The official answer is retained.]

1. We know
$$\sin a = \sin\phi \sin\delta + \cos\phi \cos\delta \cos H.$$
(3 points) At rise, $a = 0$, giving $\cos H = -\tan\phi \tan\delta$. (1 point) Solve for H to get 5 hours 49 minutes 59 seconds. (3 points) Then $t_{\text{rise}} = 12\,\text{pm} - H = 6{:}10\,\text{am}$. (2 points)

2. Since the stars are very close, use planar geometry,
$$d = \sqrt{(\Delta\alpha)^2 + (\Delta\delta)^2}.$$
(1 point) Substitute $\Delta\alpha = 0.08376\,\text{rad}$ (2 points) and $\Delta\delta = 0.01927\,\text{rad}$. (2 points) This gives $d = 0.08595\,\text{rad}$, or 4.92°. (1 point) Answers rounding to 4.9° are acceptable. Answers in degrees or subunits thereof for any part are acceptable. Correct solutions with spherical trigonometry also receive full points.

Solution 1.4 Where is Theodor?

Problem p. 2

(a) Take $\lambda_S = 180^\circ$ on 22 September (also accept 23 September). With D the number of days since the chosen equinox,
$$\lambda_S = 0^\circ/180^\circ + \frac{D}{365.25} \times 360^\circ.$$
(0.5 point) The two choices give $\lambda_S = 207.6^\circ$ or 206.6°. (0.5 point)
$$\sin\delta = \sin\beta\cos\varepsilon + \cos\beta\sin\varepsilon\cos(90^\circ - \lambda) = \sin\varepsilon\sin\lambda \quad (\beta = 0^\circ).$$
(1 point) Therefore $\delta = -10^\circ 38'$ or $-10^\circ 16'$. (0.5 point) Using $\cos\alpha = \cos\lambda/\cos\delta$ (0.5 point) gives $\alpha = 13^\text{h}42^\text{m}$ or $13^\text{h}38^\text{m}$. (1 point) Award 75% for computing α directly with uniform motion on the equator, and 25% for Euclidean geometry in place of spherical trigonometry.

(b)
$$\begin{aligned}\cos(90^\circ - h) &= \cos(90^\circ - \varphi)\cos(90^\circ - \delta) \\ &\quad + \sin(90^\circ - \varphi)\sin(90^\circ - \delta)\cos H, \\ 0 &= \sin\varphi\sin\delta + \cos\varphi\cos\delta\cos H, \\ \cos H &= -\tan\varphi\tan\delta.\end{aligned}$$
(2 points) Writing the last formula directly receives full credit. Local time is
$$t_l = H + 12^\text{h} + n - L + \eta + \text{D.S.T.}$$
The other terms are constant during a day, so $\Delta t_l \approx \Delta H$. (0.5 point) The two solutions of $\cos x = a$ in $[0, 2\pi]$ are $\arccos a$ and $2\pi - \arccos a$, giving $t = 2\arccos(-\tan\varphi\tan\delta)$ after conversion to hours. The four durations are
$$t_1 = 10^\text{h}42^\text{m}, \quad t_2 = 12^\text{h}35^\text{m}, \quad t_3 = 11^\text{h}31^\text{m}, \quad t_4 = 10^\text{h}16^\text{m}.$$
(1 point) Theodor is in city 1, Kutaisi. (0.5 point) Full credit is also awarded for calculating both solutions for H for each city and taking their time difference.

(c) For the Sun to be directly overhead, $h_S = 90^\circ$ requires $\varphi = \delta$. (0.5 point) Hence $\varphi = -10^\circ 38'$. (0.5 point) With $L = 12^\mathrm{h} - T_\mathrm{GMT}$ (0.5 point) and $T_\mathrm{GMT} = H + 12^\mathrm{h} - L$, the longitude is $37^\circ 47'$ W, or $37^\circ 1'$ W using 23 September. (0.5 point)

Solution 1.5 Ceahlau Mountain

Problem p. 3

(a) A circumpolar star never sets below the horizon. At sea level, the limiting declination is

$$\delta_m = 90^\circ - \varphi = 90^\circ - 46^\circ 53' 57'' = 43^\circ 6' 3''.$$

(1 point) Considering atmospheric refraction $\alpha = 34'$, $\delta_m = 42^\circ 32' 3''$. Thus Almach is not circumpolar at sea level. (1 point) At 1907 m the horizon drops, with

$$\cos\theta = \frac{R}{R+h}, \qquad \theta = \arccos\frac{R}{R+h} \approx 1.4^\circ.$$

(2 points) The source then gives $\delta_m = 90^\circ - \varphi - \alpha = 41^\circ 8'$, so Almach is circumpolar from Mount Ceahlău. (1 point) *The printed formula omits the horizon-dip term θ. Its numerical result $41^\circ 8'$ includes both the $34'$ refraction and the approximately 1.4° horizon dip.*

(b)

$$\theta = \frac{1.22\lambda}{D} \quad \Rightarrow \quad D = \frac{1.22\lambda}{\theta}.$$

(2 points) With $\theta = 10'' = 4.84 \times 10^{-5}$ rad, $D \approx 14$ mm. (1 point)

(c) Pogson's law gives $E_1/E_2 = 10^{0.4(m_2 - m_1)}$, hence $E_1 = 10.537 E_2$. (2 points) The total brightness is $E = E_1 + E_2 = 11.537 E_2$. (1 point) The printed key writes

$$\frac{E_1}{E} = 2.5^{m - m_1}, \qquad m = 2.17.$$

(1 point) *The base 2.5 in the printed equation is inconsistent with Pogson's law stated above.*

(d)

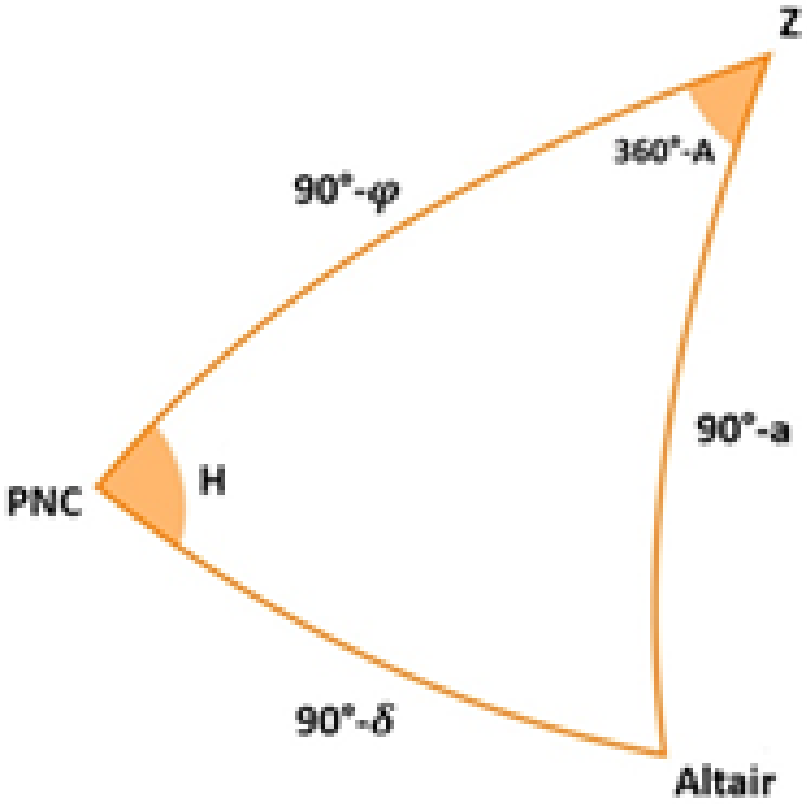
$$M = m + 5 - 5\log_{10} d.$$

(1 point) $d = 355$ ly $= 108.896$ pc. (1 point) Thus $M = -3.04$. (1 point)

Solution 1.6 The Flying Aquila

Problem p. 4

(a) Draw the spherical triangle North Celestial Pole–Zenith–Star.



The spherical law of cosines gives

$$\sin a_{\text{Altair}} = \sin\varphi \sin\delta_{\text{Altair}} + \cos\varphi \cos\delta_{\text{Altair}} \cos H_{\text{Altair}}.$$

At rising, neglecting horizon effects and refraction, $a_{\text{Altair}} = 0°$. Hence

$$\cos H_{\text{Altair}} = -\tan\varphi \tan\delta_{\text{Altair}}, \qquad H_{\text{Altair}} = \pm 6^{\text{h}}31^{\text{m}}.$$

(2 points) Altair rises $6^{\text{h}}31^{\text{m}}$ before crossing the meridian. (1 point) On the meridian $T_{\text{sidereal}} = \alpha_{\text{Altair}} = 19^{\text{h}}51^{\text{m}}$. (1 point) Thus $T_{\text{sidereal}} = 19^{\text{h}}51^{\text{m}} - 6^{\text{h}}31^{\text{m}} = 13^{\text{h}}20^{\text{m}}$. (1 point)

(b) At rising, $T_{\text{local}} = 22{:}39{:}43$ and $H_{\text{Sun}} = 10{:}39{:}43$. (1 point) The sidereal time is $H_\gamma = 13^{\text{h}}20^{\text{m}}$. (0.5 point) Therefore $\alpha_{\text{Sun}} = 2^{\text{h}}40^{\text{m}}17^{\text{s}}$. (1 point) This is 40.75 days after the spring equinox, (0.5 point) so the date is 1 May. (1 point)

(c) The Sun's right ascension changes negligibly in a few hours, so $\Delta T_{\text{local}} \approx \Delta T_{\text{sidereal}}$. When the stars have the same altitude,

$$\begin{aligned} T_{\text{sidereal},2} &\approx T_{\text{sidereal},1} + T_{\text{local},2} - T_{\text{local},1} \\ &= 13^{\text{h}}20^{\text{m}} + 2^{\text{h}}43^{\text{m}} - 22^{\text{h}}40^{\text{m}} = 17^{\text{h}}23^{\text{m}}. \end{aligned}$$

(1 point) For each star,

$$\begin{aligned} \sin a_{\text{Altair}} &= \sin\varphi \sin\delta_{\text{Altair}} + \cos\varphi \cos\delta_{\text{Altair}} \cos H_{\text{Altair}}, \\ \sin a_{\text{Arcturus}} &= \sin\varphi \sin\delta_{\text{Arcturus}} + \cos\varphi \cos\delta_{\text{Arcturus}} \cos H_{\text{Arcturus}}. \end{aligned}$$

(1 point) Equating the two altitudes gives

$$\cos H_{\text{Arcturus}} = \frac{\sin\varphi \sin\delta_{\text{Altair}} + \cos\varphi \cos\delta_{\text{Altair}} \cos H'_{\text{Altair}} - \sin\varphi \sin\delta_{\text{Arcturus}}}{\cos\varphi \cos\delta_{\text{Arcturus}}}.$$

The key gives $H_{\text{Arcturus}} \in \{3^{\text{h}}10^{\text{m}}36^{\text{s}}, 20^{\text{h}}49^{\text{m}}30^{\text{s}}\}$. The western solution is $3^{\text{h}}10^{\text{m}}36^{\text{s}}$. (3 points) Accept $3^{\text{h}}9^{\text{m}}$–$3^{\text{h}}12^{\text{m}}$. Now

$$\alpha_{\text{Arcturus}} = T_{\text{sidereal},2} - H_{\text{Arcturus}} \approx 14^{\text{h}}12^{\text{m}}.$$

(1 point) Accept $14^{\text{h}}10^{\text{m}}$–$14^{\text{h}}18^{\text{m}}$.

Chapter 2

Time and Calendars

Theory problems

Problem 2.1 Moon phases

Junior IOAA 2023, Volos, Greece Theory problems Q5 10 points Solution p. 12

The first full moon for 2024 in Volos is on the 25th of January at 19:54. At what date and time will the last full moon for 2024 (visible from Volos) occur?

Problem 2.2 Longitude and local time in Nepal

Junior IOAA 2024, Kathmandu, Nepal Theory problems T3 15 points Solution p. 12

Nepal has a unique timezone of UTC +5:45, with the meridian for the country passing through the peak of Mt. Gaurishankar in the Himalayas.

1. Knowing that Mt. Gaurishankar is about 105 km due East from Kathmandu, estimate Kathmandu's longitude.
2. Binod, who stays in Kathmandu, sees a bright star at meridian at exactly 8:35 pm as per his watch. Calculate Kathmandu's local time for this star's meridian crossing.
3. Binod wants to tell about this bright star to his friend Ravi who lives in Ranchi, a city in India that has the same longitude as Kathmandu, but a timezone of UTC +5:30. What is the difference in local time and time in the watch for Ravi? What time will the same star cross the meridian as per Ravi's watch?

Supplied examination data. Kathmandu's latitude is 27.7172°N and Earth's mean radius is 6378 km.

Problem 2.3 The lunar calendar and Dashain

Junior IOAA 2024, Kathmandu, Nepal Theory problems T6 15 points Solution p. 12

The festival of Dashain is one of the most important celebrations in the Nepali calendar. The final day of the festival coincides with a Waxing Gibbous Moon that is exactly 10 days old. The festival always happens in a fixed month of lunar calendar, which has 12 months. Answer the following questions regarding the estimated timing of the Dashain festival.

1. In 2024, the final day is scheduled to fall on the 12th of October. Assuming that the Moon

is exactly 10 days old at 11 am, predict on which date the final day of Dashain occurred in 2023.

2. Predict on which day Kojagrat Purnima, which is the Full Moon immediately after Dashain, will fall in 2025.

Supplied examination data. The lunar mean synodic month is 29.53059 days.

Solutions

Solution 2.1 Moon phases

Problem p. 10

The lunar month, from full moon to full moon, is 29.53 days. A value of 29.5 days is acceptable. (3 points)

Twelve lunar months correspond to 354.36 days, which gets us to 7/1/2025. Thus, we need to account for 11 lunar months, 324.83 days, which corresponds to 15:49 on 15 December 2024. If 29.5 days is used, the result is 7:54 am on 15 December 2024. (7 points)

Solution 2.2 Longitude and local time in Nepal

Problem p. 10

[The official longitude calculation multiplies by cos(latitude), and its Ranchi clock time is inconsistent with the stated watch offsets. The published expressions and answers are retained.]

(a) The longitude of Gaurishankar is

$$\lambda_g = \frac{5.75}{12} \times 180^\circ = 86.25^\circ\text{E}.$$

(3 points) The official calculation gives

$$\lambda_k = \lambda_g - \frac{105\cos\phi}{6378}\frac{180^\circ}{\pi} = 85.415^\circ\text{E}.$$

(4 points)

(b) The official time difference is

$$t_{\text{diff}} = \frac{105\cos\phi}{6378}\frac{12\times 3600}{\pi}\,\text{s} = 200.4\,\text{s} = 3\,\text{min}\ 20.4\,\text{s}.$$

(3 points) Then $t_{\text{Kathmandu}} = 8{:}35\,\text{pm} - t_{\text{diff}} = 8{:}31{:}40\,\text{pm}$. Answers of 8:32 pm or 8:31:40 pm are accepted. (1 point)

(c) Ravi has the same local time as Kathmandu. On his watch, however, his time zone is UTC +5:30. Thus the difference is $15 - t_{\text{diff}}$ minutes, or 11 minutes 40 (39.6) seconds. (2 points) As per his watch, therefore, the occultation will occur at 8:16:40 pm, or 8:17 pm. (2 points)

Solution 2.3 The lunar calendar and Dashain

Problem p. 10

1. The mean synodic period of the Moon is 29.53059 days. Dashain happened 12 synodic periods earlier. (1 point)

$$12 \times 29.53059 = 354.36708\,\text{days} = 354\,\text{days}\ 8\,\text{hours}\ 48\,\text{minutes}.$$

(3 points) Because 2024 is a leap year, there are 366 days between the two 12 Octobers. (1 point) Dashain in 2023 will be 354 days before 12 October 2024. (3 points) In other words, it was 12 days after 12 October 2023, which was 24 October 2023. (2 points)

2. Dashain in 2025 is 354 days 8 hours 48 minutes after 11 am on 12 October 2024. (1 point) This is 11 days before 12 October 2025, at around 7:48 pm on 1 October 2025. (2 points) Four days and 12 hours after this is 7:48 am on 6 October 2025. Therefore, Kojagrat Purnima falls on 6 October 2025. (2 points)

Chapter 3

The Solar System

Theory problems

Problem 3.1 The Sun as Seen from Saturn

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q6 Solution p. 18

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

It is known that: the distance Saturn–Sun is 9.54 times greater than the distance Earth–Sun; the angular diameter of the Sun's disk, seen from Earth, is $32'$. The angular diameter of the Sun seen from Saturn, is

(a) $\alpha_{\text{Sun}-\text{Saturn}} \approx 0.001$ radian.

(b) $\alpha_{\text{Sun}-\text{Saturn}} \approx 0.002$ radian.

(c) $\alpha_{\text{Sun}-\text{Saturn}} \approx 0.003$ radian.

(d) $\alpha_{\text{Sun}-\text{Saturn}} \approx 0.004$ radian.

Problem 3.2 Solar eclipse

Junior IOAA 2023, Volos, Greece Theory problems Q6 10 points Solution p. 18

Assume that one day the Moon's apogee coincides with the Earth's perihelion. An observer, at an appropriate location on the Earth, observes the eclipse at zenith at its maximum. What fraction of the solar disk area is covered? The maximum distance between the centers of the Earth and the Moon is 405,000 km, while the minimum distance between the Earth and the Sun is 147,100,000 km.

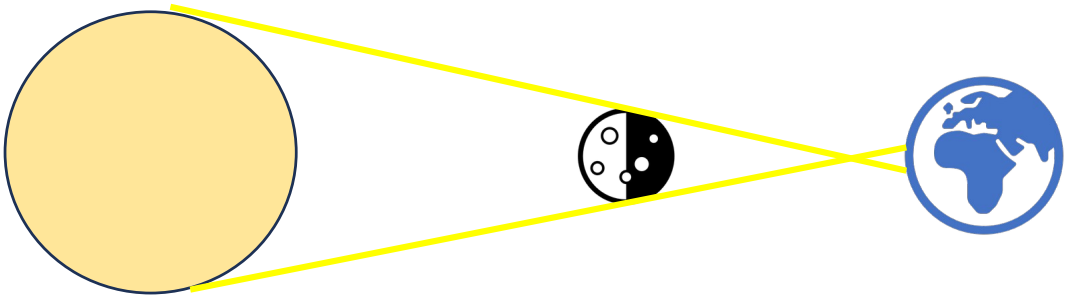


Figure not to scale.

Data from the accompanying constants sheet. The mean radii of the Sun, Moon and Earth are respectively 695,500 km, 1,737 km and 6,378 km.

Problem 3.3 Lunar Eclipse

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.7 30 points Solution p. 19

On September 7th 2025, there was a total lunar eclipse, visible from Australia, Asia, Africa and Europe. A detailed lunar eclipse figure, obtained from the website eclipsewise.com of eclipse predictions by Fred Espenak, is shown below. Based on the figure, respond to the following items.

(a) Measure the inclination between the orbit of the Moon around the Earth and the Ecliptic plane. The orbit of the Moon is assumed to be circular.

(b) During the total eclipse phase, the moon appears red in colour. Which of the following factors are necessary to explain this red colour? Write their item letters in the answer sheet.

A) Density of atmosphere

B) Albedo of Moon

C) Orbital radius of Moon

D) Albedo of Earth

E) Effective temperature of the Sun

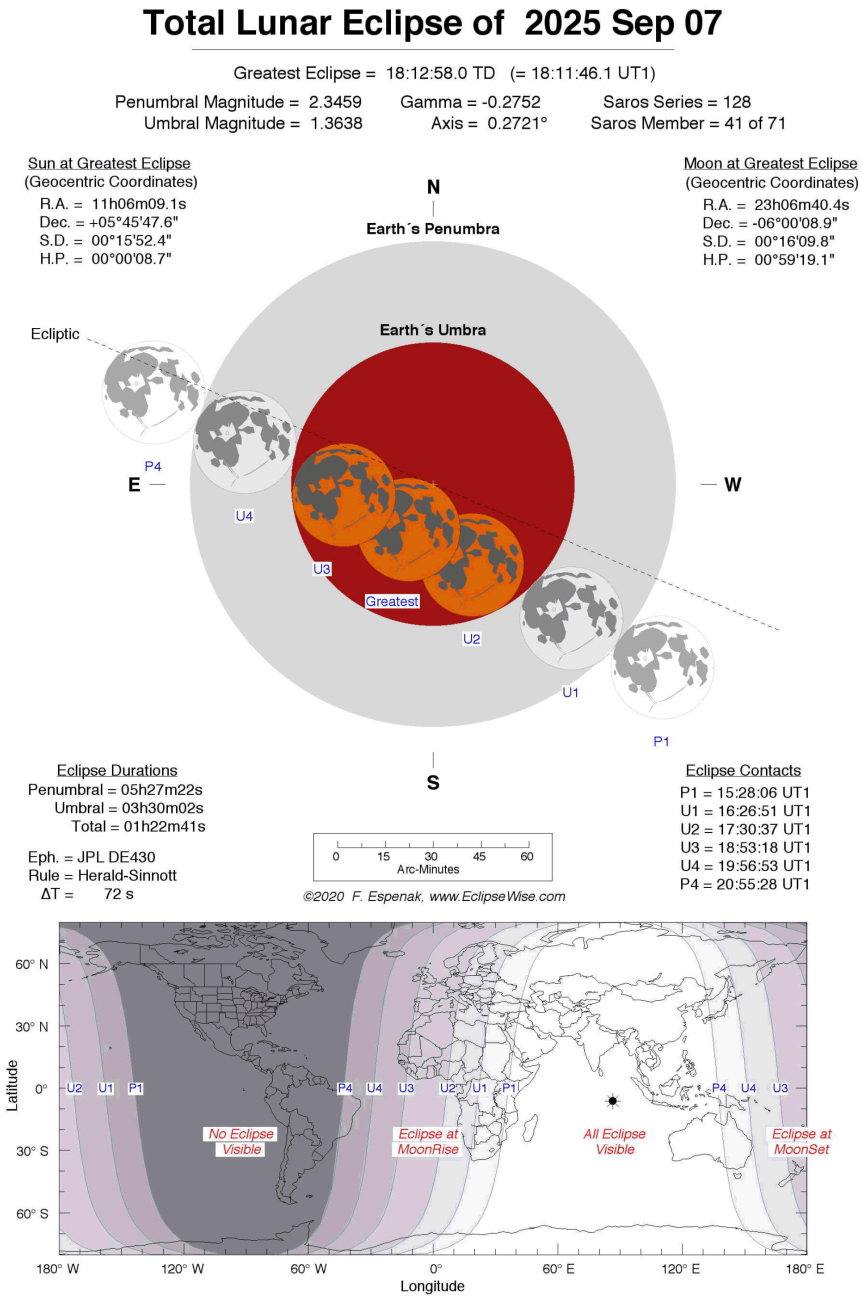
(c) Measure the angular diameter of the Moon. Knowing the value of the radius of the Moon, calculate the Earth–Moon distance during the eclipse, in kilometres.

(d) Measure the angular diameter of Earth's umbra and the angular diameter of Earth's penumbra.

(e) What is the instantaneous angular velocity of the Moon relative to the observer, during the total eclipse? Give the answer in arcminutes per hour ($'$/h).

(f) On 26 June 2029, a central total lunar eclipse will take place. This means that the centre of the disk of the Moon will pass through the axis of Earth's umbral shadow. If the angular velocity is assumed to be the same for the two eclipses, by what percentage will the 2029 total lunar eclipse last longer than the one in 2025 (percentage to be calculated of the totality duration in 2025)?

(g) Calculate the total duration of the eclipse in 2029 (from Penumbral first contact to Penumbral last contact).



COURTESY OF *21ST CENTURY CANON OF Lunar ECLIPSES*, FRED ESPENAK, ASTROPIXELS PUBLISHING, 2020

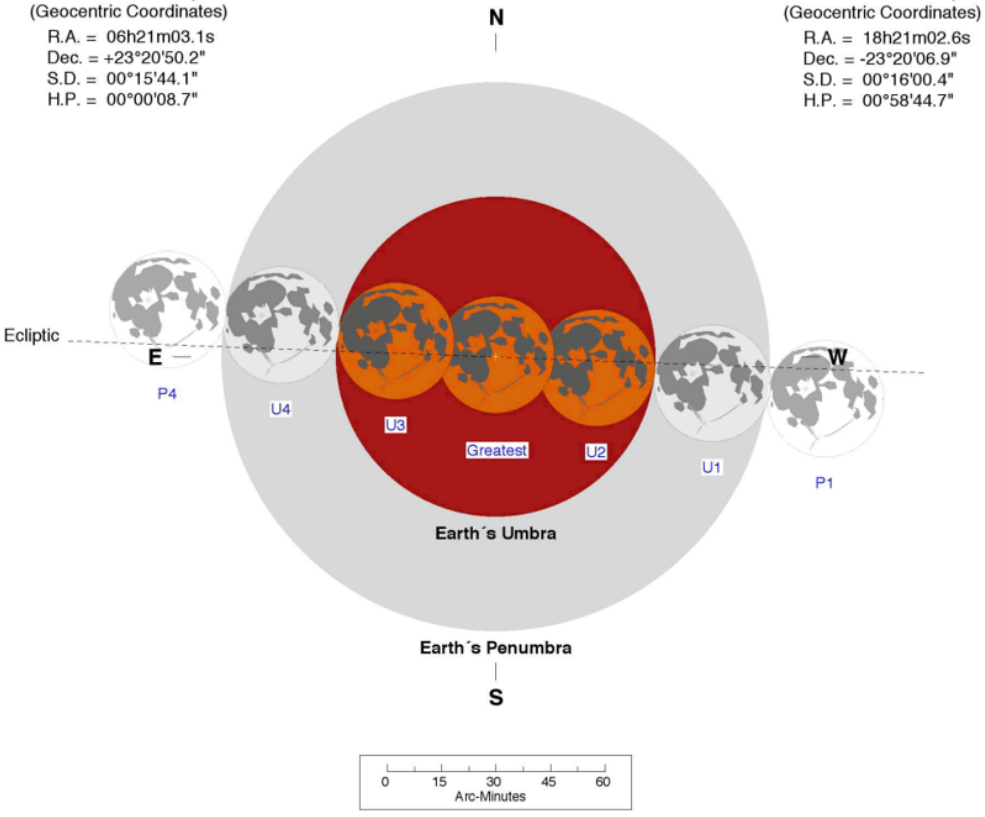
Total Lunar Eclipse of 2029 Jun 26

Greatest Eclipse = 03:23:22.5 TD (= 03:22:08.8 UT1)

Penumbral Magnitude = 2.8282 Gamma = 0.0124 Saros Series = 130
Umbral Magnitude = 1.8452 Axis = 0.0121° Saros Member = 35 of 71

Sun at Greatest Eclipse
(Geocentric Coordinates)
R.A. = 06h21m03.1s
Dec. = +23°20'50.2"
S.D. = 00°15'44.1"
H.P. = 00°00'08.7"

Moon at Greatest Eclipse
(Geocentric Coordinates)
R.A. = 18h21m02.6s
Dec. = -23°20'06.9"
S.D. = 00°16'00.4"
H.P. = 00°58'44.7"



Solutions

Solution 3.1 The Sun as Seen from Saturn

Problem p. 14

The Sun's angular diameter as observed from Saturn is 9.54 times smaller than its angular diameter from Earth.

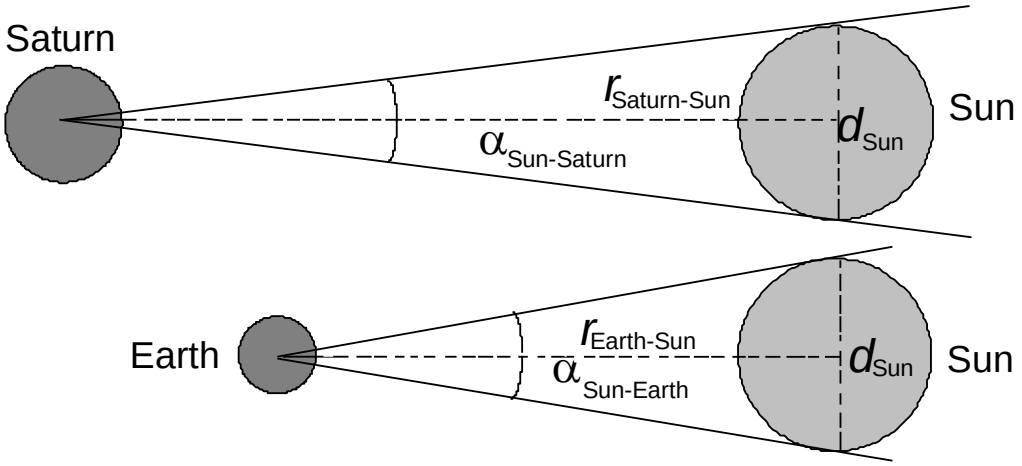


Fig. 1

Using $d_{\text{Sun}} = r_{\text{Saturn}-\text{Sun}}\alpha_{\text{Sun}-\text{Saturn}} = r_{\text{Earth}-\text{Sun}}\alpha_{\text{Sun}-\text{Earth}}$,

$$\begin{aligned}\alpha_{\text{Sun}-\text{Saturn}} &= \frac{1}{9.54}\alpha_{\text{Sun}-\text{Earth}} = \frac{32'}{9.54} \\ &\approx 3.35' \approx 0.00097\,\text{rad} \approx 0.001\,\text{rad}.\end{aligned}$$

The answer is (a).

Solution 3.2 Solar eclipse

Problem p. 14

The angular diameter of the Sun during perihelion is

$$d_1 = 2 \times \frac{695{,}500}{147{,}100{,}00 - 6{,}378} = 2 \times 0.004731 = 0.009463 \text{ rad},$$

while the angular diameter of the Moon during apogee is

$$d_2 = 2 \times \frac{1{,}737}{405{,}000 - 6{,}378} = 2 \times 0.004358 = 0.008715 \text{ rad}.$$

(6 points)

The fraction of the area covered is the ratio of the squares of these values,

$$r = \left(\frac{d_2}{d_1}\right)^2 = 0.8482 = 84.82\%.$$

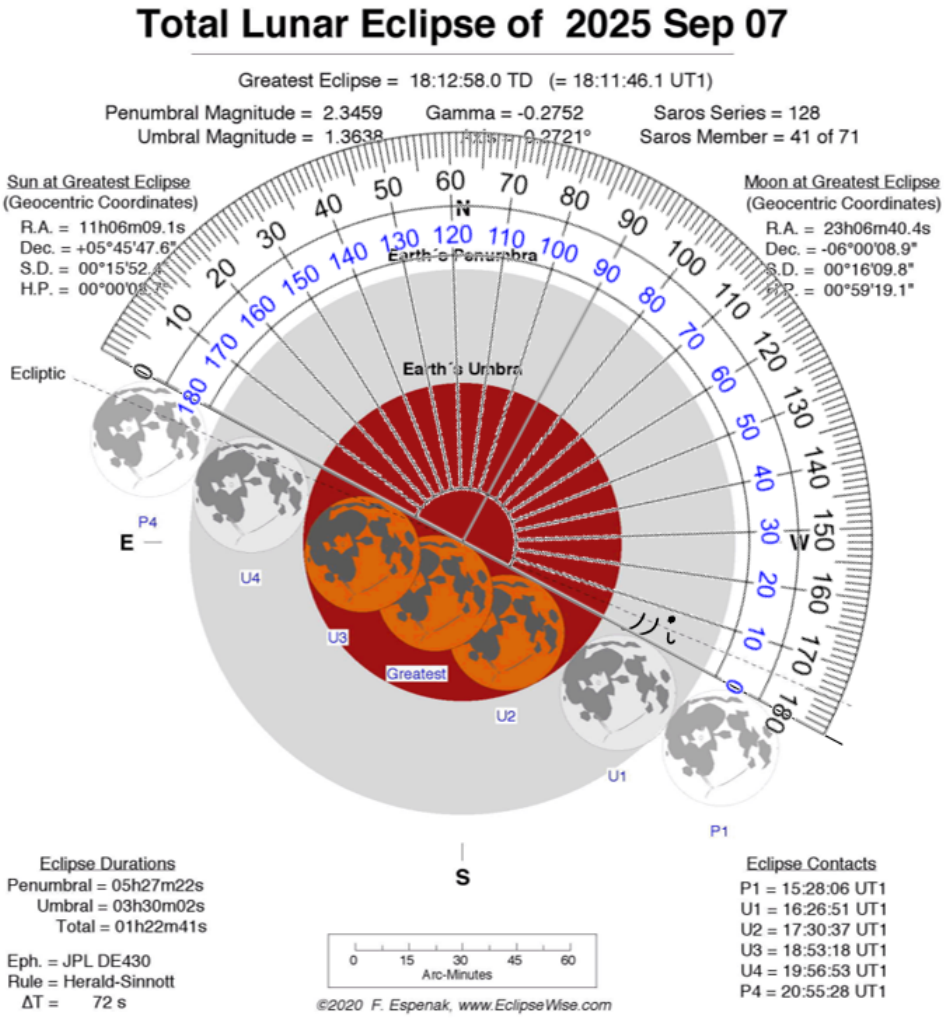
(4 points)

Solution 3.3 Lunar Eclipse

Problem p. 15

(a) The measured inclination is about 5°–6°. (4 points)

(b) A) and C). Award 2 points for each correct option and deduct 1 point for each wrong option, with a minimum of zero. (4 points)



(c) The figure scale is 2.5 cm to 60′. Distances in centimetres may change with scaling, but the angular answers remain the same. The prose gives the measured Moon diameter as 1.6 cm, while the printed calculation is

$$d_{\text{Moon}} = \frac{1.3\,\text{cm} \times 60'}{2.5\,\text{cm}} = 31.2'.$$

(4 points) Accept 31′–32′. The angular radius is

$$r_{\text{Moon}} = \frac{d_{\text{Moon}}}{2} = 15.6' \approx 16'.$$

(1 point) With $R_{\text{Moon}} = D_{\text{Earth–Moon}} \sin r_{\text{Moon}} = 1737\,\text{km}$,

$$D_{\text{Earth–Moon}} = \frac{R_{\text{Moon}}}{\sin r_{\text{Moon}}} = \frac{1737}{\sin 16'} \approx 373200\,\text{km}.$$

(2 points) *The printed marking note says "Answers between 1730 km and 1790 km are accepted (corresponding to* $15.5'$ *and* $16'$*)." This numerical range is inconsistent with the Earth–Moon distance just calculated.*

(d) The measured diameters of the umbra and penumbra are 3.7 cm and 6.3 cm. Thus

$$d_{\text{umbra}} = \frac{3.7\,\text{cm} \times 60'}{2.5\,\text{cm}} = 89' = 1°29', \qquad d_{\text{penumbra}} = \frac{6.3\,\text{cm} \times 60'}{2.5\,\text{cm}} = 151' = 2°31'.$$

(6 points) Accept $88'$–$90'$ and $150'$–$153'$ respectively.

(e) Between U2 and U3,

$$\Delta t = t_{U3} - t_{U2} = 18^{\text{h}}53^{\text{m}}18^{\text{s}} - 17^{\text{h}}30^{\text{m}}37^{\text{s}} = 1^{\text{h}}22^{\text{m}}41^{\text{s}}.$$

(0.5 point) The measured path is 1.9 cm, corresponding to $1.9 \times 60'/2.5 = 45.6'$. (0.5 point) Accept $45'$–$47'$. The angular velocity is

$$\omega_{\text{Moon}} = \frac{45.6'}{1^{\text{h}}22^{\text{m}}41^{\text{s}}} \approx 33.1'/\text{h}.$$

(1 point) Accept $32.5'$/h–$34.1'$/h.

(f) In the 2029 figure, 2.7 cm represents $60'$ and the totality path is 2.5 cm. Hence

$$\Delta l_{\text{totality}} = \frac{2.5\,\text{cm} \times 60'}{2.7\,\text{cm}} = 55.56'.$$

(1.5 points) Accept $55'$–$57'$. The duration is

$$\Delta t_{\text{totality}} = \frac{55.56'}{33.1'/\text{h}} = 1.69\,\text{h} \approx 1^{\text{h}}41^{\text{m}}.$$

(0.5 point) Accept 1.6–1.75 hours. The source also prints the range ($1^{\text{h}}36^{\text{m}}$–$1^{\text{h}}75^{\text{m}}$). The difference from 2025 is $1^{\text{h}}41^{\text{m}} - 1^{\text{h}}22^{\text{m}}41^{\text{s}} = 18^{\text{m}}19^{\text{s}}$. (0.5 point) Therefore

$$\frac{\Delta t_{\text{totality,2029}} - \Delta t_{\text{totality,2025}}}{\Delta t_{\text{totality,2025}}} = \frac{18^{\text{m}}19^{\text{s}}}{1^{\text{h}}22^{\text{m}}41^{\text{s}}} \approx 22.15\%.$$

(0.5 point) Accept 15.7%–26.6%.

(g) The P1–P4 path is measured as 8.2 cm, giving

$$\Delta l_{\text{eclipse}} = \frac{8.2\,\text{cm} \times 60'}{2.7\,\text{cm}} = 3.037° \approx 3°2'13''.$$

(2 points) Accept 3°–3.1°. The eclipse duration is

$$\Delta t_{\text{eclipse}} = \frac{3.037 \times 60'}{33.1'/\text{h}} \approx 5.5\,\text{h} \approx 5^{\text{h}}30^{\text{m}}.$$

(2 points) Accept $5^{\text{h}}17^{\text{m}}$–$5^{\text{h}}42^{\text{m}}$.

Chapter 4

Orbital Mechanics

Theory problems

Problem 4.1 The Escape Velocity

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q3 Solution p. 30

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

It is given: $V_0 \approx 30\,\mathrm{km\,s^{-1}}$, the speed of the Earth in its circular orbit around the Sun; $v_0 \approx 7.9\,\mathrm{km\,s^{-1}}$, the speed of a terrestrial satellite orbiting the Earth in a very low circular orbit (the first cosmic speed). It is known that

$$\frac{M_{\mathrm{Earth}}}{R_{\mathrm{Earth}}} \ll \frac{M_{\mathrm{Sun}}}{R_{\mathrm{Earth-Sun}}}.$$

The variation of the kinetic energy of the body, in relation to the Sun, is neglected during the motion of the body from the surface of the Earth to the limit of the gravitational attraction of the Earth.

The approximate minimum escape velocity of a body B, relative to the Sun, launched at a distance from the Sun equal to 1 AU, so that it leaves the Solar System forever, is

(a) $v_B \approx 45.52\,\mathrm{km\,s^{-1}}$.

(b) $v_B \approx 32.32\,\mathrm{km\,s^{-1}}$.

(c) $v_B \approx 22.22\,\mathrm{km\,s^{-1}}$.

(d) $v_B \approx 42.42\,\mathrm{km\,s^{-1}}$.

Problem 4.2 Balls Suspended inside a Terrestrial Satellite

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q7 Solution p. 31

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

An artificial satellite moves around the Earth in a circular orbit with radius r. We observe the satellite at the position and orientation shown in the figure. Inside the satellite are suspended, by very light wires, four identical spherical balls, each with mass m, such that the balls (b) and (c) are symmetrical to the ball (a), the balls (a) and (d) are symmetrical to the ball (c), and the difference of their distances to the center of the Earth is $\Delta r = 2d$.

It is given: M, the mass of the Earth; G, gravitational attraction constant.

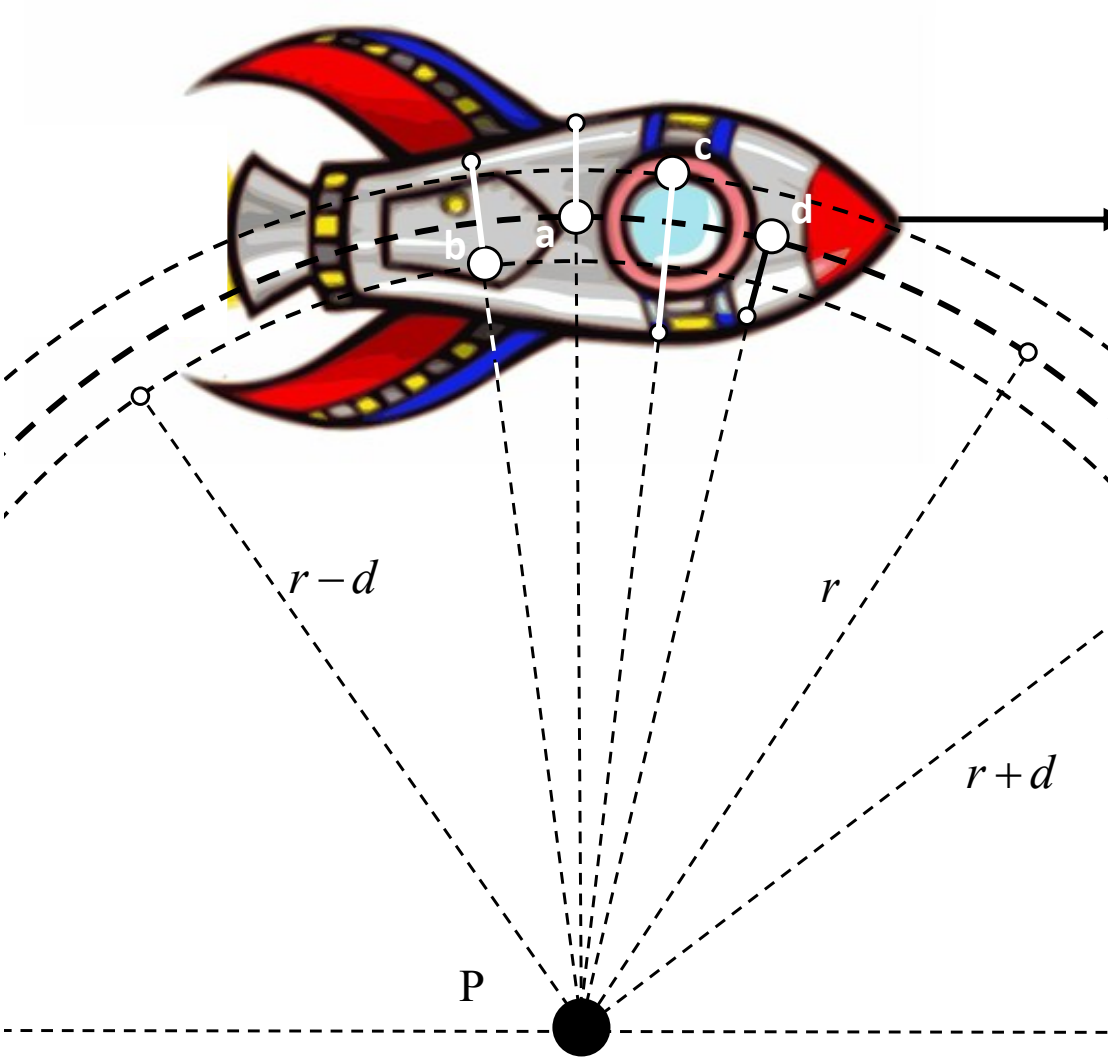


Fig. 1

The tension in each suspension wire, is

(a) $T_{(a)} = 0$, $T_{(b)} = 3G\dfrac{mMd}{r^3} > 0$, $T_{(c)} = 3G\dfrac{mMd}{r^3} > 0$, $T_{(d)} = 0$.

(b) $T_{(a)} = G\dfrac{mMd}{r^3}$, $T_{(b)} = G\dfrac{mMd}{r^3} > 0$, $T_{(c)} = G\dfrac{mMd}{r^3} > 0$, $T_{(d)} = G\dfrac{mMd}{r^3}$.

(c) $T_{(a)} = 3G\dfrac{mMd}{r^3} > 0$, $T_{(b)} = 0$, $T_{(c)} = 0$, $T_{(d)} = 3G\dfrac{mMd}{r^3} > 0$.

(d) $T_{(a)} = 2G\dfrac{mMd}{r^3} > 0$, $T_{(b)} = 2G\dfrac{mMd}{r^3} > 0$, $T_{(c)} = 2G\dfrac{mMd}{r^3} > 0$, $T_{(d)} = 2G\dfrac{mMd}{r^3} > 0$.

Problem 4.3 Spaceship to the Sun

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q8 Solution p. 31

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

In a future program, NASA plans to launch a spacecraft, aimed directly at the Sun, without a human crew, to gather information, on its way to the Sun, both about all the inner planets and, in particular, about the Sun. They are known: the distance Earth–Sun, $r_{ES} = 1.5 \times 10^{11}\,\mathrm{m}$; the period of the Earth's rotation around the Sun, $T_E = 3.15 \times 10^7\,\mathrm{s}$; the radius of the Sun, $R_S \approx r_{ES}/200$.

If the launch of the spacecraft will be done in such a way that its motion relative to the Sun is a free fall, then the approximate duration of the Earth–Sun flight is

(a) $t \approx 3.6 \times 10^5\,\mathrm{s}$.

(b) $t \approx 6.6 \times 10^6\,\mathrm{s}$.

(c) $t \approx 5.6 \times 10^6\,\mathrm{s}$.

(d) $t \approx 4.6 \times 10^6\,\mathrm{s}$.

Problem 4.4 Zodiac IOAA-J 2022

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q11 100 points Solution p. 32

In their apparent motions, the Moon and the other large planets in our solar system do not stray far from the plane of the ecliptic. Their apparent trajectories described on the celestial sphere remain contained in a region that extends symmetrically on both sides of the ecliptic, having a total width of about 18°. Figure 1 shows the heliocentric orbit of the Earth in the ecliptic plane, and Figure 2 the equivalent apparent geocentric orbit of the Sun relative to the Earth.

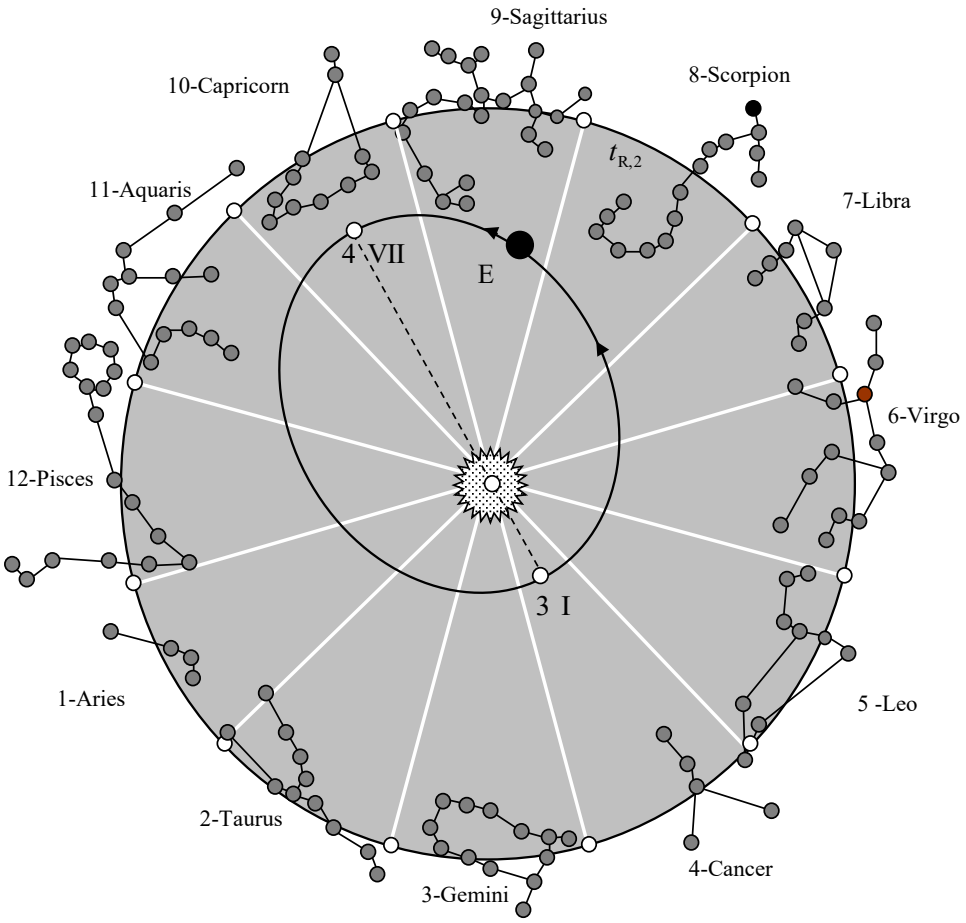


Figure 1, reproduced in enlarged format!
This image will be worked on and will be handed in with the competition sheets!

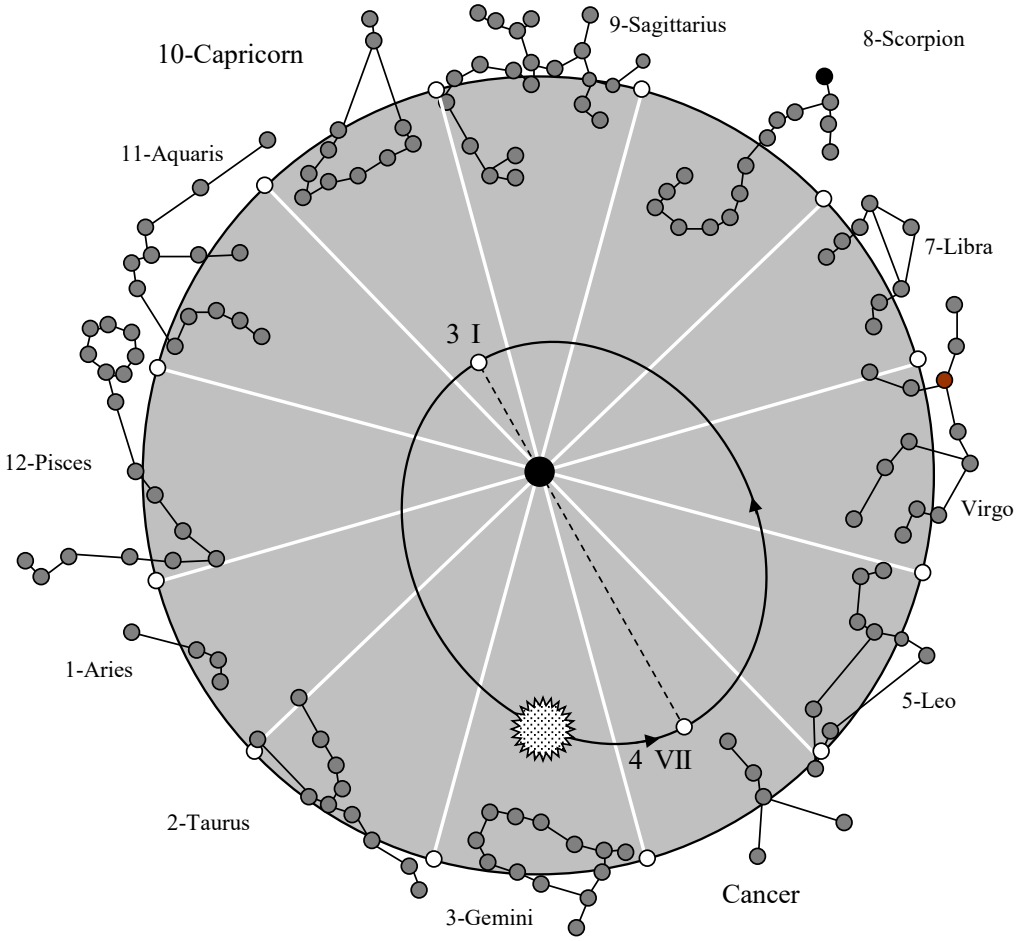


Figure 2, reproduced in enlarged format!
This image will be worked on and will be handed in with the competition sheets!

The time intervals of the evolution of the true Sun in each of the 12 constellations of the Zodiac are as follows.

1) Aries	21 III–20 IV	2) Taurus, 21 IV–20 V
3) Gemini	21 V–20 VI	4) Cancer, 21 VI–22 VII
5) Leo	23 VII–22 VIII	6) Virgo, 23 VIII–22 IX
7) Libra	23 IX–22 X	8) Scorpio, 23 X–21 XI
9) Sagittarius	22 XI–21 XII	10) Capricorn, 22 XII–21 I
11) Aquarius	22 I–19 II	12) Pisces, 20 II–21 III

The International Olympiad on Astronomy and Astrophysics for Juniors, Edition I, was to take place in Suceava, in Romania, on March 28–April 3, 2020. The event, proposed by Romania, was established at the International Olympiad of Astronomy and Astrophysics, 13th Edition, held in Hungary on August 2–10, 2019. But, given the known conditions, IOAA-jr, Edition I, is taking place in Romania only now, in November 2022!

Hint: All tasks requiring direct measurements on drawings will be performed on the enlarged pictures in source pages 5 (Figure 1) and 6 (Figure 2), reproduced above. Both pages are part of the answer sheets.

(A) Identify the following. [30 points]

1. The sign of the OIAA development, 13th Edition, Hungary, August 2–10, 2019.
2. The sign when the OIAA-J was to take place, Edition I, March 28–April 3, 2020, Romania (Edition I, proposal).
3. The sign of the OIAA-J, Edition I, November 2022.

(B) Each specified Olympic event can be considered to have taken place on the day when the Sun, viewed from Earth, was in the middle of the angular range corresponding to its sign. [20 points]

1. Determine the time interval between any two of the events, $\Delta t_{H-R,1}$, $\Delta t_{R,1-R,2}$, and $\Delta t_{H-R,2}$.
2. Estimate by direct measurements with a protractor the angles between the Earth's position vectors $\vec{r}_H$, $\vec{r}_{R,1}$, $\vec{r}_{R,2}$ relative to the Sun and the apse line Aph–Ph, namely α_H, $\alpha_{R,1}$, and $\alpha_{R,2}$.

(C) Determine the area swept out by the Earth's position vector relative to the Sun as follows. [20 points]

1. From $\vec{r}_{\text{Hungary}}$ to $\vec{r}_{\text{Romania},1}$.
2. From $\vec{r}_{\text{Romania},1}$ to $\vec{r}_{\text{Romania},2}$.
3. From $\vec{r}_{\text{Hungary}}$ to $\vec{r}_{\text{Romania},2}$.

(D) Determine the following. [30 points]

1. The Earth–Sun distance $r_{R,1}$ on the day when IOAA for Juniors, Edition I, should have taken place in Suceava in 2020. The Earth's orbit has semimajor axis $a = 149597500\,\text{km}$ and semiminor axis $b = 149580670\,\text{km}$.
2. The Earth's acceleration $a_{E,1}$ on that day, compared with the acceleration $g_{S,1}$ in the Sun's gravitational field at distance $r_{R,1}$.
3. The components of the Earth's velocity $\vec{v}_{R,1}$ parallel and perpendicular to the ellipse's major axis, $v_\parallel$ and $v_\perp$.

The Earth's angular momentum relative to the Sun and the total mechanical energy of the Earth–Sun system are

$$L = M_E b\sqrt{\frac{GM_S}{a}}, \qquad E = -G\frac{M_E M_S}{2a}.$$

It is given: $G = 6.67 \times 10^{-11}\,\text{N}\,\text{m}^2\,\text{kg}^{-2}$, $M_S = 1.989 \times 10^{30}\,\text{kg}$, and the Earth's revolution period $T = 365.256$ days.

Problem 4.5 The Rocket that Destroys the Threatening Asteroid

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q12 100 pointsSolution p. 37

With the help of a rocket R, launched from a spaceship N, an asteroid A must be destroyed, approaching the Earth threateningly, coming right in the direction of the center of the Earth, as shown in Figure 1. The figure shows the positions of the spacecraft and asteroid when the rocket is launched. The spacecraft has constant Earth-relative velocity $\vec{v}_N$ along the spacecraft–asteroid line. The asteroid has constant Earth-relative velocity $\vec{v}_A$, making an angle α with the spacecraft's direction. During the rocket's flight until impact, consider the Earth at rest relative to the Sun, and the motions of the rocket R, spacecraft N, and asteroid A rectilinear and uniform.

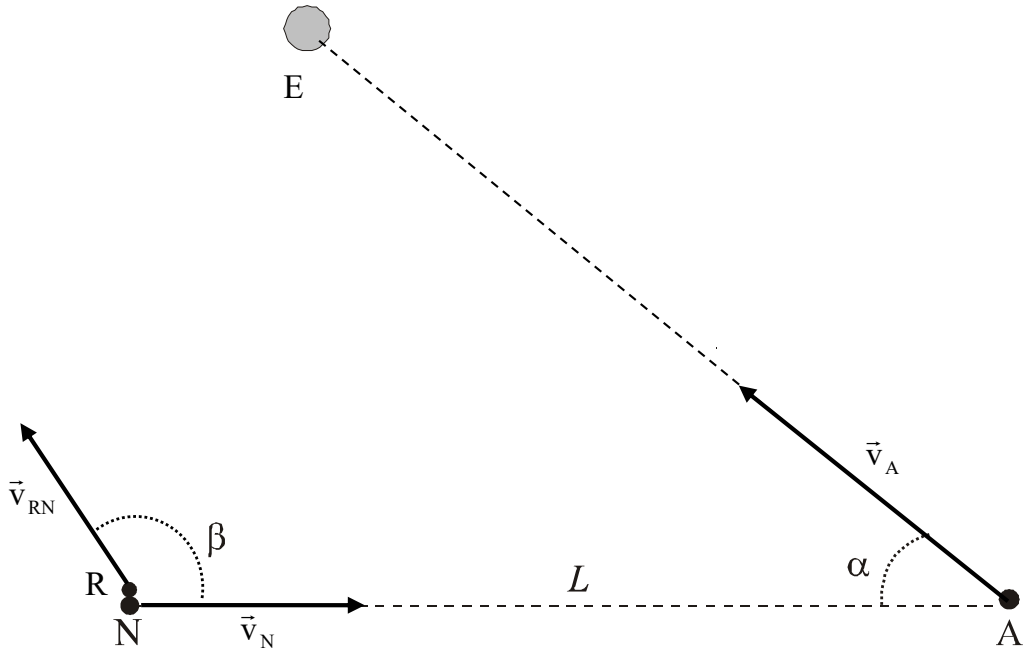


Fig. 1

We assume that the direction of absolute displacement of rocket R is perpendicular to the direction of absolute movement of asteroid A. Determine the angle β at which rocket R was launched relative to the direction of motion of spacecraft N.

Problem 4.6 Planet orbits

Junior IOAA 2023, Volos, Greece Theory problems Q2 10 points Solution p. 38

Two planets, A and B, orbit a star S in circular, coplanar orbits, in the same direction. The orbital radius of planet A is 9 times smaller than the orbital radius of planet B. The orbital period of planet A is 10 Earth days. An astronomer on Earth realizes that at 10:00 am on 26/9/2023, an observer on planet B would observe a transit of planet A in front of star S. When is the next transit due to planet A to be seen by an observer on planet B? All dates/times are given on Earth times. The masses of planets A and B are very small compared to the mass of star S.

Problem 4.7 Satellites

Junior IOAA 2023, Volos, Greece Theory problems Q12 40 points Solution p. 39

The term geostationary (GEO) refers to a satellite whose orbit is on the equatorial plane, its orbit is circular, and its orbital period is equal to Earth's period of revolution around its axis. For this problem, all orbits around Earth are considered circular. An important characteristic of the orbit of a satellite is the elevation angle of the satellite, which is defined as the angle at which the satellite is visible above the local horizon for an observer on Earth.

(A) What is the maximum latitude the geostationary satellite is visible at? [12 points]

Geostationary satellites, because of their distance from the center of the Earth, observe only a fraction of the Earth's surface. If a satellite needs to view a larger fraction of the Earth, it must change its orbit using rockets.

(B) The satellite moves from its location and observes the Earth having an angular diameter of

10°. What is the distance of the satellite from the center of the Earth? [10 points]

(C) How much does the total (kinetic and potential) energy of the satellite need to change to move from the geostationary orbit to the one described in question (B)? The satellite has a mass of 1000 kg. [10 points]

(D) What is the ratio of the gravitational acceleration g on the Earth's surface compared to the gravitational acceleration at the location described in question (B)? [8 points]

Data from the accompanying constants sheet. $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$, $M_E = 5.972 \times 10^{24}\,\mathrm{kg}$, $R_E = 6378$ km, and one sidereal day is $23^{\mathrm{h}}56^{\mathrm{m}}04^{\mathrm{s}}$.

Problem 4.8 Energy and speed in an elliptical orbit

Junior IOAA 2024, Kathmandu, Nepal Theory problems T2 15 points Solution p. 40

Starting from the law of conservation of energy, derive an expression for the total mechanical energy of a body of mass m on an elliptical orbit, in terms of G, the mass of the object being orbited M, and the semi-major axis of the orbit a. Use your derivation to then find the ratio of the speeds at perihelion and aphelion, for an object that has an orbit with eccentricity e.

Problem 4.9 Altitude of a geostationary satellite

Junior IOAA 2024, Kathmandu, Nepal Theory problems T5 15 points Solution p. 40

Nepal has ambitions to launch a geostationary satellite of its own. If the satellite is to maintain a geostationary orbit such that, when viewed from Kathmandu, it is at the prime meridian. What would be its altitude when observed from Kathmandu? You can neglect refraction due to the atmosphere and other smaller effects. *Supplied examination data.* Kathmandu's latitude is 27.7172°N, Earth's mean radius is 6378 km and mass is 5.972×10^{24} kg, $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$, and one sidereal day is $23^{\mathrm{h}}56^{\mathrm{m}}04^{\mathrm{s}}$.

Problem 4.10 Free fall inside a gas cloud

Junior IOAA 2024, Kathmandu, Nepal Theory problems T9 15 points Solution p. 41

Suppose you are travelling on a spacecraft that encounters a uniformly spherically distributed gas cloud with radius R, and volume density ρ. At your moment of approach, you decelerate till you are at rest with respect to the gas cloud and then turn off your engines. You may neglect gas friction in this problem. **Note.** Gravitational potential inside the uniform solid sphere is given by

$$V(r) = -\frac{GM}{R^3}\left(\frac{3R^2}{2} - \frac{r^2}{2}\right),$$

where r is the distance from the center of the gas cloud. You may also find the following fact useful. For an equation of the form $a = -k^2 r$, where k is some constant and a is acceleration, the time period of the oscillation is given by $T = 2\pi/k$.

1. Once inside the gas cloud, how long will it take you to return to your original position if you are in free fall inside the cloud?

2. Start from the law of conservation of energy. Suppose you are tired of being inside the gas cloud after investigating it for a while and want to escape it (to infinity). What is your required speed if you are initially located in the center of the cloud and are at rest?

Problem 4.11 Speed of Asteroid

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.2 10 points Solution p. 41

An asteroid moves in an elliptical orbit around the Sun. Its maximum orbital speed is 19.00 km/s and its minimum orbital speed is 14.00 km/s.

(a) Compute the eccentricity of the orbit.

(b) What is the speed of the asteroid when it is exactly at either of the end points of the minor axis of its orbit?

(c) A comet from near the outer edge of our Solar System reaches perihelion with an orbital speed of 90.00 km/s. At this instant, the asteroid is at aphelion and the comet, Sun, and asteroid are aligned (comet–Sun–asteroid). Compute the distance between the comet and the asteroid.

Problem 4.12 Roche Limit, Hill Sphere, Lagrange Points

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.8 40 points Solution p. 42

The following astronomical data are given.

- The masses of Mars M_M and the Sun $M_\odot$ are expressed relative to Earth's mass $M_\oplus$, with $M_M = 0.107 M_\oplus$ and $M_\odot = 333054 M_\oplus$.
- The mean Mars–Sun distance is $r_{MS} = 1.524\,\text{au}$ and Mars has a circular orbit.
- The mean Earth–Sun distance is $r_{ES} = 1\,\text{au} = 1.496 \times 10^{11}\,\text{m}$ and Earth has a circular orbit.
- Mars's moon Phobos has mass $M_{\text{Phobos}} = 1.5 \times 10^{-8} M_M$.
- Earth's sidereal period is $P_{\text{Sid}\oplus} = 1$ year $= 365.2564$ days.
- Phobos has average radius $R_{\text{Phobos}} = 12.5\,\text{km}$ and orbital radius $d_{\text{Phobos}M} = 9380\,\text{km}$.
- Phobos is spiraling inward towards Mars at the constant rate of 1.8 cm/yr.
- You may use $(1+x)^n \approx 1 + nx$ when $x \ll 1$.
- The Roche limit is the boundary inside which the tidal force of the planet on a small mass is more than its self-gravity.

(a) Derive the formula for the L1 Lagrange point (distance between secondary body and the L1 point) and calculate its numerical value for the Sun–Mars system. Find the ratio of Phobos's orbital radius to this distance.

(b) Derive the formula for the L2 Lagrange point (distance between secondary body and the L2 point) and calculate its numerical value for the Mars–Phobos system.

(c) Derive the formula for the Roche limit of the Mars–Phobos system. How long would it take Phobos to reach this limit? It is known that it approaches the surface of the planet during its rotation and will either collide with it or begin to disintegrate, forming a ring. The satellite is considered to be solid.

(d) Calculate the distance l between the Lagrange points L4 and L5 for the Sun–Mars system.

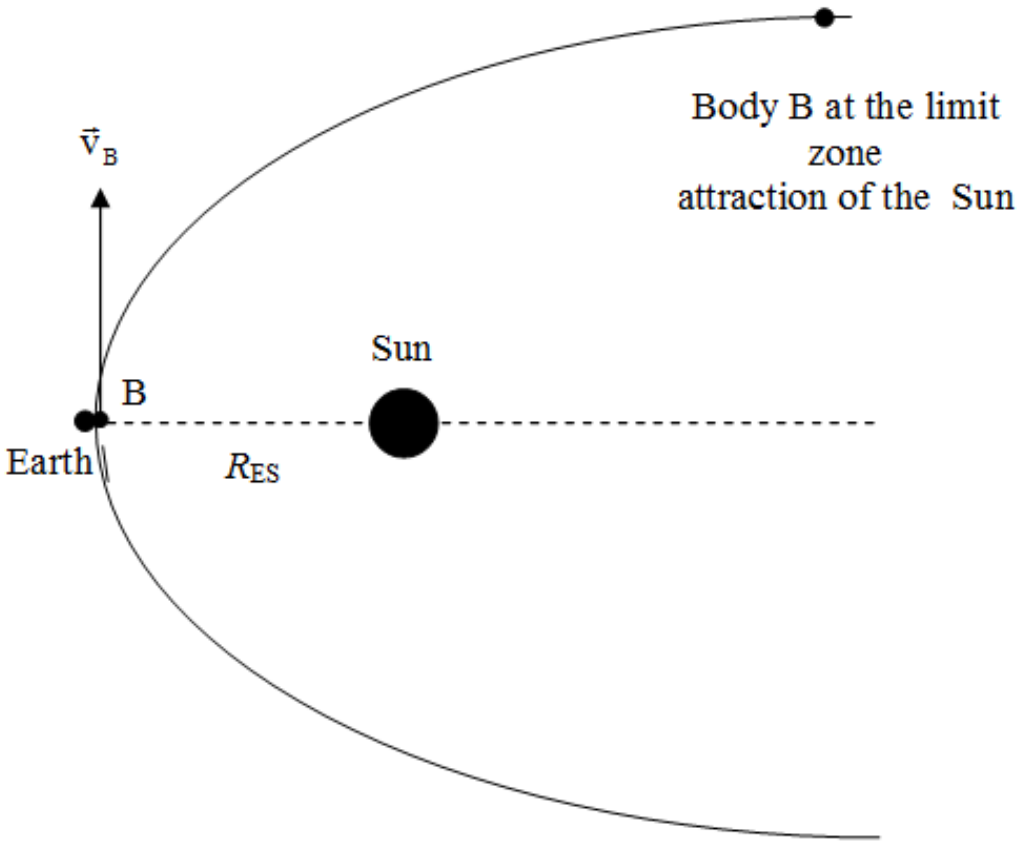
Solutions

Solution 4.1 The Escape Velocity

Problem p. 21

[The S3 file restates the problem as an Earth-relative third cosmic speed, while its derivation and final 42.42 km/s result use the Sun-relative speed asked for in the question paper.]

Let v_B be the speed of body B at launch, relative to the Sun, such that the body reaches the limit of the Sun's gravitational attraction at rest.



Conservation of mechanical energy gives

$$\frac{mv_B^2}{2} - K\frac{mM_E}{R_E} - K\frac{mM_S}{R_{ES}} = 0.$$

Here K denotes the gravitational constant. With the supplied assumption $M_E/R_E \ll M_S/R_{ES}$,

$$\frac{mv_B^2}{2} - K\frac{mM_S}{R_{ES}} = 0, \qquad v_B = \sqrt{\frac{2KM_S}{R_{ES}}} = \sqrt{2}\,V_0.$$

The Earth's circular orbital speed is

$$V_0 = \frac{2\pi R_{ES}}{T_{ES}} \approx 30\,\mathrm{km\,s^{-1}}, \qquad v_B \approx 42.42\,\mathrm{km\,s^{-1}}.$$

The trajectory relative to the Sun is parabolic, with the Sun at its focus. The answer is (d).

Solution 4.2 Balls Suspended inside a Terrestrial Satellite

Problem p. 21

The spacecraft keeps a fixed orientation and is a rigid body, so all its points have the same angular velocity ω. The orbital equation gives

$$K\frac{m_{NC}M}{r^2} = m_{NC}\omega^2 r, \qquad \omega^2 = \frac{KM}{r^3}.$$

The balls' centripetal accelerations are

$$a_{cp(a)} = a_{cp(d)} = \omega^2 r, \quad a_{cp(b)} = \omega^2(r-d), \quad a_{cp(c)} = \omega^2(r+d).$$

For (a), gravity alone supplies the centripetal force,

$$T_{(a)} = K\frac{mM}{r^2} - m\omega^2 r = 0.$$

For (b), use $d \ll r$ and $(1-d/r)^{-2} \approx 1 + 2d/r$,

$$F_{ag(b)} = K\frac{mM}{(r-d)^2} \approx K\frac{mM}{r^2}\left(1+\frac{2d}{r}\right),$$

$$T_{(b)} = F_{ag(b)} - m\omega^2(r-d) = 3K\frac{mMd}{r^3} > 0.$$

For (c), $(1+d/r)^{-2} \approx 1 - 2d/r$,

$$F_{ag(c)} \approx K\frac{mM}{r^2}\left(1-\frac{2d}{r}\right), \qquad T_{(c)} = m\omega^2(r+d) - F_{ag(c)} = 3K\frac{mMd}{r^3} > 0.$$

For (d),

$$T_{(d)} = m\omega^2 r - K\frac{mM}{r^2} = 0.$$

Thus (a) and (d) are weightless with untensioned wires, while (b) and (c) have tensioned wires. The answer is (a), with $K = G$.

Solution 4.3 Spaceship to the Sun

Problem p. 23

Imagine a launch with initial Sun-relative speed slightly lower than the Earth's orbital speed. The spacecraft follows an ellipse with the Sun at the focus opposite the launch point. As the initial speed decreases, the ellipse becomes increasingly elongated. For an orbit tangent to the Sun,

$$r_{\min} = R_S \approx \frac{r_{ES}}{200}.$$

Its semiminor axis is negligible compared with the semimajor axis. The trajectory can therefore be approximated by a straight free fall to the Sun. The initial Sun-relative speed must be zero, so the Earth-relative launch velocity is opposite to the Earth's orbital velocity with the same magnitude,

$$v_{ES} = \frac{2\pi r_{ES}}{T_E} = \frac{2\times 3.14\times 1.5\times 10^{11}}{3.15\times 10^7} \approx 3\times 10^4\,\mathrm{m\,s^{-1}} = 30\,\mathrm{km\,s^{-1}}.$$

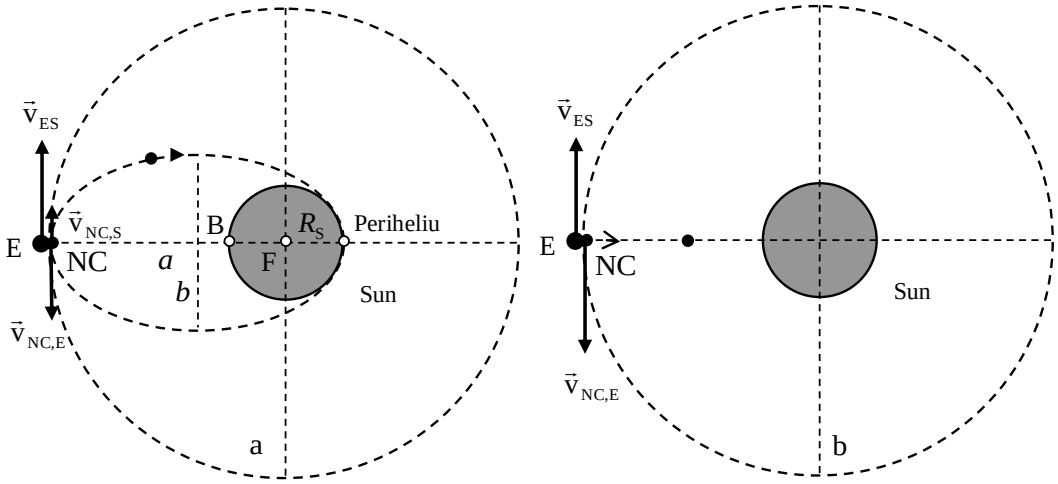


Fig. 1

Kepler's third law gives

$$\frac{T_E^2}{r_{ES}^3} = \frac{T_{NC}^2}{(r_{ES}/2)^3}, \qquad T_{NC} = \frac{T_E}{2\sqrt{2}}.$$

The fall is half the period,

$$t = \frac{T_{NC}}{2} = \frac{T_E}{4\sqrt{2}} \approx 5.6 \times 10^6\,\text{s}.$$

The answer is (c).

Solution 4.4 Zodiac IOAA-J 2022

Problem p. 23

[The supplied official key leaves the 2022 midpoint date and a separate time interval blank. Its printed 223-day interval disagrees with its stated dates; the subsequent swept areas use the source intervals. The numerical values are retained as printed.]

(A) The positions of the Earth on its orbit and the Sun's projections into the zodiac belt at the three events are shown in Figures 3 and 4.

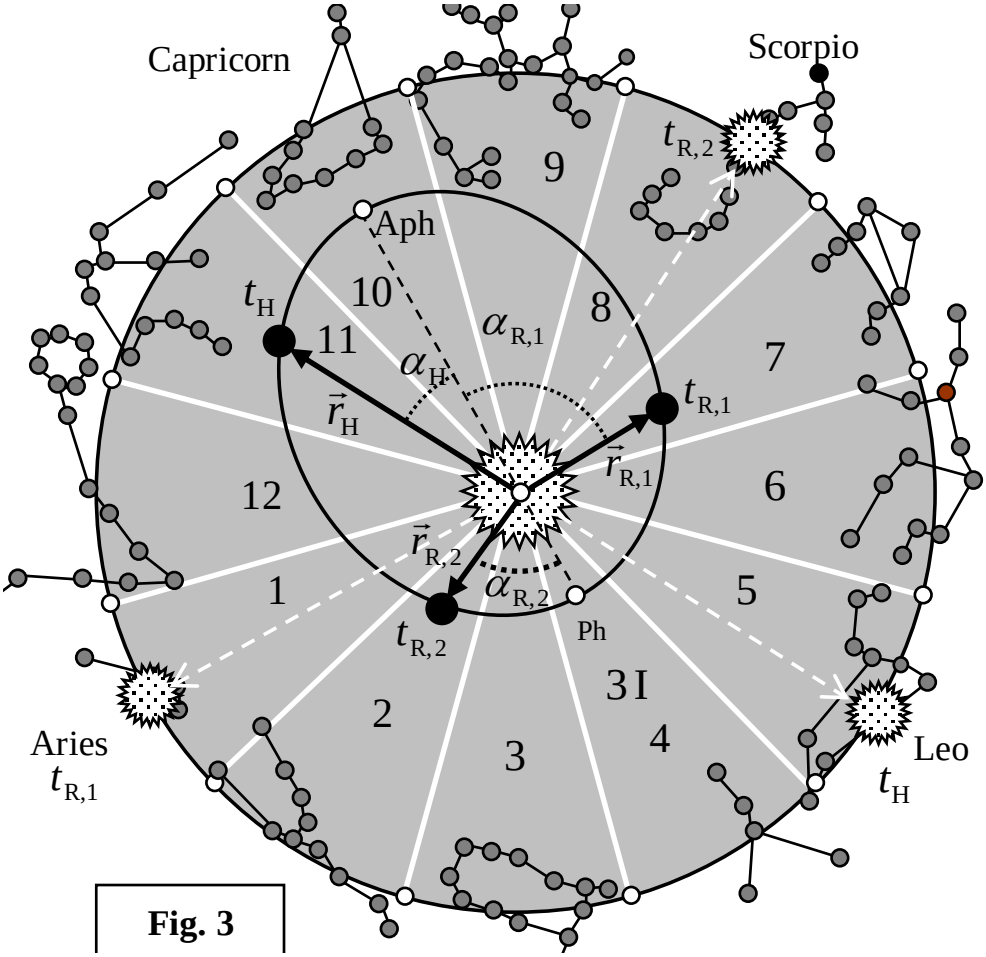

Capricorn
Scorpio
Aries
$t_{R,1}$
Leo
t_H
$t_{R,2}$
Aph
Ph
t_H
$t_{R,1}$
$t_{R,2}$
$\alpha_{R,1}$
α_H
$\alpha_{R,2}$
$\vec{r}_H$
$\vec{r}_{R,1}$
$\vec{r}_{R,2}$
1
2
3
4
5
6
7
8
9
10
11
12
3 I

Fig. 3

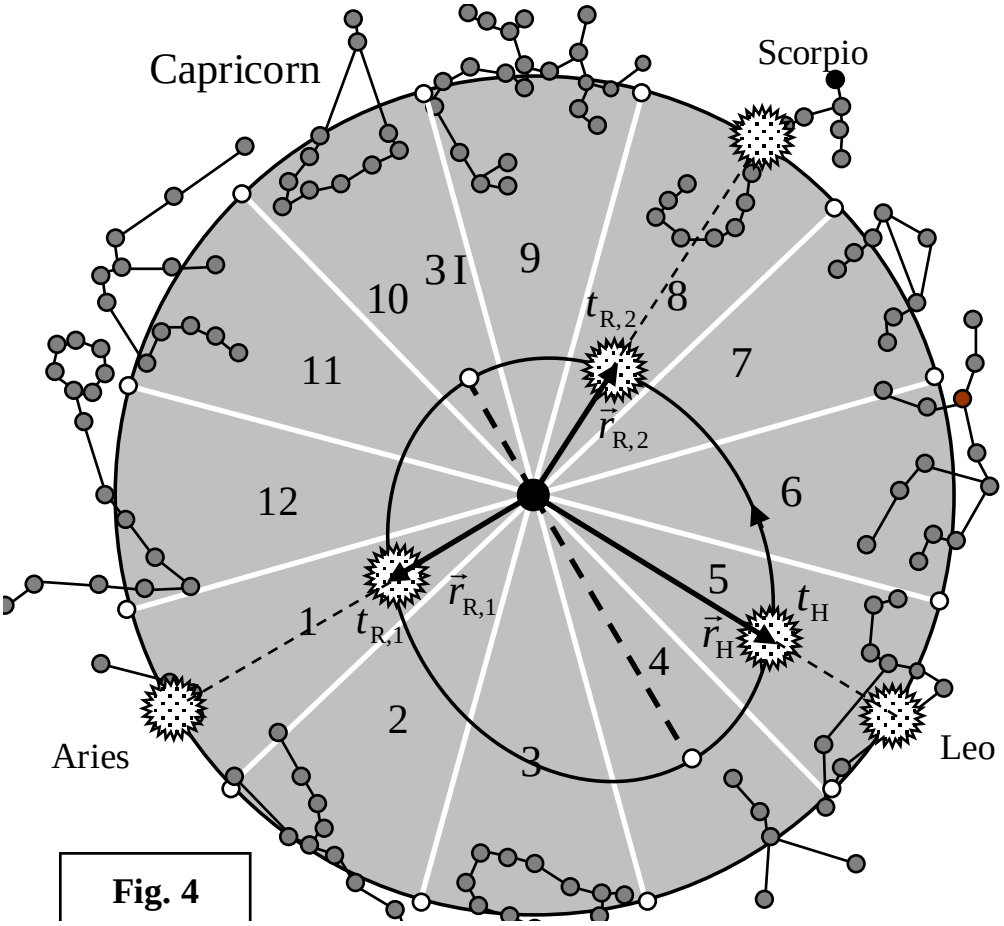


Fig. 4

The source identifies Leo for the Hungary announcement, with midpoint August 6, 2019; Aries for the proposed Romanian event, with midpoint April 4, 2020; and Scorpio for November 2022.

(B) The source lists

$$t_H = 6 \text{ August } 2019, \qquad t_{R,1} = 4 \text{ April } 2020,$$

and leaves $t_{R,2}$ blank. It gives

$$\begin{aligned} \Delta t_{H-R,1} &= t_{R,1} - t_H = 223 \text{ days}, \\ \Delta t_{R,1-R,2} &= t_{R,2} - t_{R,1} = [\text{blank in source}], \\ \Delta t_{H-R,2} &= 223 \text{ days} + 886 \text{ days} = 1109 \text{ days}. \end{aligned}$$

Direct protractor measurements in Figure 3 give

$$\alpha_H \approx 30^\circ, \qquad \alpha_{R,1} \approx 90^\circ, \qquad \alpha_{R,2} \approx 60^\circ,$$

so $\vec{r}_{R,1}$ is perpendicular to the apse line.

(C) The areal velocity is constant,

$$\Omega = \frac{\Delta A}{\Delta t} = \text{constant}, \qquad \frac{A_{\text{ellipse}}}{T} = \frac{\Delta A}{\Delta t}, \qquad A_{\text{ellipse}} = \pi ab.$$

Using the intervals printed in the solution, $T = 365.256$ days, $a = 149597500\,\text{km}$, $b =$

149580670 km, and $\pi = 3.14$,

$$\Delta A_{H-R,1} = \frac{223}{365.256}\pi ab = 4.2897 \times 10^{16}\,\mathrm{km}^2,$$
$$\Delta A_{R,1-R,2} = \frac{886}{365.256}\pi ab = 1.704 \times 10^{17}\,\mathrm{km}^2,$$
$$\Delta A_{H-R,2} = \frac{1109}{365.256}\pi ab = 2.133 \times 10^{17}\,\mathrm{km}^2.$$

(D) The ellipse geometry is shown in Figure 5.

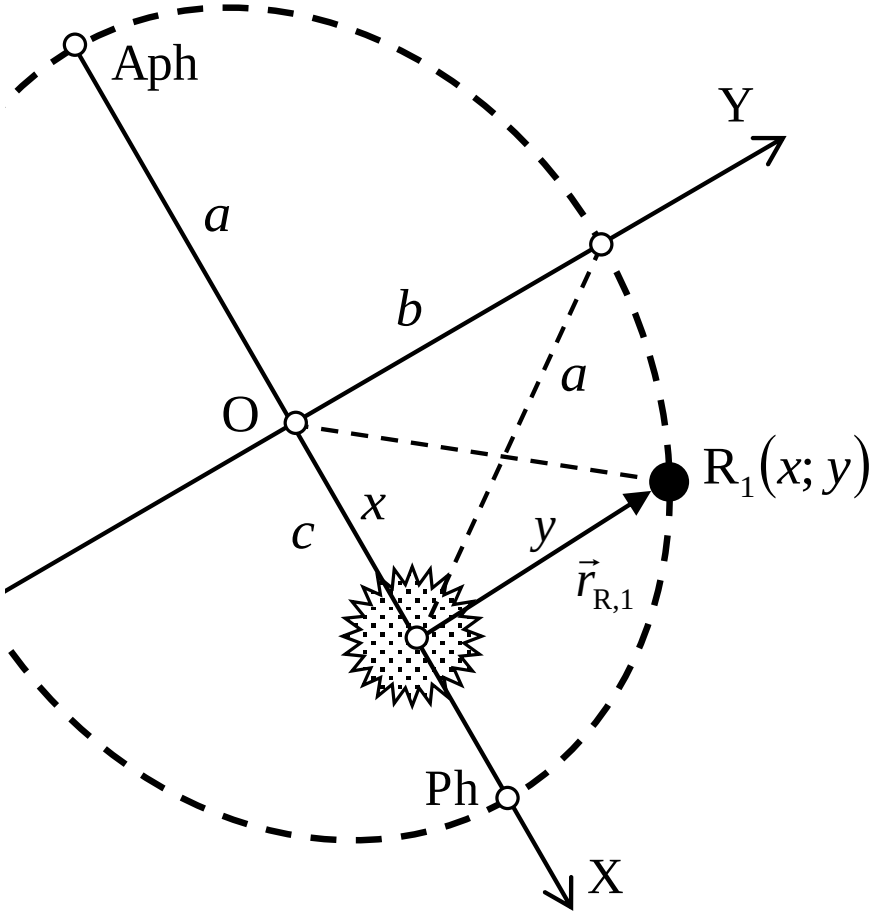


Fig. 5

1. Since $x = c$, $y = r_{R,1}$, and $c^2 = a^2 - b^2$,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad \frac{a^2 - b^2}{a^2} + \frac{r_{R,1}^2}{b^2} = 1,$$

so

$$r_{R,1} = \frac{b^2}{a} = 149563841.9\,\mathrm{km}.$$

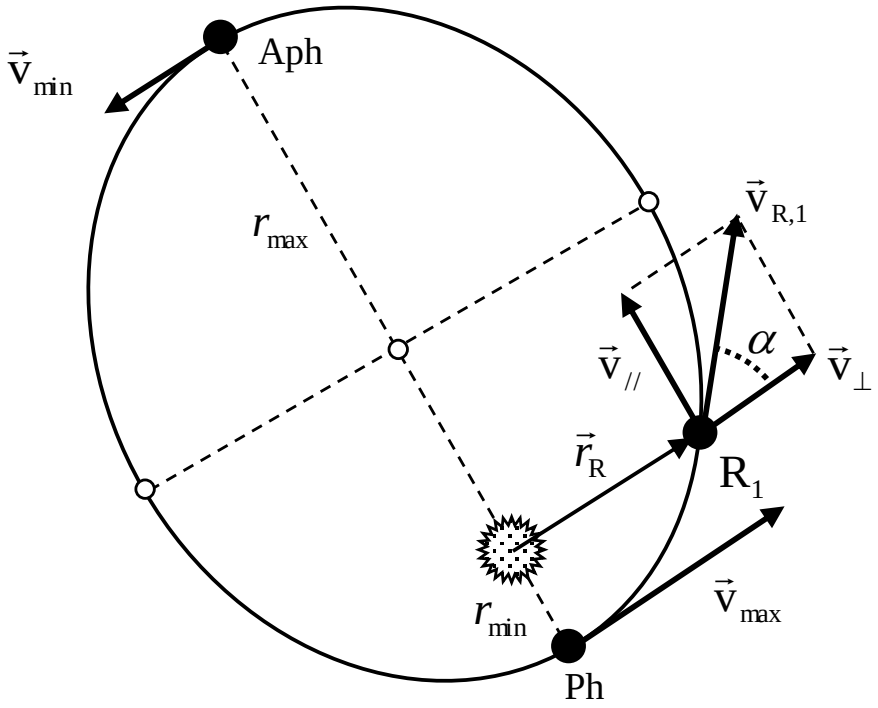
2. The force equation and solar gravitational acceleration are

$$K\frac{M_E M_S}{r_{R,1}^2} = M_E a_{E,1}, \qquad a_{E,1} = \frac{KM_S}{r_{R,1}^2} \approx 0.006\,\mathrm{m\,s^{-2}},$$

$$g_{S,1} = \frac{KM_S}{r_{R,1}^2} = a_{E,1}.$$

Here the solution denotes the gravitational constant by K.

3. Figure 6 shows the velocity components.



Conservation of angular momentum gives

$$L_{R,1} = r_{R,1} M_E v_{R,1} \sin\alpha = r_{R,1} M_E v_{\parallel}, \qquad L = M_E b \sqrt{\frac{K M_S}{a}}.$$

With $r_{R,1} = b^2/a$,

$$v_{\parallel} = \frac{a}{b}\sqrt{\frac{K M_S}{a}} = 29782.9\,\mathrm{m\,s^{-1}} = 29.7829\,\mathrm{km\,s^{-1}}.$$

Conservation of mechanical energy gives

$$\frac{M_E v_{R,1}^2}{2} - K\frac{M_E M_S}{r_{R,1}} = -K\frac{M_E M_S}{2a},$$

$$v_{R,1} = \sqrt{K M_S\left(\frac{2}{r_{R,1}} - \frac{1}{a}\right)} = 29.7862\,\mathrm{km\,s^{-1}},$$

and the source concludes

$$v_{\perp} = \sqrt{v_{R,1}^2 - v_{\parallel}^2} = 0.4433\,\mathrm{km\,s^{-1}}.$$

Solution 4.5 The Rocket that Destroys the Threatening Asteroid

Problem p. 26

[The supplied solution describes minimum-time interception, whose perpendicular geometry is assumed directly in Q12. Its relative-velocity subtraction and squared-speed expression use v_A where the spacecraft velocity v_N would be expected. The derivation and final expression are reproduced as printed.]

The supplied solution formulates the perpendicular-intercept geometry as a minimum-time interception. Figure 2 shows the absolute rocket path NC perpendicular to the asteroid path AC.

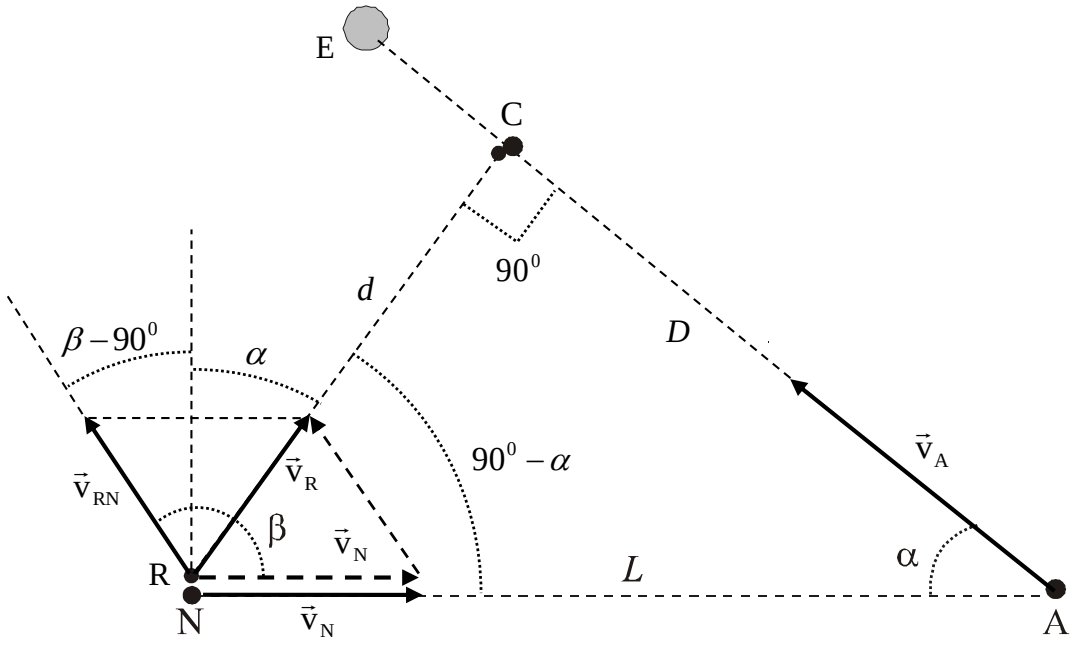


Fig. 2

With $L = NA$, the distances traveled are

$$D = v_A t_{\min} = L\cos\alpha, \qquad d = v_R t_{\min} = L\sin\alpha.$$

Thus

$$t_{\min} = \frac{L\cos\alpha}{v_A} = \frac{L\sin\alpha}{v_R}, \qquad v_R = v_A\tan\alpha.$$

The source then prints

$$\vec{v}_{RN} = \vec{v}_R - \vec{v}_A, \qquad v_{RN}^2 = v_R^2 + v_A^2 - 2v_R v_N\cos(90^\circ - \alpha) = v_R^2 + v_A^2 - 2v_R v_N\sin\alpha.$$

The transverse components in the figure give

$$v_R\cos\alpha = v_{RN}\cos(\beta - 90^\circ) = v_{RN}\sin\beta,$$

so

$$\sin\beta = \frac{v_R}{v_{RN}}\cos\alpha = \frac{v_A}{v_{RN}}\sin\alpha.$$

Substituting $v_R = v_A\tan\alpha$ into the source's squared-speed expression,

$$\begin{aligned} v_{RN}^2 &= v_A^2\frac{\sin^2\alpha}{\cos^2\alpha} + v_A^2 - 2v_A v_N\frac{\sin^2\alpha}{\cos\alpha} \\ &= \frac{v_A^2}{\cos^2\alpha} - 2v_A v_N\frac{\sin^2\alpha}{\cos\alpha}, \end{aligned}$$

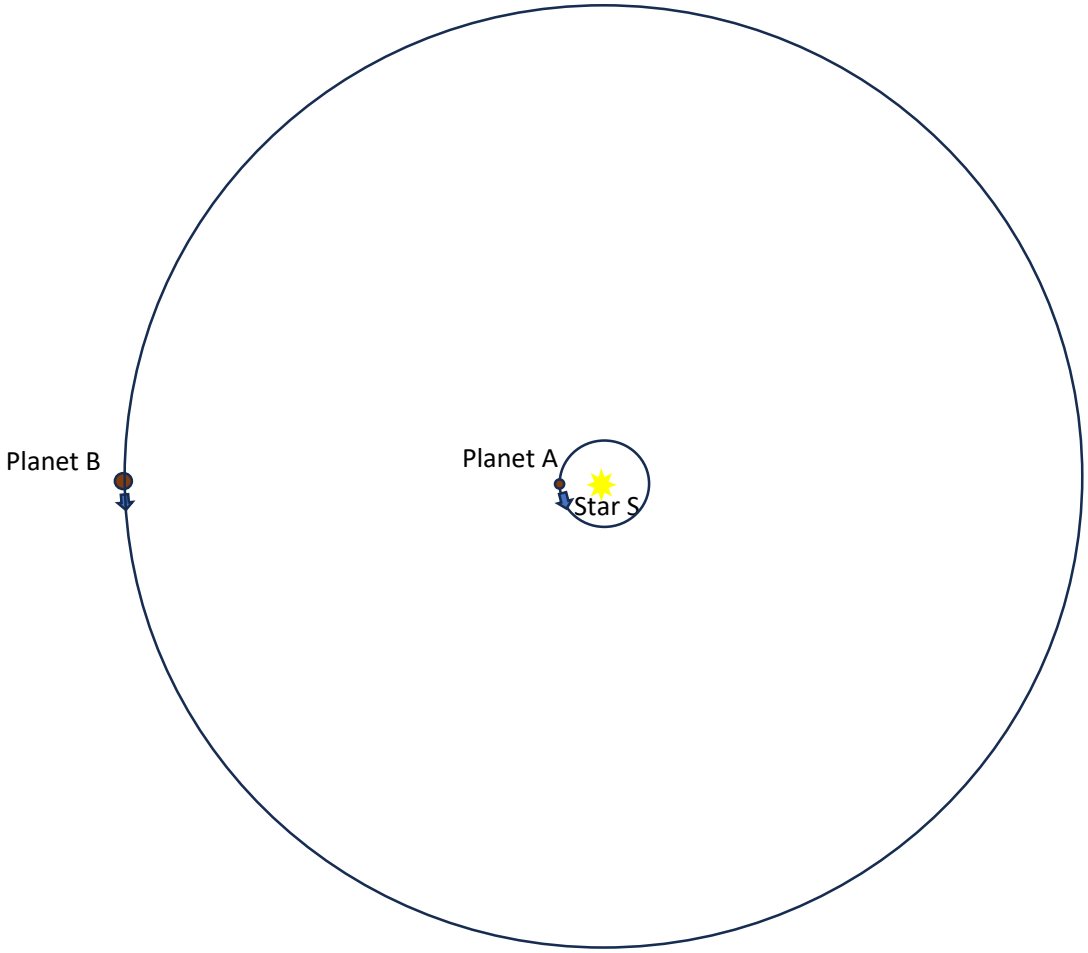
$$v_{RN} = \frac{v_A}{\cos\alpha}\sqrt{1 - 2\frac{v_N}{v_A}\sin^2\alpha\cos\alpha}.$$

The official final result is

$$\boxed{\sin\beta = \frac{\sin\alpha\cos\alpha}{\sqrt{1 - 2(v_N/v_A)\sin^2\alpha\cos\alpha}}}.$$

Solution 4.6 Planet orbits

Problem p. 27



The ratio of the orbits is 9. According to Kepler's law, the ratio of the squares of the periods is equal to the ratio of the cubes of the radii of the orbits,

$$\left(\frac{r_1}{r_2}\right)^3 = \left(\frac{P_1}{P_2}\right)^2.$$

(3 points)

Substituting the numbers, we obtain the period of planet 2,

$$P_2 = P_1\left(\frac{r_2}{r_1}\right)^{3/2} = 10 \times 3^3 = 270 \text{ days}.$$

(2 points)

The angular velocity is $\omega = 2\pi/P$ and the orbital phase is $\varphi = \omega t$. A transit will occur when $\varphi_1 - \varphi_2 = 2k\pi$, with k an integer. The first one will occur for $k = 1$. Therefore,

$$2\pi t \left(\frac{1}{10} - \frac{1}{270}\right) = 2\pi, \qquad t = 10.3846 \text{ days}.$$

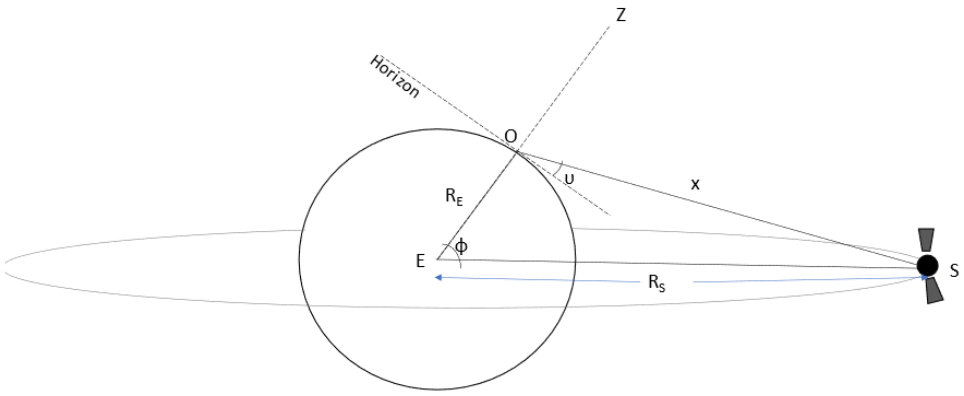
This is 10 days, 9 hours and 14 minutes after the original transit, thus on 6 October 2023, 7:14 pm. (5 points)

Solution 4.7 Satellites

Problem p. 27

(A) First evaluate the radius of the geostationary orbit using the sidereal period $T = 23^{\text{h}}56^{\text{m}}4^{\text{s}}$,

$$T^2 = \frac{4\pi^2 R_S^3}{GM_E} \quad \Longrightarrow \quad R_S = 4.2163 \times 10^7 \text{ m} = 42163 \text{ km}.$$



In the figure, E is the center of Earth, S the satellite, O the observer on Earth, and Z the observer's zenith. The latitude is φ and the elevation angle is υ. $EO = R_E$ is Earth's radius, $ES = R_S$ the satellite's orbital radius, and $OS = x$ the observer–satellite distance. The limits of the visible latitude range occur when $\upsilon = 0$. The sine law gives

$$\frac{\sin \widehat{EOS}}{R_S} = \frac{\sin \widehat{OSE}}{R_E}, \qquad \frac{\sin 90^\circ}{R_S} = \frac{\sin(180^\circ - (90^\circ + \varphi))}{R_E}.$$

Hence $\varphi = \cos^{-1}(R_E/R_S) \simeq 81.3^\circ$. The maximum latitudes are 81.3° N and 81.3° S. (12 points)

(B) Find the location where $\widehat{OSE} = 5^\circ$ and $\upsilon = 0$. The triangle SOE is right-angled, with $\widehat{EOS} = 90^\circ$. Therefore,

$$\sin \widehat{OSE} = \frac{R_E}{R_S} = \sin 5^\circ = 0.0872, \qquad R_S = \frac{6378}{0.0872} = 73179 \text{ km}.$$

(10 points)

(C) The total energy is

$$E_{\text{tot}} = \tfrac{1}{2}mv^2 - G\frac{M_E m}{r}.$$

The centripetal acceleration relation $v^2/r = GM/r^2$ gives

$$E_{\text{tot}} = -\frac{1}{2}G\frac{M_E m}{r}.$$

Substitution at the two locations gives $E_{\text{tot},1} = -4.727 \times 10^9$ J and $E_{\text{tot},2} = -2.723 \times 10^9$ J. The energy has increased by 2.00×10^9 J. (10 points)

(D) Gravitational acceleration is inversely proportional to the square of the radius, $g = GM/r^2$. Thus

$$\frac{g_E}{g_B} = \left(\frac{r_B}{r_E}\right)^2 = \left(\frac{73179000}{6378000}\right)^2 = 131.6.$$

(8 points)

Solution 4.8 Energy and speed in an elliptical orbit

Problem p. 28

[The official key displays the reciprocal of the speed ratio requested in the question. Its relation is retained.]

The perihelion distance is $r_p = a(1-e)$ and the aphelion distance is $r_a = a(1+e)$. (2 points) The law of conservation of energy gives

$$E = -\frac{GMm}{r_p} + \frac{mv_p^2}{2} = -\frac{GMm}{r_a} + \frac{mv_a^2}{2}.$$

(2 points) From conservation of angular momentum,

$$mv_a r_a = mv_p r_p, \qquad v_a = v_p \frac{r_p}{r_a}.$$

(1 point) Substitute velocities and rearrange the first equation. An alternative expression making v_p the subject is also correct.

$$v_a^2 \left(1 - \frac{r_a^2}{r_p^2}\right) = 2GM \left(\frac{1}{r_a} - \frac{1}{r_p}\right).$$

(3 points) Substitute into the equation for the total mechanical energy,

$$E = -\frac{GMm}{r_a} + \frac{GMmr_p^2}{r_p^2 - r_a^2} \left(\frac{r_p - r_a}{r_a r_p}\right).$$

After simplifying or any equivalent formulation,

$$E = -\frac{GMm}{2a} \left(\frac{2}{1+e} - \frac{1-e}{1+e}\right).$$

(3 points) Finally, $E = -GMm/(2a)$. (2 points) Substituting r_a and r_p in the energy equations gives $v_a/v_p = (1-e)/(1+e)$. One can also use angular momentum directly. (2 points)

Solution 4.9 Altitude of a geostationary satellite

Problem p. 28

[The question says prime meridian, while the official solution assumes the satellite is on Kathmandu's local meridian.]

Firstly calculate the geocentric orbit height using Newton's law of gravitation and $T = 23$ hours 56 minutes 4 seconds.

$$\frac{GMm}{d^2} = \frac{mv^2}{d}, \qquad v = \frac{2\pi d}{T}.$$

(3 points) Solving gives $d = (GMT^2/(4\pi^2))^{1/3}$. (1 point) Substitute the correct values to obtain $d = 4.216 \times 10^7$ m. (1 point) Construct a triangle. The law of sines gives

$$\frac{\sin(90^\circ + a)}{d} = \frac{\sin(90^\circ - (\phi + a))}{r},$$

where r is Earth's radius, a the altitude, and ϕ Kathmandu's latitude. (5 points) Simplifying,

$$\cos a \, \frac{r}{d} = \cos\phi \cos a - \sin\phi \sin a,$$

$$1 - \tan\phi \tan a = \frac{r}{d\cos\phi}.$$

(2 points) Rearranging and solving,

$$a = \tan^{-1}\left(\frac{1 - r/(d\cos\phi)}{\tan\phi}\right).$$

(2 points) Plugging in the values gives $a = 57.64^\circ$. (1 point)

Solution 4.10 Free fall inside a gas cloud

Problem p. 28

[The official escape-energy equation omits a mass factor in its internal-potential term. The published equation and final speed are retained.]

1. The acceleration due to gravity becomes $a = -\frac{4}{3}G\pi\rho r$. (2 points) This is $a = -k^2 r$ with $k = \sqrt{\frac{4}{3}G\pi\rho}$. It is the equation for a harmonic oscillator, as given in the hint. Its period is $T = 2\pi/k$. (1 point) Thus $T = \sqrt{3\pi/(G\rho)}$. (2 points)

2. From the expression for V, the potential difference between $r = R$ and $r = 0$ is $\Delta V_{\text{int}} = GM/(2R)$. (3 points) Finally, to escape,

$$\frac{1}{2}mv_{\text{esc}}^2 - \frac{4}{3}Gm\rho\pi R^2 - \Delta V_{\text{int}} = 0.$$

(2 points) Collecting terms gives $v_{\text{esc}} = 2R\sqrt{G\pi\rho}$. (5 points)

Solution 4.11 Speed of Asteroid

Problem p. 29

(a) If a is the semi-major axis, M_S the mass of the Sun and G the gravitational constant,

$$v_{\max} = \sqrt{\frac{GM_S}{a}\frac{1+e}{1-e}}, \qquad v_{\min} = \sqrt{\frac{GM_S}{a}\frac{1-e}{1+e}}.$$

(1.5 points) Then $k = v_{\max}/v_{\min} = (1+e)/(1-e)$ and $k(1-e) = 1+e$, so

$$e = \frac{k-1}{k+1} = \frac{v_{\max} - v_{\min}}{v_{\max} + v_{\min}}.$$

(0.5 point) Numerically, $e = (19-14)/(19+14) \approx 0.15$. (1 point)

(b) At distance r from the Sun,

$$v(r) = \sqrt{GM_S\left(\frac{2}{r} - \frac{1}{a}\right)}.$$

(1 point) At either co-vertex, the asteroid is at the intersection between the ellipse and its semi-minor axis, so $r = a$. (1 point) Thus

$$v(a) = \sqrt{\frac{GM_S}{a}} = \sqrt{v_{\max} v_{\min}} \approx 16.31\,\text{km/s}.$$

(1 point)

(c) For a comet from the outer edge of the Solar System, the orbit can be assumed parabolic. Its total energy is zero, so

$$v = \sqrt{\frac{2GM_\odot}{d}}, \qquad d_{\min} = \frac{2GM_\odot}{v_{\max}^2}.$$

(1.5 points) Numerically,

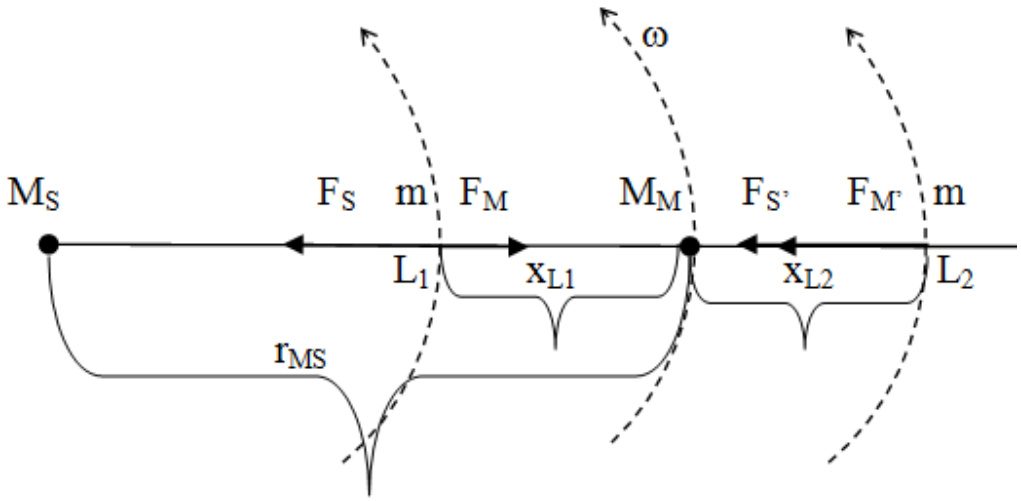
$$d_{\min} = \frac{2 \times 6.673 \times 10^{-11} \times 1.989 \times 10^{30}}{(9 \times 10^4)^2} \approx 0.22\,\text{au}.$$

(0.5 point) For the asteroid, $a = 3.34\,\text{au}$ and $r_{\max} = a(1+e) = 3.84\,\text{au}$. (1.5 points) The comet–asteroid distance is $D = d_{\min} + r_{\max} = 0.22 + 3.84 = 4.06\,\text{au}$. (0.5 point)

Solution 4.12 Roche Limit, Hill Sphere, Lagrange Points

Problem p. 29

(a) For the Lagrange points L1 and L2, use the following diagram.



For Mars in its orbit around the Sun, $F_{\text{cf}} = F_{SM}$,

$$M_M \omega^2 r_{MS} = G\frac{M_M M_\odot}{r_{MS}^2}.$$

(2 points) Hence $\omega = \sqrt{GM_\odot / r_{MS}^3}$. (1 point) For a mass m at L1,

$$m\omega^2(r_{MS} - x_{L1}) = G\frac{mM_\odot}{(r_{MS} - x_{L1})^2} - G\frac{mM_M}{x_{L1}^2}.$$

(2 points) Substituting ω^2 and rearranging,

$$\frac{M_\odot}{r_{MS}^3}(r_{MS}-x_{L1}) = \frac{M_\odot}{(r_{MS}-x_{L1})^2} - \frac{M_M}{x_{L1}^2},$$
$$M_\odot\left(1-\frac{x_{L1}}{r_{MS}}\right)^3 = M_\odot - M_M\left(\frac{r_{MS}}{x_{L1}}-1\right)^2.$$

Since $x_{L1} \ll r_{MS}$, use

$$\left(1-\frac{x_{L1}}{r_{MS}}\right)^3 \approx 1-3\frac{x_{L1}}{r_{MS}}$$

(2 points) and $r_{MS}/x_{L1} - 1 \approx r_{MS}/x_{L1}$. (1 point) Thus

$$3M_\odot\frac{x_{L1}}{r_{MS}} = M_M\left(\frac{r_{MS}}{x_{L1}}\right)^2, \qquad x_{L1}^3 = \frac{M_M}{3M_\odot}r_{MS}^3,$$

so $x_{L1} = r_{MS}\sqrt[3]{M_M/(3M_\odot)}$. (2 points) Numerically,

$$x_{L1} = 1.524\times1.496\times10^{11}\sqrt[3]{\frac{0.107M_\oplus}{3\times333054M_\oplus}} = 1.08\times10^9\,\mathrm{m} = 1.08\times10^6\,\mathrm{km}.$$

(2 points) The ratio is $d_{\mathrm{Phobos}M}/x_{L1} \approx 9\times10^{-3}$. (1 point)

(b) For a mass m at L2,

$$m\omega^2(r_{MS}+x_{L2}) = G\frac{mM_\odot}{(r_{MS}+x_{L2})^2} + G\frac{mM_M}{x_{L2}^2}.$$

(2 points) As before,

$$M_\odot\left(1+\frac{x_{L2}}{r_{MS}}\right)^3 = M_\odot + M_M\left(\frac{r_{MS}}{x_{L2}}+1\right)^2.$$

For $x_{L2} \ll r_{MS}$,

$$\left(1+\frac{x_{L2}}{r_{MS}}\right)^3 \approx 1+3\frac{x_{L2}}{r_{MS}}$$

(2 points) and $r_{MS}/x_{L2}+1 \approx r_{MS}/x_{L2}$. (1 point) Therefore $3M_\odot x_{L2}/r_{MS} = M_M(r_{MS}/x_{L2})^2$, yielding

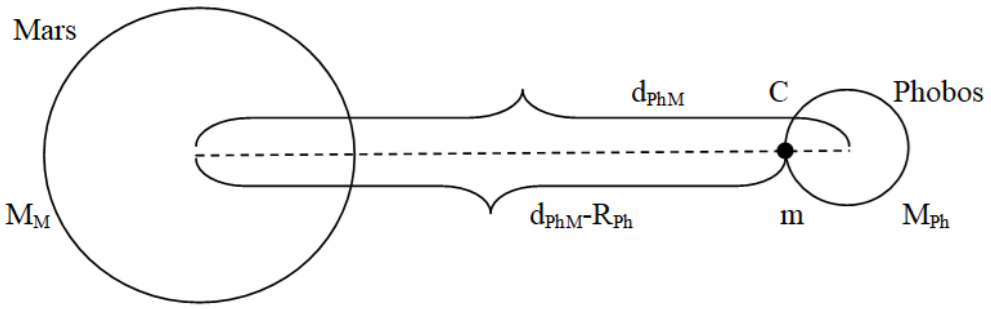
$$x_{L2} = r_{MS}\sqrt[3]{\frac{M_M}{3M_\odot}}.$$

(2 points) For Mars–Phobos,

$$x_{L2} = d_{\mathrm{Phobos}M}\sqrt[3]{\frac{M_{\mathrm{Phobos}}}{3M_M}} = 9380\sqrt[3]{\frac{1.5\times10^{-8}M_M}{3M_M}} \approx 16\,\mathrm{km}.$$

(2 points) L2 is close to the surface of Phobos, whose radius is 12.5 km.

(c)



The tidal force is the difference between the attraction of Mars on mass m at the satellite's near surface and on the same mass at its centre. Writing $d = d_{\text{Phobos}M}$ and $R = R_{\text{Phobos}}$,

$$F_{\text{tidal}} = G\frac{mM_M}{(d-R)^2} - G\frac{mM_M}{d^2}.$$

(4 points) For $R \ll d$, $(1 - R/d)^{-2} \approx 1 + 2R/d$. (2 points) Consequently

$$F_{\text{tidal}} = G\frac{mM_M}{d^2}\left(1 + \frac{2R}{d} - 1\right) = 2G\frac{mM_M R}{d^3}.$$

(1 point) Phobos's attraction is $F_{\text{grav}} = GmM_{\text{Phobos}}/R^2$. (1 point) The Roche limit satisfies $F_{\text{tidal}} = F_{\text{grav}}$. (1 point) Therefore

$$d_{\text{Roche}}^3 = 2\frac{M_M}{M_{\text{Phobos}}}R^3, \qquad d_{\text{Roche}} = R\sqrt[3]{2\frac{M_M}{M_{\text{Phobos}}}}.$$

(2 points) Numerically, $d_{\text{Roche}} = 12.5\sqrt[3]{2/(1.5\times10^{-8})} = 6386\,\text{km}$. (2 points) The inward distance is $\Delta d = 9380 - 6386 = 2994\,\text{km}$. (1 point) The required time is

$$\Delta t = \frac{\Delta d}{v} = \frac{2994000}{1.8\times10^{-2}}\,\text{yr} = 1.67\times10^8\,\text{yr}.$$

(1 point)

Alternative solution. Model Phobos as two spheres, each of mass $M_{\text{Phobos}}/2$ and density equal to that of Phobos. (3 points) Their radius follows from $2(4\pi r^3/3) = 4\pi R^3/3$, giving $r = R/\sqrt[3]{2}$. (1 point) The mutual gravitational force is

$$F_{\text{grav}} = G\frac{M_{\text{Phobos}}^2}{4(2r)^2}.$$

(2 points) The difference in the force due to Mars is

$$\Delta F = G\frac{M_M M_{\text{Phobos}}}{2(d-r)^2} - G\frac{M_M M_{\text{Phobos}}}{2(d+r)^2}.$$

(2 points) Applying the approximation,

$$\begin{aligned}\Delta F &\approx G\frac{M_M M_{\text{Phobos}}}{2d^2}\left(1 + \frac{2r}{d} - 1 + \frac{2r}{d}\right)\\ &= G\frac{2M_M M_{\text{Phobos}} r}{d^3}.\end{aligned}$$

(2 points) Equating the forces gives

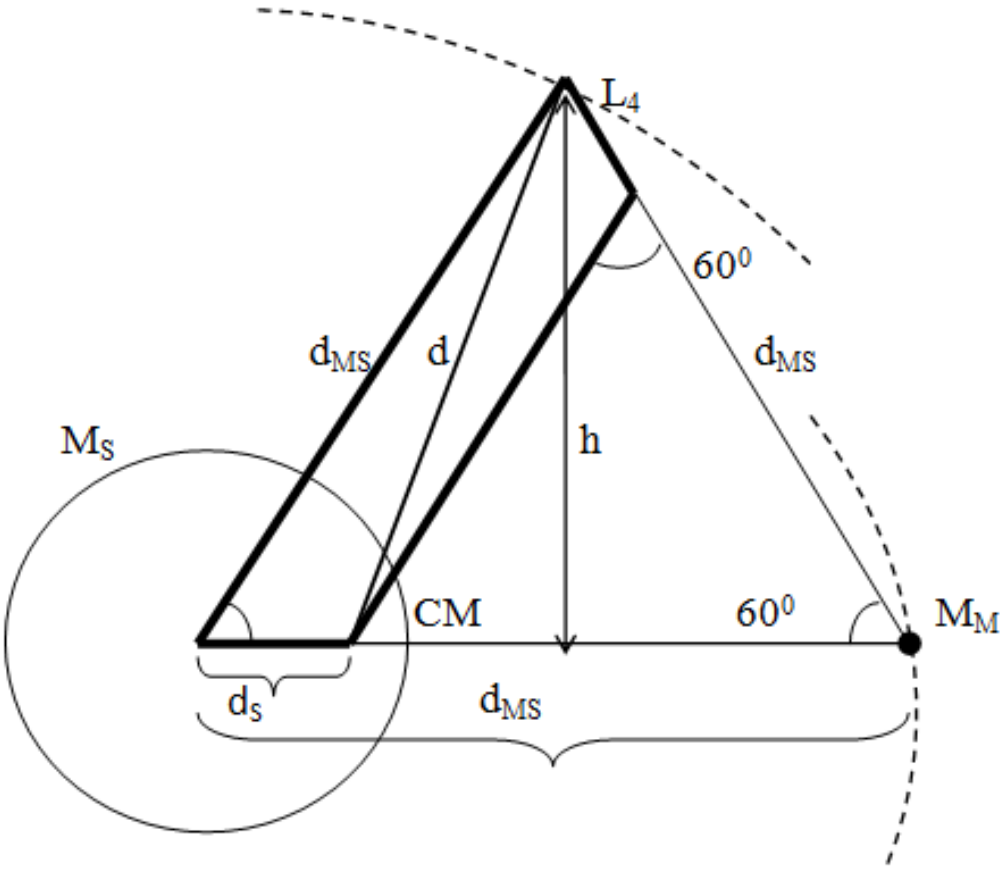
$$G\frac{2M_M M_{\text{Phobos}} r}{d_{\text{Roche}}^3} = G\frac{M_{\text{Phobos}}^2}{4(2r)^2}.$$

(2 points) The printed key continues with

$$\frac{32r^3}{d_{\text{Roche}}^3} = \frac{M_{\text{Phobos}}}{M_M}, \qquad d_{\text{Roche}} = 2R\sqrt[3]{2\frac{M_M}{M_{\text{Phobos}}}}.$$

(1 point) This gives $d_{\text{Roche}} = 12772\,\text{km}$. (1 point) On this model Phobos is already inside the Roche limit. (1 point) The official key awards full credit for this alternative model.

(d) Let d_S be the distance from the Sun's centre to the centre of mass, and d the distance from the centre of mass to L4.



The diagram gives

$$h^2 = d_{MS}^2 - \frac{d_{MS}^2}{4}, \qquad h = 1.32\,\text{au}, \qquad l = 2h = 2.64\,\text{au}.$$

(3 points)

Chapter 5

Light and Telescopes

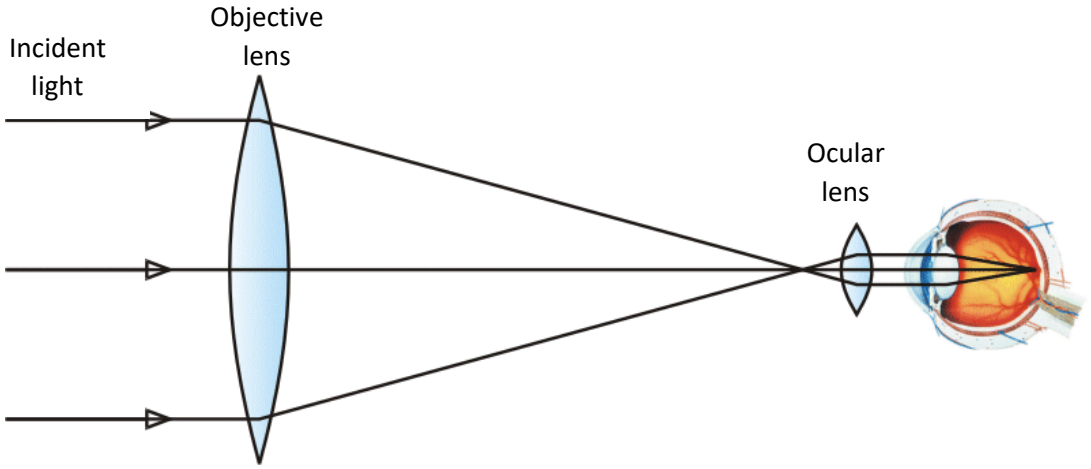
Theory problems

Problem 5.1 The Stars Observed with the Telescope

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

The diameter of the objective lens of a telescope, represented in the drawing, is $D_{\text{objective telescope}} = 300\,\text{mm}$, and the diameter of the pupil of the observer's eye is $D_{\text{pupil eye}} = 6\,\text{mm}$. The light gathering power, coming from a star, due to this telescope, is



(a) $g_c = 2500$.

(b) $g_c = 1500$.

(c) $g_c = 1000$.

(d) $g_c = 3000$.

Problem 5.2 The Terrestrial Satellite

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q4 Solution p. 49

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

A small satellite (a luminous sphere) that orbits the Earth, in an equatorial circular orbit at the altitude h, is photographed with a camera on the ground, the lens of which is a converging lens with focal length f. The photography is made when the satellite aligns with the camera and the centre of the Earth.

It is given: the mass of the Earth, M; the gravitational constant, G; the radius of the Earth, R. During the photographic exposure the proper rotation of the Earth is neglected.

The length of the image on the photograph, after developing the photographic plate, placed in the focal plane of the objective lens, as indicated by the drawing, if the exposure time of the camera was Δt, is given by the expression

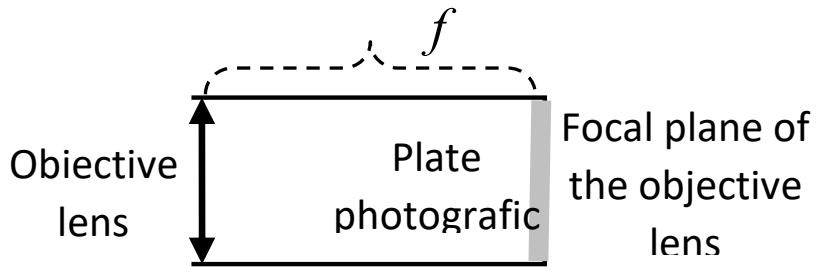


Fig. 1

(a) $l = \dfrac{f}{h-f}\sqrt{G\dfrac{M}{R+h}}\,\Delta t.$

(b) $l = \dfrac{f}{h+f}\sqrt{G\dfrac{M}{R+h}}\,\Delta t.$

(c) $l = \dfrac{f}{h-f}\sqrt{G\dfrac{M}{h-R}}\,\Delta t.$

(d) $l = \dfrac{f}{h-f}\sqrt{2G\dfrac{M}{R+h}}\,\Delta t.$

Problem 5.3 Telescope

Junior IOAA 2023, Volos, Greece Theory problems Q8 10 points Solution p. 50

The focal length of a telescope is 900 mm and its aperture is 4 inches. We use an eyepiece whose focal length is 25 mm. What is the magnification m of this telescope and what is its focal ratio f? One inch is 2.54 cm.

Problem 5.4 Telescope limiting magnitude and interstellar extinction

Junior IOAA 2024, Kathmandu, Nepal Theory problems T1 15 points Solution p. 50

A student used a telescope of diameter $D = 100\,\text{cm}$ to observe a distant star whose absolute magnitude is -0.5. If we consider the limiting magnitude of the eye as 6 and the diameter of the pupil d to be 7 mm, then answer the following questions.

1. If no interstellar medium (ISM) is present, find the maximum distance to which the student can detect this star.
2. In reality, an ISM of extinction factor $0.05\,\text{mag}\,\text{kpc}^{-1}$ is present between the star and the Earth. Find the maximum distance to which the student can detect the star.

Note. When an equation cannot be solved analytically, we use iterative methods to find an approximate numerical solution. As an example, suppose you want to find out the value of $x = x_{\text{fin}}$, for which $x_{\text{fin}} = f(x_{\text{fin}})$. We start with some value $x = x_0$ and proceed as follows.

- Step 1. Find $f(x_0)$ and call it x_1.
- Step 2. Find $f(x_1)$ and call it x_2.
- Step 3. Find $f(x_2)$ and call it x_3.

Keep repeating this iteration process till x_n is almost the same as x_{n-1}. That is your desired x_{fin} value.

Problem 5.5 Exploring NEAs

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.4 10 points Solution p. 50

Near-Earth Asteroids (NEAs) are small rocky bodies orbiting the Sun in the vicinity of Earth's orbit. A science mission designed to study a NEA target is operating a spacecraft that will make its closest approach to the target at a distance of $\Delta_c = 100\,\text{m}$. One of the instruments aboard the spacecraft is a CCD camera called the Asteroid Framer (AF), which can acquire images in the V-band at an effective wavelength of $\lambda = 550\,\text{nm}$. After the mission launched, the operations team conducted initial tests by acquiring images of the Earth using the AF camera with an aperture diameter of $D = 10\,\text{mm}$. At that time, the spacecraft was at a distance of $d = 1.5 \times 10^6\,\text{km}$ from Earth. The Earth was observed on the images within a photometric aperture of $\alpha_\oplus = 60$ pixels in diameter.

(a) Compute the pixel scale in arcseconds per pixel.

(b) Estimate the linear diameter of the smallest morphological feature that can be observed on the surface of the asteroid at the closest approach.

The saturation test on initial images showed Earth was just over-exposed at 10 s exposure time and just under-exposed at 0.5 s (under-exposed means signal-to-noise ratio < 5). Using this data, the operations team can determine the exposure time interval allowed to observe the NEA target, which has the expected albedo of $A = 0.10$. Ignore phase angle.

Assume that the detected signal is proportional to the object's surface brightness and the exposure time, and that the surface brightness is proportional to the albedo divided by the square of its distance from the Sun.

(c) Determine the corresponding exposure time interval for which the planned observations of the NEA target at the closest approach will be successful without any under-exposure or over-exposure. At that time, the target will be located at $r_{\text{NEA}} = 1.20\,\text{au}$ from the Sun.

Solutions

Solution 5.1 The Stars Observed with the Telescope

Problem p. 46

[The official S1 solution stops after the light-gathering formula without substituting the values or explicitly giving an answer letter.]

The degree of collection of light from a star, g_c, by a telescope is defined as the ratio of the area of the circle whose diameter is the objective diameter and the area of the observer's pupil,

$$g_c = \frac{\frac{\pi}{4}D^2_{\text{objective telescope}}}{\frac{\pi}{4}D^2_{\text{pupil eye}}} = \left(\frac{D_{\text{objective telescope}}}{D_{\text{pupil eye}}}\right)^2, \qquad D_{\text{objective telescope}} > D_{\text{pupil eye}}, \quad g_c > 1.$$

Solution 5.2 The Terrestrial Satellite

Problem p. 47

During the exposure interval, the satellite travels the arc $AB = v\Delta t$. Equating gravity to the centripetal force gives

$$K\frac{mM}{r^2} = \frac{mv^2}{r}, \qquad v = \sqrt{\frac{KM}{r}} = \sqrt{\frac{KM}{R+h}}.$$

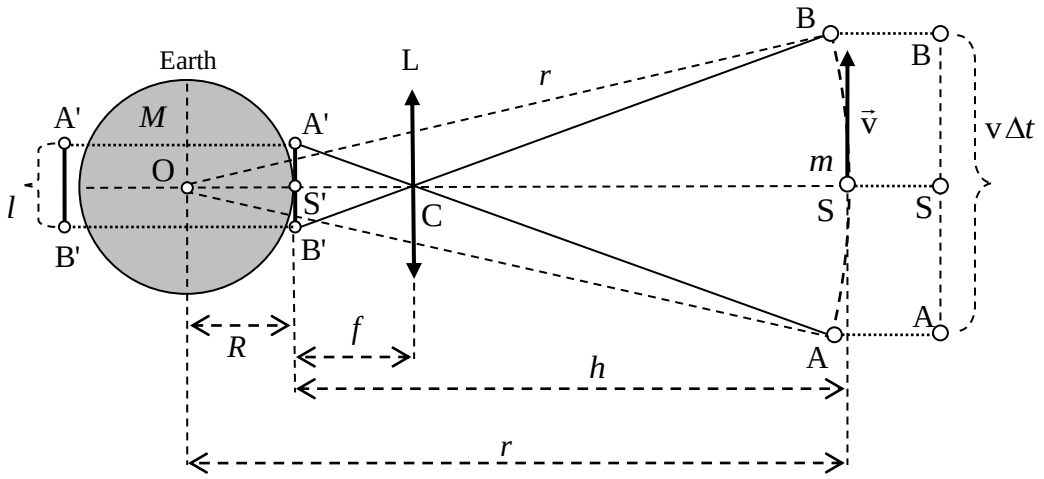


Fig. 1

The short arc can be treated as a line segment. The image forms in the focal plane, and the similar triangles ACS and $A'CS'$ give

$$\frac{A'S'}{AS} = \frac{CS'}{CS}, \qquad \frac{l/2}{v\Delta t/2} = \frac{f}{h-f}.$$

Consequently,

$$l = \frac{f}{h-f}v\Delta t = \frac{f}{h-f}\sqrt{\frac{KM}{R+h}}\,\Delta t.$$

The answer is (a), with $K = G$.

Solution 5.3 Telescope

Problem p. 47

The magnification is the ratio of the focal length of the telescope to the focal length of the eyepiece,

$$m = \frac{900}{25} = 36.$$

(5 points)

The focal ratio f is the ratio of the focal length to the aperture,

$$f = \frac{900}{40 \times 2.54} = 8.9.$$

(5 points)

Solution 5.4 Telescope limiting magnitude and interstellar extinction

Problem p. 48

[The official extinction solution has a sign inconsistency between its printed intermediate magnitude equation and subsequent iteration formula. Both expressions are retained.]

(a) We know $m_t - m_e = -2.5\log(F_t/F_e)$. (1 point) Since $F_{\rm lim} \propto 1/D^2$, $m_t - m_e = 5\log(1000/7)$. (1 point) Thus $m_t = 10.8 + 6 = 16.8$. (1 point) Using the distance modulus,

$$m = M + 5\log\left(\frac{r}{10\,\mathrm{pc}}\right), \qquad \frac{r}{10} = 10^{3.454}.$$

So $r = 29\,\mathrm{kpc}$ is the maximum distance to which the student can detect the star using the telescope. (2 points)

(b) $m' = m + A = 16.8 + \alpha r$. (2 points) Now $m' = M + 5\log(r/(10\,\mathrm{pc}))$, so

$$16.8 + 0.00005r = -0.5 + 5\log r - 5.$$

(1 point) Therefore $r = 10^{4.46 - 0.00001r}$. (2 points) This equation cannot be solved analytically. Iterate, taking the previous result of 29 kpc as a good initial approximation. (2 points) Then $r = 18.73\,\mathrm{kpc}$. (3 points)

Solution 5.5 Exploring NEAs

Problem p. 48

(a) The pixel scale is the ratio of the angular dimension to its diameter in pixels,

$$s = \frac{2R_\oplus/d}{\alpha_\oplus}.$$

(1 point) Numerically,

$$s = \frac{2 \times 6.378 \times 10^3}{1.50 \times 10^6} \frac{180 \times 3600}{\pi} \frac{1}{60} = 29\,\mathrm{arcsec/pix}.$$

(1 point)

(b) Rayleigh's criterion gives

$$\theta = 1.22\frac{\lambda}{D} = \frac{1.22 \times 550 \times 10^{-9}}{0.01}\frac{180 \times 3600}{\pi} = 13.8\,\text{arcsec}.$$

(0.5 point) Thus $\theta < s \times (1\,\text{pix})$. (0.5 point) A detectable feature should be comparable to one pixel's linear projection, $l = s \times (1\,\text{pix}) \times \Delta_c$. (1 point) Solutions using 2 or 3 pixels receive full credit. Numerically,

$$l = 29\,\text{arcsec} \times \frac{\pi}{180 \times 3600} \times 100\,\text{m} = 1.4\,\text{cm}.$$

(1 point)

(c) Signal per pixel is proportional to surface brightness and exposure, $S \propto Bt$. (1 point) For reflected sunlight, $B \propto A/r^2$. (1 point) Hence

$$\frac{B_{\text{NEA}}}{B_\oplus} = \frac{A_{\text{NEA}} r_\oplus^2}{A_\oplus r_{\text{NEA}}^2}.$$

(1 point) Surface brightness is independent of the distance between the sensor and the extended target. Both Earth and the NEA are extended objects. For similar signal levels,

$$t_{\text{NEA}} = t_{\text{Earth}} \frac{A_\oplus r_{\text{NEA}}^2}{A_{\text{NEA}} r_\oplus^2}.$$

(1 point) Numerically,

$$t_{\text{NEA}} = t_{\text{Earth}} \frac{0.30}{0.10} \left(\frac{1.20}{1.00}\right)^2 = 4.32 t_{\text{Earth}}.$$

(0.5 point) The exposure interval is 2.16 s–43.2 s. (0.5 point)

Chapter 6

The Sun and Stars

Theory problems

Problem 6.1 The Moon, Seen on the Way to the Moon

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q9 Solution p. 59

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

The full Moon, whose linear diameter is $d = 3436\,\text{km}$, at Earth in the distance $r_0 = 384000\,\text{km}$, has the apparent magnitude $m_0 = -12.7$. The apparent magnitude of the Sun, seen from Earth, is $m_S = -26.84$. The distance, r, from the full Moon, where a cosmonaut is, on the way of his spaceship to the Moon, the magnitude of the full Moon should be 50% of the magnitude of the Sun as seen from Earth, is

(a) $r \approx 198243\,\text{km}$.

(b) $r \approx 320732\,\text{km}$.

(c) $r \approx 132521\,\text{km}$.

(d) $r \approx 275633\,\text{km}$.

Problem 6.2 The Luminosity of a Star

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q10 Solution p. 59

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

The luminosity of a star Σ is $L_\Sigma = 100L_S$, where L_S is the luminosity of the Sun, and the star's surface temperature is $T_\Sigma = \frac{1}{2}T_S$, where T_S is the Sun's surface temperature. Ratio between the radius of the star and the radius of the Sun is

(a) $R_\Sigma/R_S = 40$.

(b) $R_\Sigma/R_S = 30$.

(c) $R_\Sigma/R_S = 60$.

(d) $R_\Sigma/R_S = 50$.

Problem 6.3 Unresolved binary star

Junior IOAA 2023, Volos, Greece Theory problems Q3 10 points Solution p. 60

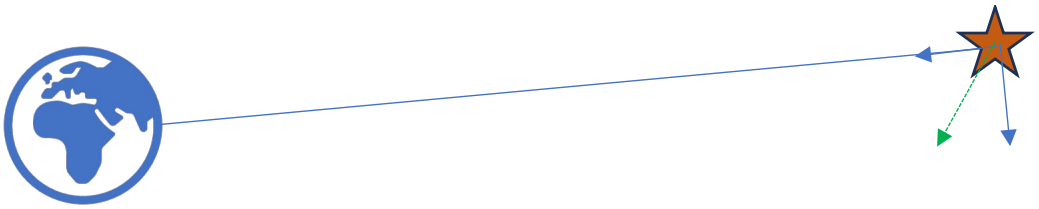
An unresolved binary star consists of two members whose apparent magnitudes are $m_1 = 2.0$ mag and $m_2 = 4.5$ mag. What is the combined apparent magnitude of this binary system?



Problem 6.4 Parallax and proper motion

Junior IOAA 2023, Volos, Greece Theory problems Q7 10 points Solution p. 60

The parallax of Barnard's star is 0.552 arcseconds and its proper motion is $\mu = 10.31$ arcseconds/year. From its spectrum, an observer noticed that the absorption line Hα is seen at 656.044 nm instead of 656.281 nm. What is the total speed of Barnard's star relative to this observer?



Data from the accompanying constants sheet. $c = 2.998 \times 10^8\,\mathrm{m\,s^{-1}}$, $1\,\mathrm{pc} = 3.086 \times 10^{16}\,\mathrm{m} = 206{,}265\,\mathrm{au}$, and one tropical year is 365.2422 solar days.

Problem 6.5 Sirius A and B

Junior IOAA 2023, Volos, Greece Theory problems Q11 40 points Solution p. 60

Sirius is the brightest star of the night sky during the winter of the northern hemisphere and the summer of the southern hemisphere. It has an apparent visual magnitude $m_V = -1.47$ mag. Sirius B is the binary companion of Sirius A and its magnitude is $m_V = 8.5$ mag. Their orbital period is 50 years and the parallax of the system is 350 milliarcseconds, while their angular separation is 7 arcseconds. You can assume that Sirius A has a surface temperature of $T = 10{,}000$ K. The temperature of Sirius B is estimated to be 25,000 K. Both stars have a bolometric correction equal to -0.4. An absolute bolometric magnitude of zero corresponds to a luminosity of 3×10^{28} W. The orbit is assumed to be circular, and the line of sight is perpendicular to the orbital plane of the system.

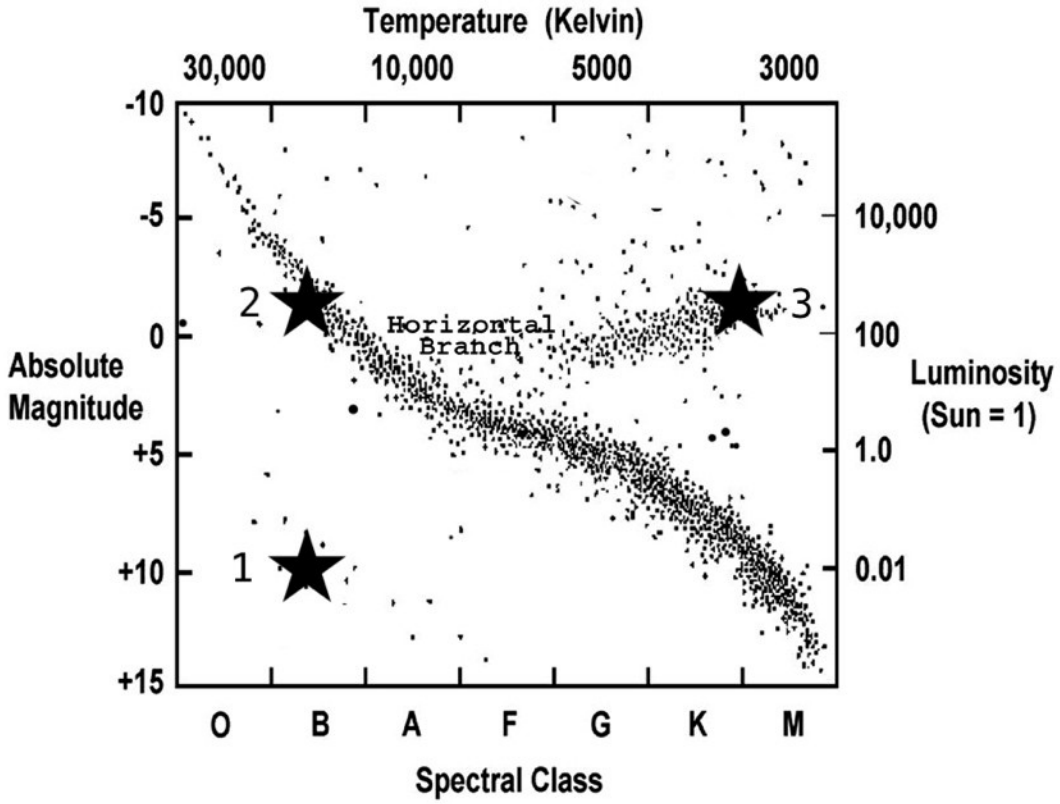
(A) Calculate the luminosity and radius of Sirius A and B. [12 points]

(B) What is the orbital separation of the system in meters? [8 points]

(C) What is the total mass of the system? [8 points]

(D) Assuming that the mass of Sirius A is twice the mass of Sirius B, what is the mean density of Sirius B? [6 points]

(E) In the H-R diagram shown in question 10, which star number (1 / 2 / 3) will be the closest to the star Sirius B? [6 points]



Data from the accompanying constants sheet. $\sigma = 5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$, $1\,\mathrm{pc} = 3.086 \times 10^{16}\,\mathrm{m}$, $1\,\mathrm{au} = 1.496 \times 10^{11}\,\mathrm{m}$, and $M_\odot = 1.988 \times 10^{30}\,\mathrm{kg}$.

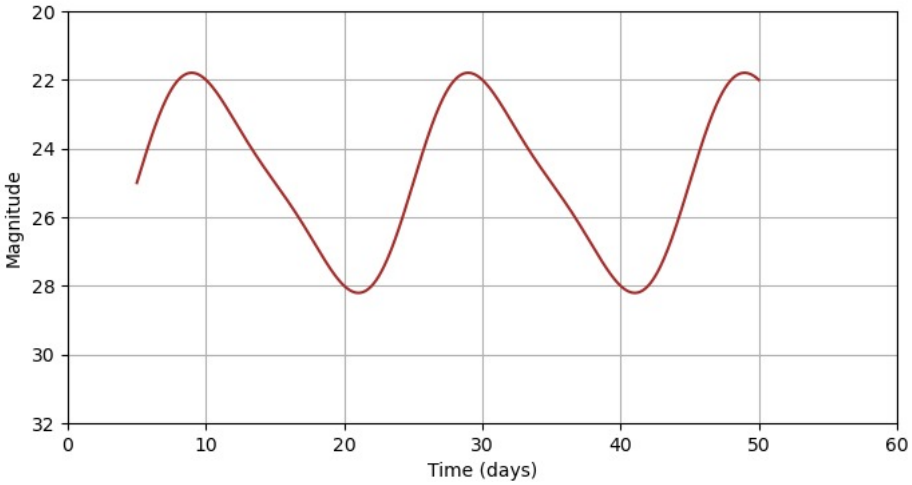
Problem 6.6 A Cepheid period-luminosity relation

Junior IOAA 2024, Kathmandu, Nepal Theory problems T4 15 points Solution p. 61

In Cosmology, Cepheid variables are called standard candles as they can be used to estimate distances. Edwin Hubble used these standard candles to settle the Shapley–Curtis debate by measuring the distance to the Andromeda Galaxy. Cepheid variables are variable stars whose brightness vary periodically. Henrietta Leavitt discovered the relation between the periodicity and average luminosity of these stars, and this relation can be used to derive the equation linking Absolute magnitude and Period. For Cepheid variables, luminosity L varies with period P as $L \propto P^{2.5}$, where P is in days.

1. Find the relation between Absolute Magnitude M and P for this type of variable stars.

2. You might notice that it is a linear equation between M and a function of P. If the unknown constant of the given equation is $-1.5\,\text{mag}$, then find the distance to the given Cepheid Variable.



Problem 6.7 A spectroscopic binary system

Junior IOAA 2024, Kathmandu, Nepal Theory problems T10 15 points Solution p. 62

Spectroscopic binary star systems are those systems in which individual stars of the system can't be resolved visually due to their low angular separation. But we can detect that the system is binary due to shifts in their spectral line caused by Doppler shift. In double lined spectroscopic binaries, we can detect two spectra, one for each star. From the change in wavelength of the spectrum over a period of time, we can find the radial velocity curve. Suppose an astronomer observed a hypothetical spectroscopic binary system having a circular orbit as shown in Figure 1. The plane of the orbit makes an angle θ with respect to the plane of the sky. Star 1 (mass m_1) is located on the outer orbit and Star 2 (mass m_2) is located in the inner orbit. The observed radial velocity graph is shown in Figure 2.

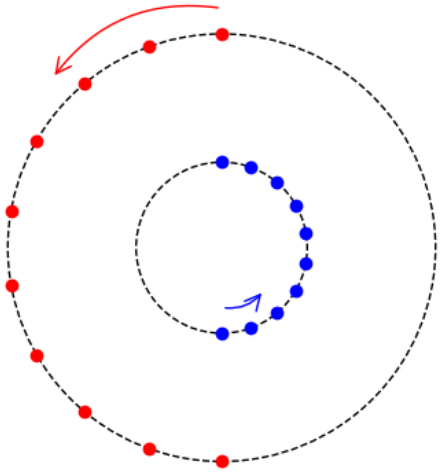


Figure 1. A top-view of the binary system.

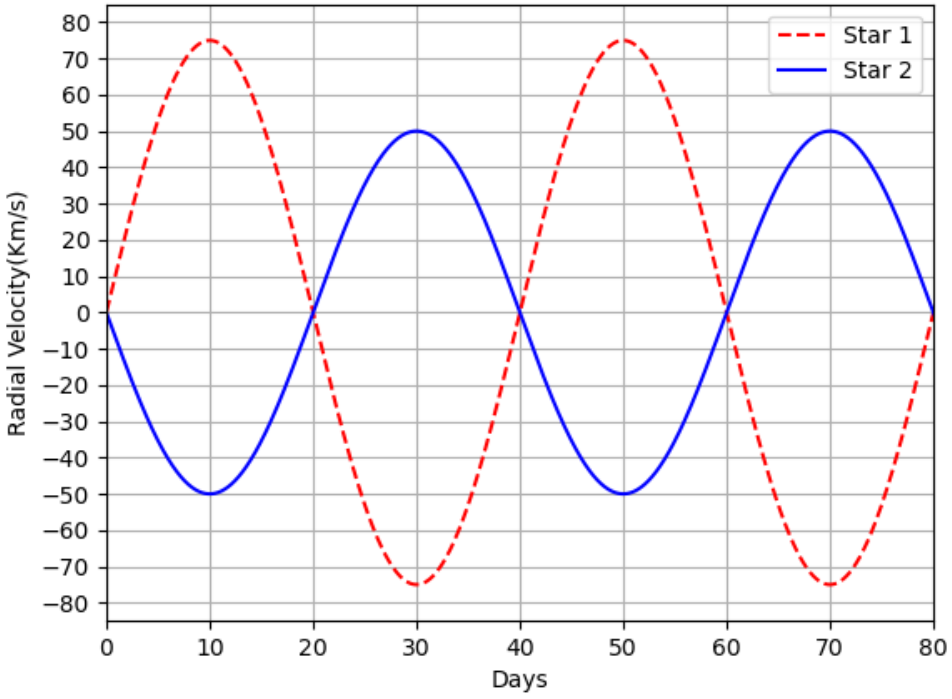


Figure 2. Radial Velocity Plots for the Binary Stars

1. The binary mass function is an expression which depends upon only the observable quantities such as radial velocity and period. Find the binary mass function for both star 1 and star 2.

2. Use your expression in (1) to estimate the masses for the stars. Is your estimate exact, a

lower-limit or an upper-limit?

Supplied examination data. $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ and $M_\odot = 1.988 \times 10^{30}\,\mathrm{kg}$.

Problem 6.8 A radiating interstellar dust cloud

Junior IOAA 2024, Kathmandu, Nepal Theory problems T12 15 points Solution p. 62

Interstellar dust is heated up by hot ionizing radiation from young blue stars. They then re-emit the UV light. A dust cloud in a dusty galaxy was recently studied by a researcher. Make the simplifying approximation that you can treat the dust cloud as a blackbody. Help him to answer the following questions.

1. Find the wavelength where the dust emission peaks $\lambda_{\max}$ if it has a temperature of $T = 50\,\mathrm{K}$.
2. The researcher determines that the dust has a total luminosity $L_{\mathrm{dust}} = 1.6 \times 10^{11} L_\odot$. Estimate the size of the dust cloud, assuming a spherical distribution.
3. If the cosmological redshift of the galaxy z is 0.01, estimate the angular size θ of the dust cloud.
4. What aperture telescope is minimally needed if we want to resolve the dust cloud?

Supplied examination data. $\lambda_m T = 2.898 \times 10^{-3}\,\mathrm{m\,K}$, $L_\odot = 3.826 \times 10^{26}\,\mathrm{W}$, $T_{\mathrm{eff},\odot} = 5778\,\mathrm{K}$, $R_\odot = 695500\,\mathrm{km}$, $c = 2.998 \times 10^8\,\mathrm{m\,s^{-1}}$, $H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and $1\,\mathrm{pc} = 3.086 \times 10^{16}\,\mathrm{m}$.

Problem 6.9 Binary System

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.1 10 points Solution p. 63

Two stars A and B, that form a non-eclipsing binary system, have peaks of their blackbody radiation spectra at 650 nm and 400 nm, respectively. We also know that the radius of star A is twice the radius of star B.

(a) Compute the difference between the absolute magnitudes of the stars A and B.

(b) If the combined absolute magnitude of the binary system is 0.00, find the absolute magnitude of each star.

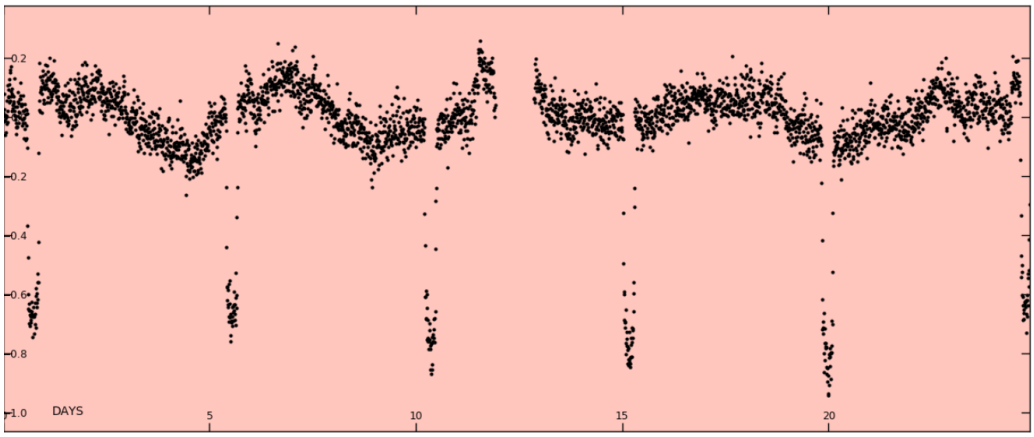
Problem 6.10 Lightcurves

Junior IOAA 2025, Piatra Neamț, Romania Theory problems Th.9 40 points Solution p. 64

(A) Transiting Exoplanet. The figure below presents the light curve of an exoplanet transit. On the X-axis are represented the days elapsed since the first observation, and on the Y-axis is shown the percentage of the average luminosity by which the brightness of the system increases or decreases relative to the average brightness of the star, that is,

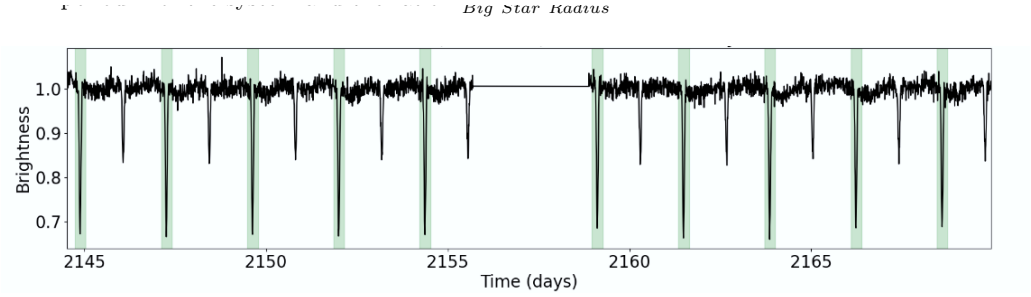
$$100 \times \frac{L(t) - L_{\mathrm{average}}}{L_{\mathrm{average}}}.$$

Determine the period P of the planet and the ratio Planet Radius/Star Radius.



On the Y-axis, 0.2 means 0.2% and 1.0 means 1%.

(B) Eclipsing Binary Star System. Another type of object with an interesting light curve is an eclipsing binary star system. The lightcurve of such an object is presented below, with the X-axis representing days since the start of the observing mission, and the Y-axis representing the relative brightness of the system, that is, $L(t)/L_{\text{average}}$. We know that the smaller star is hotter than the larger star. Calculate the period P of the system and the ratio Small Star Radius/Big Star Radius.



On the Y-axis, 0.7 means 70% and 1.0 means 100%.

Solutions

Solution 6.1 The Moon, Seen on the Way to the Moon

Problem p. 52

[The supplied S9 file solves a different version requiring the Moon's brightness to equal the Sun's brightness, giving 568.88 km. Q9 instead asks for 50% of the Sun's magnitude. The solution also initially prints 348000 km for the lunar distance, then substitutes 384000 km.]

In the supplied solution, the required magnitude change is

$$\Delta m = m_S - m_0 = -26.84 - (-12.7) = -14.14, \qquad m = m_S = -26.84.$$

For luminous intensity I, solid angle Ω, flux ϕ, and area A,

$$I = \frac{\phi}{\Omega}, \quad \Omega = \frac{A}{\Delta^2}, \quad E = \frac{\phi}{A} = \frac{I}{\Delta^2} = \frac{L}{4\pi\Delta^2}.$$

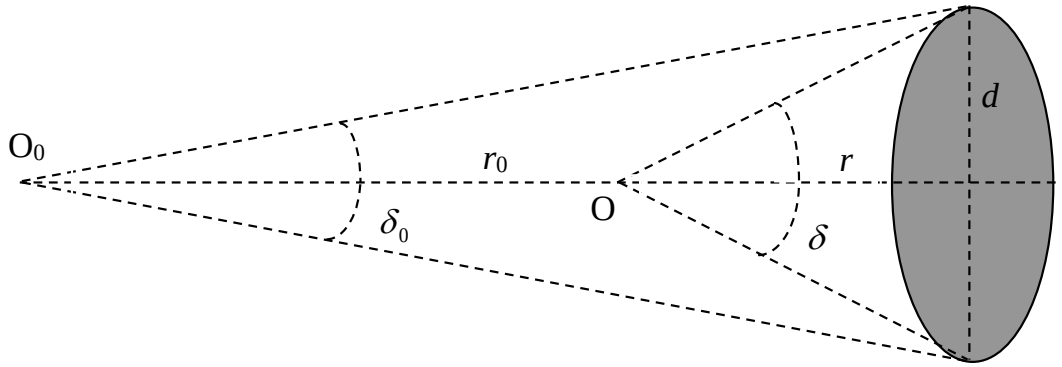


Fig. 1

Pogson's relation gives

$$\log\frac{E}{E_0} = -0.4(m - m_0), \qquad E_0 = \frac{I}{r_0^2}, \quad E = \frac{I}{r^2},$$

$$2\log\frac{r_0}{r} = -0.4\Delta m, \qquad \log\frac{r_0}{r} = 0.2 \times 14.14 = 2.828.$$

Thus $10^{2.828} \approx 675$ and

$$r = \frac{r_0}{675} = \frac{384000\,\text{km}}{675} \approx 568.88\,\text{km}.$$

Solution 6.2 The Luminosity of a Star

Problem p. 52

The luminosity is the total radiation emitted per unit time through the star's entire surface at all wavelengths. The Stefan–Boltzmann law gives

$$L_\Sigma = 4\pi R_\Sigma^2 \sigma T_\Sigma^4, \qquad L_S = 4\pi R_S^2 \sigma T_S^4, \qquad L_\Sigma = 100 L_S.$$

Hence

$$\frac{R_\Sigma^2}{R_S^2} = 10^2 \frac{T_S^4}{T_\Sigma^4} = 10^2 \times 2^4, \qquad \frac{R_\Sigma}{R_S} = 40.$$

The answer is (a).

Solution 6.3 Unresolved binary star

Problem p. 53

The apparent magnitude of a star is defined as

$$m = -2.5 \log_{10} \left(\frac{F_x}{F_{x,0}} \right).$$

(4 points)

Solve for the flux F_x and add the fluxes corresponding to each star. This gives

$$m_b = -2.5 \log_{10} \left(10^{-0.4 m_1} + 10^{-0.4 m_2} \right) = 1.89 \approx 1.9 \text{ mag}.$$

(6 points)

Solution 6.4 Parallax and proper motion

Problem p. 53

The radial and transverse velocities of Barnard's star can be extracted from the given information. For the radial velocity, use the non-relativistic Doppler formula,

$$v_r = \frac{\Delta\lambda}{\lambda} c = \frac{656.044 - 656.281}{656.281} \times 2.998 \times 10^5 = -108.27 \text{ km s}^{-1}.$$

(3 points)

For the tangential velocity, first evaluate the distance from the parallax, $d = 1/0.552 = 1.81\,\text{pc}$. The proper motion of 10.31 arcseconds/year corresponds to

$$v_t = \mu d = \frac{10.31}{206265} \times 1.81 \times 3.086 \times 10^{13} \times \frac{1}{365.2422 \times 24 \times 3600} = 88.55 \text{ km s}^{-1}.$$

(4 points)

The speed follows from Pythagoras' theorem,

$$v = \sqrt{108.27^2 + 88.55^2} = 139.87 \text{ km s}^{-1}.$$

(3 points)

Solution 6.5 Sirius A and B

Problem p. 53

(A) Use the relation between apparent and absolute magnitude,

$$m - M = 5 \log d - 5.$$

The distance from the parallax is $d = 1/0.350 = 2.86\,\text{pc}$, so Sirius A has

$$M_V = -1.47 - 5 \log 2.86 + 5 = 1.25 \text{ mag}.$$

Taking the bolometric correction into account gives $M_B = 1.25 - 0.4 = 0.85$ mag. For Sirius B the same process gives $M_B = 0.85 + (8.5 + 1.47) = 10.82$. Use

$$M_V = -2.5 \log \frac{L}{L_0}, \qquad L = L_0 \times 10^{-0.4 M_V}.$$

Substitution gives $L_A = 1.37 \times 10^{28}\,\text{W}$ and $L_B = 1.41 \times 10^{24}\,\text{W}$. The Stefan–Boltzmann law $L = 4\pi R^2 \sigma T^4$ then gives $R_A = 1.39 \times 10^9\,\text{m}$ and $R_B = 2.25 \times 10^6\,\text{m}$. (12 points)

(B) Convert the angular separation to radians,

$$\theta_1 = \frac{7}{206265} = 3.39 \times 10^{-5} \text{ rad.}$$

The separation is $s_1 = 3.39 \times 10^{-5} \times 2.86 = 9.71 \times 10^{-5}$ pc, or 3.00×10^{12} m. (8 points)

(C) Use $M_{\text{tot}} = a^3/P^2$, with a in astronomical units, P in years, and mass in solar masses. The separation is 20 au, giving

$$M_{\text{tot}} = \frac{20.0^3}{50^2} = 3.20\, M_\odot.$$

(8 points)

(D) Sirius B has mass $\frac{1}{3} \times 3.20 = 1.07\, M_\odot$ and radius 2.25×10^6 m from part (A). Its mean density is $\bar{\rho} = 4.4 \times 10^{10}\,\text{kg}\,\text{m}^{-3}$. (6 points)

(E) Its radius is comparable to Earth's while its mass is close to the Sun's. It is therefore a white dwarf, closer to star 1. (6 points)

Solution 6.6 A Cepheid period-luminosity relation

Problem p. 54

(a) $L \propto P^{2.5}$ implies $L = KP^{2.5}$, so

$$\log L = \log K + 2.5 \log P.$$

(2 points) Also $\log L = \log F + \log 4 + \log \pi + 2 \log r$. The magnitude relation is $m_1 - m_2 = -2.5 \log(F_1/F_2)$. (1 point) For ease of calculation use Vega as reference star and let m be the apparent magnitude of the Cepheid variable. Then $-m/2.5 = \log F_1 - \log F_0$. (1 point) Combining this with $m = M + 5 \log(r/(10\,\text{pc}))$ gives

$$M = -6.25 \log P + C,$$

where $C = -2.5(\log K - \log 4 - \log F_0 - \log \pi) + 5 \log 10$. (3 points)

(b) The magnitude–luminosity relation has $C = -1.5$. (1 point) Suppose m is the average magnitude corresponding to the luminosity $\frac{1}{2}(L_{\max} + L_{\min})$. (1 point) Then

$$m - m_{\text{ref}} = -2.5 \log \left(\frac{(L_1 + L_2)/2}{L_{\text{ref}}} \right).$$

(1 point) Equivalently,

$$m = m_{\text{ref}} + 2.5 \log 2 - 2.5 \log \left(\frac{L_1 + L_2}{L_{\text{ref}}} \right).$$

Since

$$\frac{L_{\text{tot}}}{L_{\text{ref}}} = \frac{L_1}{L_{\text{ref}}} + \frac{L_2}{L_{\text{ref}}} = (10^{-0.4m_1} + 10^{-0.4m_2}) 10^{0.4m_{\text{ref}}},$$

(1 point) substitution gives

$$m = 2.5 \log 2 - 2.5 \log(10^{-2m_1/5} + 10^{-2m_2/5}).$$

(1 point) From the graph $m_{\max} = 28$, $m_{\min} = 22$, so $m = 22.75$, and the period is 20 days. (1 point) The absolute magnitude relation gives $M = -9.63$. (1 point) Substitute into $m = M + 5 \log(r/(10\,\text{pc}))$ to obtain $r = 29.92$ Mpc. (1 point)

Solution 6.7 A spectroscopic binary system

Problem p. 55

1. We know $V_1 = \omega a_1$ and $V_2 = \omega a_2$. (1 point) From Kepler's third law, $P^2 = 4\pi^2 a^3/(GM)$, where $M = m_1 + m_2$ and $a = a_1 + a_2$. (2 points) Also $P = 2\pi/\omega$, $a_1 = m_2 a/(m_1 + m_2)$ and $a_2 = m_1 a/(m_1 + m_2)$. (1 point) Replacing these values in Kepler's equation gives

$$G(m_1 + m_2) = 4\pi^2 \frac{\omega^2 (m_1 + m_2)^3}{4\pi^2 m_2^3} a_1^3,$$

$$\frac{m_2^3}{(m_1 + m_2)^2} = (\omega a_1)^3 \frac{P}{2\pi G}.$$

(2 points) When we observe a star we measure the line-of-sight component, rather than the real orbital velocity. (2 points) The required mass functions are

$$\frac{m_2^3}{(m_1 + m_2)^2} \sin^3\theta = v_{1,\mathrm{obs}}^3 \frac{P}{2\pi G},$$

(1 point)

$$\frac{m_1^3}{(m_1 + m_2)^2} \sin^3\theta = v_{2,\mathrm{obs}}^3 \frac{P}{2\pi G}.$$

(1 point) Here $v_{1,\mathrm{obs}}$ is the maximum observed orbital velocity.

2. We know $v_1/v_2 = m_2/m_1$. (1 point) From the graph $v_1/v_2 = 1.5$, so $m_2 = 1.5 m_1$. The period is 40 days and $v_{1,\mathrm{obs}} = 75\,\mathrm{km\,s^{-1}}$. For the minimum value take $\sin\theta = 1$ and put these values in the mass function. (2 points) This gives $0.54 m_1 = 1.74 M_\odot$, hence $m_1 = 3.2 M_\odot$ (1 point) and similarly $m_2 = 4.9 M_\odot$. (1 point) The values are lower limits, since $\sin\theta \leq 1$.

Solution 6.8 A radiating interstellar dust cloud

Problem p. 57

[The question says that the dust re-emits UV light, although its stated 50 K temperature and the official 57.96 micrometre peak correspond to infrared emission. The source wording is retained.]

1. $\lambda_{\max} T = 2.898 \times 10^{-3}\,\mathrm{m\,K}$. (2 points) Thus $\lambda_{\max} = 57.96\,\mu\mathrm{m}$. (1 point)

2. Use the Stefan–Boltzmann law and compare to the Sun,

$$R_{\mathrm{dust}} = R_\odot \sqrt{\frac{L_{\mathrm{dust}} T_\odot^4}{L_\odot T_{\mathrm{dust}}^4}}.$$

(2 points) Substituting gives $R_{\mathrm{dust}} = 120.4\,\mathrm{pc}$. (2 points)

3. $d_{\mathrm{dust}} = cz/H_0 = (0.01 \times 2.998 \times 10^5/70)\,\mathrm{Mpc} = 42.83\,\mathrm{Mpc}$. (3 points) Then $\theta = R_{\mathrm{dust}}/d_{\mathrm{dust}} = 0.580''$. (2 points)

4. $D > 1.22\lambda'_{\max}/\theta = 25.4\,\mathrm{m}$. (3 points) The redshift needs to be accounted for, $\lambda'_{\max} = (1+z)\lambda_{\max}$ or an equivalent expression. A correct formula using $\lambda_{\max}$ receives only 1 point.

Solution 6.9 Binary System

Problem p. 57

(a) From Wien's law, $b = T\lambda$, where b is Wien's constant, T is the surface temperature, and λ is the wavelength corresponding to the maximum peak of radiation. Hence

$$b = T_A\lambda_A = T_B\lambda_B \quad \Rightarrow \quad \frac{T_A}{T_B} = \frac{\lambda_B}{\lambda_A}.$$

(2 points) The Stefan–Boltzmann formula gives $L = 4\pi R^2\sigma T^4$, so

$$\frac{L_A}{L_B} = \left(\frac{R_A}{R_B}\right)^2\left(\frac{T_A}{T_B}\right)^4 = \left(\frac{R_A}{R_B}\right)^2\left(\frac{\lambda_B}{\lambda_A}\right)^4 = 2^2\left(\frac{400}{650}\right)^4 \approx 0.57.$$

(2 points) The luminosity–absolute magnitude relationship is

$$\frac{L_A}{L_B} = 10^{0.4(M_B-M_A)}, \qquad \Delta M = M_A - M_B = -\frac{1}{0.4}\log_{10}\frac{L_A}{L_B} = -\frac{1}{0.4}\log_{10} 0.57 \approx 0.60.$$

(2 points) Also accept 0.61, possible due to rounding.

(b) Denote by $L = L_A + L_B$ the total luminosity and by $M = 0$ the absolute magnitude of the system. Comparing with the Sun,

$$\frac{L}{L_\odot} = 10^{0.4(M_\odot - M)} = \frac{L_A}{L_\odot} + \frac{L_B}{L_\odot} = 10^{0.4(M_\odot - M_A)} + 10^{0.4(M_\odot - M_B)}.$$

(1 point) Thus

$$\begin{aligned} 10^{-0.4M} &= 10^{-0.4M_A} + 10^{-0.4M_B} \\ &= 10^{-0.4(M_B+\Delta M)} + 10^{-0.4M_B}. \end{aligned}$$

(1 point) It follows that

$$\begin{aligned} 10^{-0.4M} &= 10^{-0.4M_B}(10^{-0.4\Delta M} + 1), \\ -0.4M &= -0.4M_B + \log_{10}(10^{-0.4\Delta M} + 1), \\ M_B &= M + \frac{1}{0.4}\log_{10}(10^{-0.4\Delta M} + 1) \\ &= 0 + \frac{1}{0.4}\log_{10}(10^{-0.4\cdot 0.61} + 1) \approx 0.49. \end{aligned}$$

(1 point) Then $M_A = M_B + \Delta M = 0.49 + 0.61 = 1.10$. (1 point)

Alternative solution to (b). With the same notation,

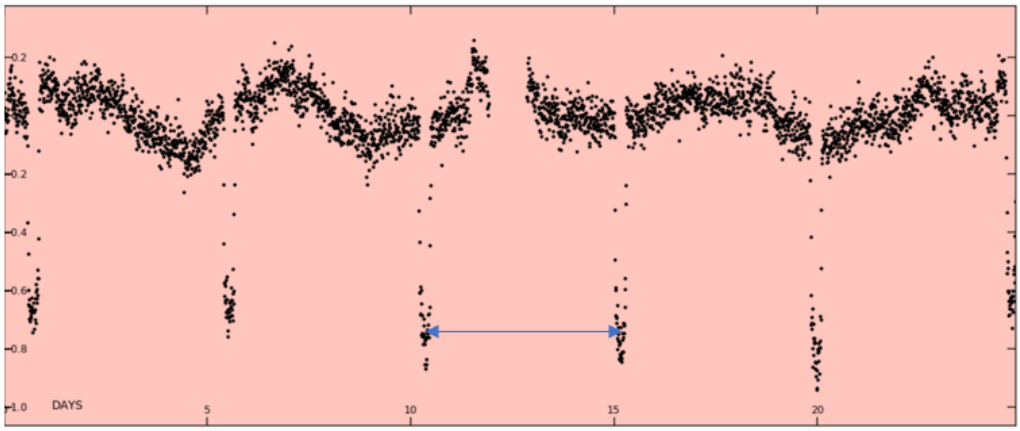
$$M - M_A = -2.5\log\frac{L_A + L_B}{L_A} = -2.5\log\left(1 + \frac{L_B}{L_A}\right).$$

(2 points) Hence $0 - M_A = -2.5\log(1 + 1/0.57)$ and $M_A = 2.5\log(2.74) = 1.10$. (1 point) Using the previously calculated difference, $M_B = M_A - \Delta M = 1.1 - 0.60 = 0.50$. (1 point)

Solution 6.10 Lightcurves

Problem p. 57

(A) Transiting Exoplanet (15 points).



Measure the distance between successive minima, then determine the X-axis scale from the 5 and 10 day marks. The key's measurements are 5 days for 2.6 cm and 2.5 cm between minima, giving

$$P = 2.5 \times \frac{5}{2.6} \approx 4.81 \text{ days}.$$

(5 points) Printing or zoom can change the scale, but should preserve the period. Answers within ± 0.1 days receive full credit.

An alternative is to use a longer baseline. From 5 to 20 days is 7.8 cm. From the first to the penultimate minimum is 10 cm, spanning four periods. Thus

$$P = \frac{10}{4}\frac{15}{7.8} \approx 4.81 \text{ days}.$$

Both methods receive full credit. Let $n\%$ be the minimum plotted value. Then

$$100\frac{L_{\text{transit}} - L_{\text{average}}}{L_{\text{average}}} = n.$$

With star radius R and planet radius r,

$$L_{\text{transit}} = \frac{L_{\text{average}}}{\pi R^2}(\pi R^2 - \pi r^2).$$

Substitution yields

$$\left(\frac{r}{R}\right)^2 = -\frac{n}{100}, \qquad \frac{r}{R} = \sqrt{\frac{|n|}{100}}.$$

(5 points) On the Y-axis, 0.6% spans 2.2 cm. A minimum's depth varies from 2.8 to 3.2 cm. The marking key accepts 2.8–3.1 cm for full credit, or an average over all minima. Using 2.9 cm,

$$|n| = 2.9 \times \frac{0.6}{2.2} \approx 0.79\%.$$

(3 points) Therefore $r/R \approx 0.09$. (2 points) The extreme values give 0.087 and 0.093, both rounding to 0.09.

(B) Eclipsing Binary Star System (25 points). The X-axis scale is 20 days to 9.1 cm and the measured period spans 1.1 cm. Thus

$$P = 1.1 \times \frac{20}{9.1} \approx 2.42 \text{ days.}$$

(6 points) Accept periods within ±0.1 days. For a cooler large star of radius R and hotter small star of radius r,

$$L_{\text{average}} = L_R + L_r.$$

At secondary minimum the smaller star transits the larger,

$$L_{\text{Secondary}} = L_r + \frac{L_R}{\pi R^2}(\pi R^2 - \pi r^2).$$

At primary minimum, the smaller star is completely eclipsed, so $L_{\text{Primary}} = L_R$. (6 points) The luminosity ratios satisfy

$$\begin{aligned} n_{\text{Primary}} &= \frac{L_R}{L_R + L_r}, \\ n_{\text{Secondary}} &= \frac{L_r + L_R(\pi R^2 - \pi r^2)/(\pi R^2)}{L_R + L_r} \\ &= 1 - \frac{L_R}{L_R + L_r}\frac{r^2}{R^2} = 1 - n_{\text{Primary}}\frac{r^2}{R^2}. \end{aligned}$$

Hence

$$\frac{r}{R} = \sqrt{\frac{1 - n_{\text{Secondary}}}{n_{\text{Primary}}}}.$$

(5 points) The Y-axis scale is 0.2 to 1.1 cm. The primary depth spans 1.8 cm, so

$$n_{\text{Primary}} = 1 - 1.8\frac{0.2}{1.1} = 1 - 0.327 \approx 0.67.$$

(3 points) The secondary depth spans 0.9 cm, so

$$n_{\text{Secondary}} = 1 - 0.9\frac{0.2}{1.1} = 1 - 0.16 \approx 0.84.$$

(3 points) Therefore

$$\frac{r}{R} = \sqrt{\frac{1 - 0.84}{0.67}} \approx 0.49.$$

(2 points) Values within ±0.02 receive full credit.

Chapter 7

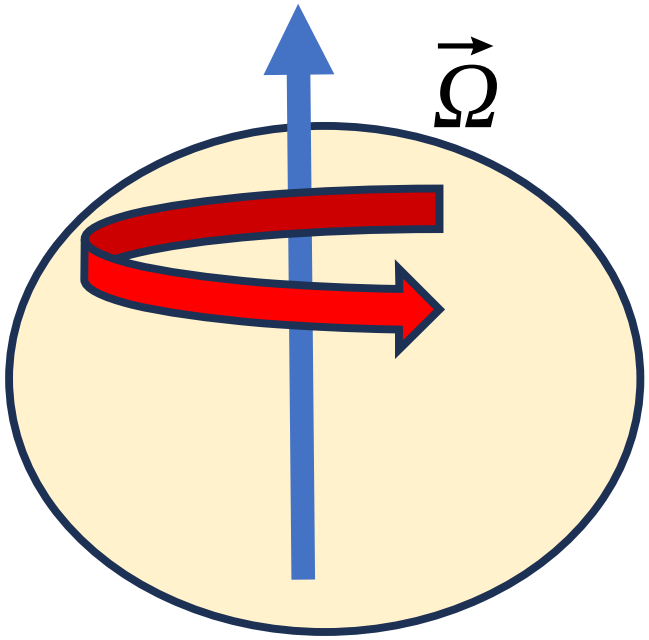
Stellar Evolution

Theory problems

Problem 7.1 Rapidly spinning neutron star

Junior IOAA 2023, Volos, Greece Theory problems Q4 10 points Solution p. 68

A neutron star is a rapidly spinning compact star. A particular neutron star has a mass of $1.40\,M_\odot$ and its radius is 10.0 km. What is the shortest possible spin period for it, before it starts shedding its mass away?

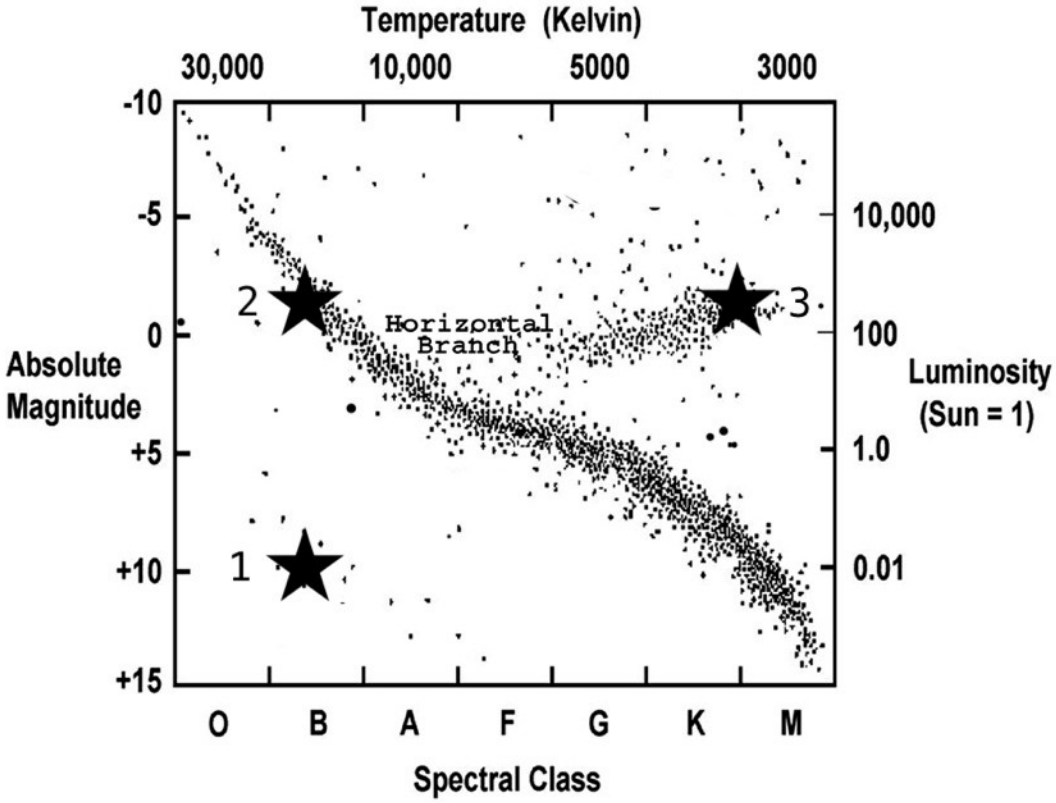


Data from the accompanying constants sheet. $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ and $M_\odot = 1.988 \times 10^{30}\,\mathrm{kg}$.

Problem 7.2 H-R diagram

Junior IOAA 2023, Volos, Greece Theory problems Q10 10 points Solution p. 68

For this question only the correct answers are required. No explanation is necessary. The image below shows the H-R diagram and three stars, 1, 2 and 3. For each statement below write "TRUE" if the statement is definitely true. Write "FALSE" if the statement is definitely false. Write "MAYBE" if the information is insufficient to draw a conclusion.



(A) Star 1 is a main sequence star.

(B) The radius of star 2 is larger than that of star 3.

(C) Star 2 is younger than the current age of the Sun.

(D) Star 3 is more massive than the Sun.

(E) Star 3 is colder than the other two stars.

Solutions

Solution 7.1 Rapidly spinning neutron star

Problem p. 66

The limit of mass shedding occurs when the centripetal/centrifugal acceleration equals the gravitational acceleration,

$$\frac{GM}{r^2} = \frac{v^2}{r}.$$

(4 points)

Substituting the values for the neutron star, the linear velocity on the equator due to rotation is $1.36 \times 10^8\,\mathrm{m\,s^{-1}}$. The period satisfies $(2\pi/P)r = v$, thus

$$P = \frac{2\pi r}{v} = 0.462 \text{ ms}.$$

(6 points)

Solution 7.2 H-R diagram

Problem p. 67

(A) FALSE. (2 points)

(B) FALSE. (2 points)

(C) TRUE. (2 points)

(D) MAYBE. (2 points)

(E) TRUE. (2 points)

Chapter 8

Galaxies and Cosmology

Theory problems

Problem 8.1 Andromeda Galaxy Rotation

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Theory problems Q5 Solution p. 71

Q1–Q10 form a 100-point multiple-choice section. The supplied paper does not state individual point allocations.

For this multiple-choice problem, choose only one correct answer. Write the chosen answer and the original problem number on the answer sheet.

The following are known: the Earth–Andromeda galaxy distance, $R = 1.42 \times 10^{11} R_0$, where R_0 is the radius of the Earth's circular orbit around the Sun; the mass of our Galaxy, $M_G = 2.5 \times 10^{11} M_0$, where M_0 is the mass of the Sun; $T_0 = 1$ terrestrial year, the period of rotation of the Earth around the Sun; the mass of the Andromeda galaxy, $M_A = 3.6 \times 10^{11} M_0$.

The period of the rotational motions of our galaxy and the Andromeda galaxy, around their common centre of mass, is

(a) 5.1×10^{10} years.

(b) 6.9×10^{10} years.

(c) 7.9×10^{10} years.

(d) 8.2×10^{10} years.

Problem 8.2 Cosmology

Junior IOAA 2023, Volos, Greece Theory problems Q9 10 points Solution p. 71

For this question only the correct answer is required. No explanation is necessary. Arrange the following events in the order in which they occurred, from earliest to the most recent.

(A) Formation of the first stars.

(B) Inflationary era of the universe.

(C) Formation of helium nuclei.

(D) Formation of neutral hydrogen.

(E) Formation of oxygen atoms.

Problem 8.3 The Cosmic Horseshoe

Junior IOAA 2024, Kathmandu, Nepal Theory problems T7 15 points Solution p. 71

The Cosmic Horseshoe is a gravitationally lensed galaxy at $z = 2.379$. It has a magnification factor $\mu = 30$. The magnification that occurs due to gravitational lensing is analogous to that provided by an optical telescope, but is due to the gravitational effects of massive bodies. Answer some questions about this galaxy to help a researcher. **Note.** AB magnitudes are converted analogously to the magnitude scale that you are familiar with. The average flux density of an object is the flux F of that object divided by the bandwidth $\Delta\nu$ of the detector.

1. At what wavelength λ_{obs} would you expect to detect the Hα emission line emitted by this galaxy? Use $\lambda_{\text{rest}} = 656.3\,\text{nm}$.
2. Source plane reconstruction is the technique by which the lensing effects on the galaxy are estimated, in order to get an understanding of the galaxy's morphology. We use the technique and now need to determine the object's physical properties. Using the Hubble Space Telescope (HST), the F606W band AB magnitude is found to be 20.5. What is the object's magnitude after correcting for lensing?
3. One can convert from AB magnitudes to flux density by using the fact that a source at 0 AB magnitude has 3631 Jy of flux density. What is (a) the apparent flux density of the Cosmic Horseshoe and (b) the magnification corrected flux density?
4. The Cosmic Horseshoe is observed to have its F606W UV continuum emission spread across an area of $13\,\text{arcsec}^2$. What is its apparent UV continuum emission surface density, in units of $\text{mag}\,\text{arcsec}^{-2}$?
5. If the angular scale at the Horseshoe's redshift is $8\,\text{kpc}\,\text{arcsec}^{-1}$, what is the radius of the galaxy, assuming that it is a face-on disk?

Problem 8.4 Critical density and tennis balls

Junior IOAA 2024, Kathmandu, Nepal Theory problems T11 15 points Solution p. 72

In a classical approximation to get a feel for expanding universe, one may assume that the universe is uniformly filled cloud with small particles, and that the critical density ρ_c is the density of the cloud at which the expansion of the universe (the Hubble–Lemaître law) stops at infinite time.

1. Derive a classical expression for ρ_c of the universe.
2. Find the value of critical density in SI units.
3. If the universe was filled with tennis balls of mass 0.145 kg each distributed uniformly (no other mass) and the mass density of this universe equal to the critical density, then how many tennis balls would be included in a sphere of size of the Sun?

Supplied examination data. $H_0 = 70\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$, $G = 6.674 \times 10^{-11}\,\text{N}\,\text{m}^2\,\text{kg}^{-2}$, $1\,\text{pc} = 3.086 \times 10^{16}\,\text{m}$, and $R_\odot = 695500\,\text{km}$.

Solutions

Solution 8.1 Andromeda Galaxy Rotation

Problem p. 69

[The supplied S5 file solves a different version asking how long a 6 m telescope at wavelength 5 × 10^-7 m needs to detect Andromeda's orbital motion. The official reasoning provides an orbital-period formula, but its final value of about 10^3 years is the detection time, not the period asked for in Q5.]

The angular resolution is

$$\varphi_0 = 1.22\frac{\lambda}{D} = 1.22\frac{5\times10^{-7}}{6} = 10.16\times10^{-8}\,\text{rad}.$$

An angular displacement of φ_0 requires

$$\tau = \frac{\varphi_0}{\omega} = \frac{\varphi_0}{2\pi}T.$$

The generalized form of Kepler's third law for the Galaxy–Andromeda and Sun–Earth systems is

$$\left(\frac{T}{T_0}\right)^2\frac{M_G+M_A}{M_0+M_P} = \left(\frac{R}{R_0}\right)^3.$$

Since $M_P \ll M_0$,

$$T^2 = T_0^2\left(\frac{R}{R_0}\right)^3\frac{M_0}{M_G+M_A}.$$

The supplied version concludes

$$\tau = 1.22\frac{\lambda}{2\pi D}T_0\left(\frac{R}{R_0}\right)^{3/2}\left(\frac{M_0}{M_G+M_A}\right)^{1/2} \approx 10^3\,\text{years}.$$

Solution 8.2 Cosmology

Problem p. 69

The correct order is B, C, D, A, E.

Solution 8.3 The Cosmic Horseshoe

Problem p. 70

1. $\lambda_{\text{obs}} = (1+z)\lambda_{\text{rest}} = 3.379\times656.3\,\text{nm}$. (1 point) Thus $\lambda_{\text{obs}} = 2217.6\,\text{nm}$. (1 point)
2. Lensing is multiplicative on the flux, so $m_{\text{corr}} = m_{\text{obs}} + 2.5\log 30 = 24.19$. (2 points)
3. $f_{\text{obs}} = 3631\times10^{-20.5/2.5}\,\text{Jy} = 2.29\times10^{-5}\,\text{Jy}$. (2 points) The corrected value is $f_{\text{corr}} = f_{\text{obs}}/\mu = 7.64\times10^{-7}\,\text{Jy}$. (2 points)
4. $\Sigma_{\text{UV}} = f_{\text{obs}}/13$. (2 points) In magnitudes, $\Sigma'_{\text{UV}} = -2.5\log(\Sigma_{\text{UV}}/3631) = 23.3\,\text{mag arcsec}^{-2}$. (2 points)
5. $r = \sqrt{13\times8^2/(\pi\mu)} = 3\,\text{kpc}$. (3 points) One point each for the correct placement of magnification factor, squaring the scale, and correct answer.

Solution 8.4 Critical density and tennis balls

Problem p. 70

1. Use the escape velocity definition $v_{\mathrm{esc}} = \sqrt{2GM/R}$ (1 point) together with Hubble's law $v = H_0 R$ (1 point) and $M = \frac{4}{3}\pi R^3 \rho_c$. Substituting gives

$$\rho_c = \frac{3H_0^2}{8\pi G}.$$

(4 points)

2. $H_0 = 2.2685 \times 10^{-18}\,\mathrm{s}^{-1}$. (2 points) This gives $\rho_c = 9.20 \times 10^{-27}\,\mathrm{kg\,m}^{-3}$. (2 points)

3. $M = \frac{4}{3}\pi\rho_c R_\odot^3$. (1 point) Then $N = M/M_{\mathrm{tennis}}$. (1 point) Substitution gives $N = 89.45$. Either 89 or 90 is acceptable. (3 points)

Chapter 9

Observational Problems

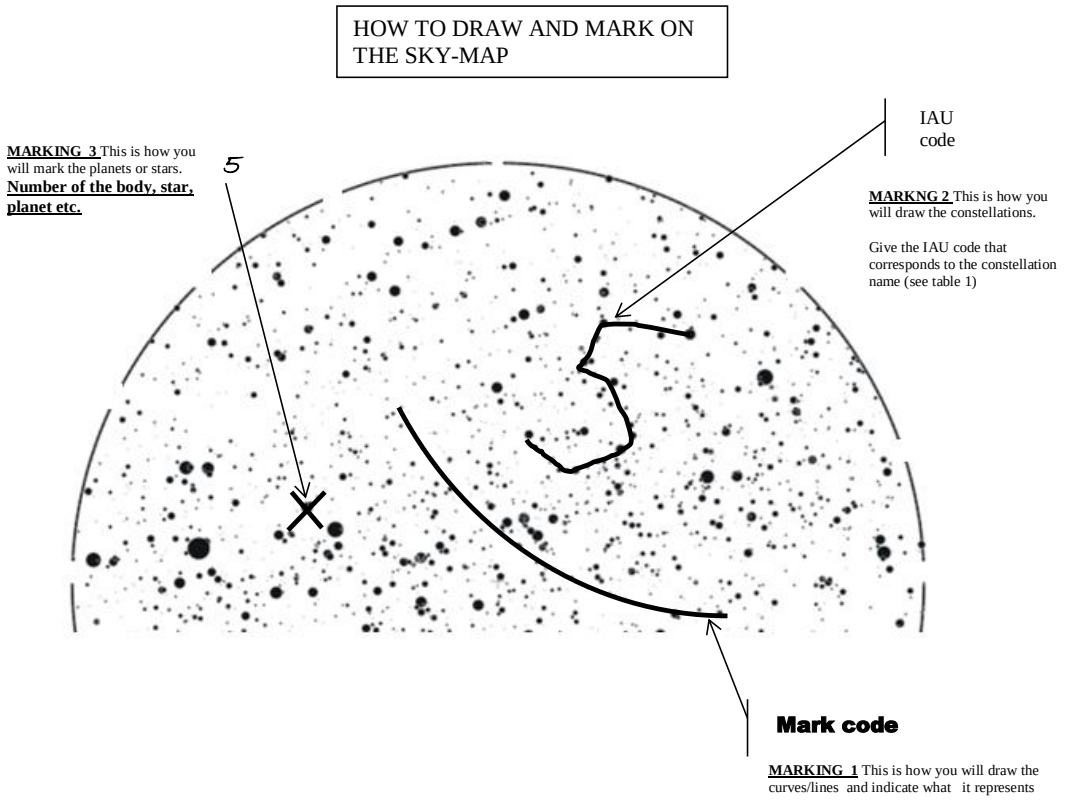
Sky maps and charts

Problem 9.1 Blind Sky Map

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Sky maps and charts Q13 Solution p. 98

You have 50 minutes to finish this part. Use only the pencil to make drawings and mark-ups on the sky map. After finishing, fill your student ID on the answer sheet and the sky map. Put the two papers in the folder, and leave the compass, ruler, and pencil on the table.

Use the sky map found in the envelope. The map is in azimuthal projection, so distance from the centre is proportional to zenith distance. The instructions for marking points, constellations, and curves are shown below.



The separately issued blind sky map and the referenced IAU-code Table 1 are absent from the

supplied question PDF. Annotated charts survive in the official solution and are reproduced there.

The map represents the sky somewhere in Romania. The zenith is at its centre.

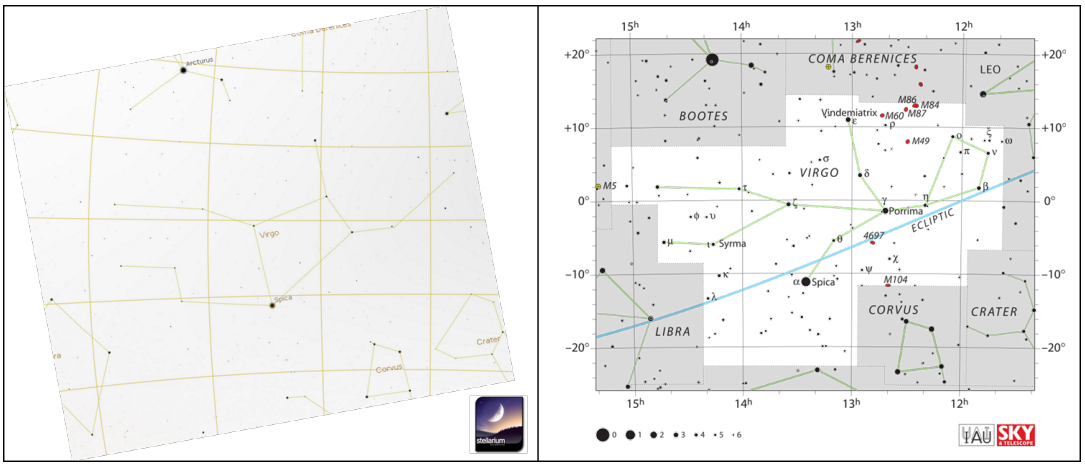
1. Find the latitude of the place and draw the greatest circumpolar circle on the map (Mark A).
2. Draw with continuous lines the celestial equator (Mark B), the ecliptic (Mark C), the galactic equator (Mark G), and the local meridian (Mark M).
3. Mark the cardinal points (N, S, E, W) and show three Solar System planets (P1, P2, P3).
4. Identify and mark the five brightest stars, with magnitudes less than +1.00, numbered from 1 (brightest) to 5 (faintest). Give their Bayer names.
5. Draw ten constellations you can identify, labelled with the corresponding IAU codes from Table 1.
6. Mark M31 (number 6), M27 (7), M15 (8), β Cet (9), and δ UMi (10).
7. Estimate the sidereal time of the map.
8. Estimate Altair's (α Aquilae) equatorial coordinates.

Problem 9.2 Navigating the night sky I

Junior IOAA 2023, Volos, Greece Sky maps and charts Q13 45 points Solution p. 99

Classroom test, part 1. You have 40 minutes in total. Two wide-field maps, around 100° wide, are provided. Map 1 is for an observer of the southern hemisphere and Map 2 for an observer of the northern hemisphere. The Milky Way is not depicted. Write your answers on the maps themselves. You can divide the available time between the two maps as you wish.

If more than the requested number of answers is provided, the following rules apply. For questions 1a, 1b, 2a and 2b there is no penalty for one extra answer, that is, marking $N + 1$ objects when N are requested. Each extra object after that loses partial points. Wrong answers do not receive negative marking. Whenever drawing constellations, use any common stick figure from atlases or planetarium software. All reasonable drawings are accepted, but keep them simple and recognisable. For example, Andromeda can be represented as the two back legs of Pegasus, rather than a person as in the IAU maps. The source gives the following two acceptable drawings of Virgo.



Map 1, southern sky.

(a) Draw stick figures for any 10 constellations on the map and write their IAU code next to them. [10 points] If you do not know the stick figure but know its area, you may provide its IAU code there. Refer to the supplied IAU list. Only choose constellations that are almost completely visible. Partially cut-off constellations are not accepted.

(b) Identify any 20 stars and provide their proper names or Bayer designations in English. [10 points] Both names are accepted but knowing both gives no extra points. The stars need not be in the constellations chosen in (a).

(c) A very bright star is missing. Mark its position with an "x". [2.5 points]

Map 2, northern sky.

(a) Draw stick figures of any 10 constellations and write their IAU codes next to them, using the same instructions as 1a. [10 points]

(b) Identify 20 stars and provide their proper names or Bayer abbreviations in English, using the same instructions as 1b. [10 points]

(c) A supernova is present. Identify and circle it. [2.5 points]

The IAU constellation list and the two maps follow.

IAU Constellations List

No	Constellation	IAU code		No	Constellation	IAU code
1	Andromeda	And		45	Lacerta	Lac
2	Antlia	Ant		46	Leo	Leo
3	Apus	Aps		47	Leo Minor	LMi
4	Aquarius	Aqr		48	Lepus	Lep
5	Aquila	Aql		49	Libra	Lib
6	Ara	Ara		50	Lupus	Lup
7	Aries	Ari		51	Lynx	Lyn
8	Auriga	Aur		52	Lyra	Lyr
9	Boötes	Boo		53	Mensa	Men
10	Caelum	Cae		54	Microscopium	Mic
11	Camelopardalis	Cam		55	Monoceros	Mon
12	Cancer	Cnc		56	Musca	Mus
13	Canes Venatici	CVn		57	Norma	Nor
14	Canis Major	CMa		58	Octans	Oct
15	Canis Minor	CMi		59	Ophiuchus	Oph
16	Capricornus	Cap		60	Orion	Ori
17	Carina	Car		61	Pavo	Pav
18	Cassiopeia	Cas		62	Pegasus	Peg
19	Centaurus	Cen		63	Perseus	Per
20	Cepheus	Cep		64	Phoenix	Phe
21	Cetus	Cet		65	Pictor	Pic
22	Chamaeleon	Cha		66	Pisces	Psc
23	Circinus	Cir		67	Piscis Austrinus	PsA
24	Columba	Col		68	Puppis	Pup
25	Coma Berenices	Com		69	Pyxis	Pyx
26	Corona Australis	CrA		70	Reticulum	Ret
27	Corona Borealis	CrB		71	Sagitta	Sge
28	Corvus	Crv		72	Sagittarius	Sgr
29	Crater	Crt		73	Scorpius	Sco
30	Crux	Cru		74	Sculptor	Scl
31	Cygnus	Cyg		75	Scutum	Sct
32	Delphinus	Del		76	Serpens	Ser
33	Dorado	Dor		77	Sextans	Sex
34	Draco	Dra		78	Taurus	Tau
35	Equuleus	Equ		79	Telescopium	Tel
36	Eridanus	Eri		80	Triangulum	Tri
37	Fornax	For		81	Triangulum Australe	TrA
38	Gemini	Gem		82	Tucana	Tuc
39	Grus	Gru		83	Ursa Major	UMa
40	Hercules	Her		84	Ursa Minor	UMi
41	Horologium	Hor		85	Vela	Vel
42	Hydra	Hya		86	Virgo	Vir
43	Hydrus	Hyi		87	Volans	Vol
44	Indus	Ind		88	Vulpecula	Vul


Map 1



Problem 9.3 Navigating the night sky II

Junior IOAA 2023, Volos, Greece Sky maps and charts Q14 45 points Solution p. 99

Classroom test, part 2. You have 40 minutes in total. Three answer-sheet tables are provided. Tables 2.1ab and 2.1c correspond to part 2.1, "Messier Objects", and Table 2.2 corresponds to part 2.2, "Bright Stars". Mark answers in these tables and divide your time as you wish. For both parts, only the first 10 answers in each column are evaluated if more are given. There is no

negative marking for wrong answers.

Part 2.1, Messier Objects. Table 2.1ab lists all 110 Messier objects, with its first row filled in as an example.

(a) Write the letter code of the correct type for any 10 objects in the corresponding "Type" column. [5 points] Choose Galaxy (GL), Nebula (NB), Globular Cluster (GC), Open Cluster (OC), or other (OT). A few Messier objects fall outside the first four categories and are classed as other.

(b) Write the IAU constellation code for any 10 objects, not necessarily the same ones as in (a), in the "Constellation" column. [5 points]

(c) Table 2.1c lists 25 objects also present in the Messier list. For any 10, write their Messier number. [10 points]

Part 2.2, Bright Stars. Table 2.2 lists the 92 brightest stars, excluding the Sun, up to 2.50 mag. It gives proper names, Bayer designations, apparent visual magnitudes m_v, and distances from the Sun in light-years. The stars are ordered alphabetically.

(d) Identify any 10 stars that are binary or members of multiple systems. Tick the "Binary or Multiple" column. [5 points] For systems, the given visual magnitude is the combined naked-eye magnitude.

(e) Identify any 10 variable stars, of any kind, by ticking the "Variable" column. [10 points] The given magnitude for a variable is its approximate time-averaged naked-eye visual magnitude.

(f) The Morgan–Keenan (MK) spectral sequence uses O, B, A, F, G, K, M from hottest to coolest. For this exercise G, K and M stars are called "cool". Identify any 10 cool stars and tick the "Cool" column. [10 points] For systems in this list, either all members have the same type (for example all cool), or their brightest component dominates the observed colour. A cool member may be much brighter than a hot member, giving a red or orange naked-eye appearance.

The blank Messier answer tables and the supplied bright-star data follow.

Table 2.1ab

M#	Type	Const.		M#	Type	Const.		M#	Type	Const.
M1	**NB**	**TAU**		M38				M75		
M2				M39				M76		
M3				M40				M77		
M4				M41				M78		
M5				M42				M79		
M6				M43				M80		
M7				M44				M81		
M8				M45				M82		
M9				M46				M83		
M10				M47				M84		
M11				M48				M85		
M12				M49				M86		
M13				M50				M87		
M14				M51				M88		
M15				M52				M89		
M16				M53				M90		
M17				M54				M91		
M18				M55				M92		
M19				M56				M93		
M20				M57				M94		
M21				M58				M95		
M22				M59				M96		
M23				M60				M97		
M24				M61				M98		
M25				M62				M99		
M26				M63				M100		

M27				M64				M101		
M28				M65				M102		
M29				M66				M103		
M30				M67				M104		
M31				M68				M105		
M32				M69				M106		
M33				M70				M107		
M34				M71				M108		
M35				M72				M109		
M36				M73				M110		
M37				M74						

Table 2.1c

Name	M#
Andromeda Galaxy	
Beehive Cluster	
Black Eye Galaxy	
Bode's Galaxy	
Butterfly Cluster	
Cat's Eye Galaxy	
Cigar Galaxy	
Crab Nebula	
Dumbbell Nebula	

Name	M#
Eagle Nebula	
Lagoon Nebula	
Le Gentil	
Omega Nebula	
Orion Nebula	
Owl Nebula	
Pinwheel Galaxy	
Pleiades	

Name	M#
Ptolemy Cluster	
Ring Nebula	
Sombrero Galaxy	
Spindle Galaxy	
Triangulum Galaxy	
Trifid Nebula	
Whirlpool Galaxy	
Wild Duck Cluster	

Table 2.2

Rank	Proper name (IAU)	Bayer designation	m_v	Distance (ly)	Binary or Multiple	Variable	Cool
2	—	α Cen	-0.27	4.34			
75	—	α Lup	2.30	550			
76	—	ε Cen	2.30	380			
79	—	η Cen	2.35	310			
82	—	κ Sco	2.39	460			
89	—	γ Cas	2.47	610			
30	—	γ Vel	1.72	840			
9	Achernar	α Eri	0.46	139			
92	Acrab	β Sco	2.50	404			
13	Acrux	α Cru	0.76	320			
22	Adhara	ε CMa	1.50	430			
14	Aldebaran	α Tau	0.86	65			
57	Algieba	γ Leo	2.08	130			
61	Algol	β Per	2.12	93			
43	Alhena	γ Gem	1.92	100			
33	Alioth	ε UMa	1.77	83			
91	Aljanah	ε Cyg	2.48	72			
40	Alkaid	η UMa	1.86	100			
72	Almach	γ And	2.26	350			
31	Alnair	α Gru	1.74	100			
29	Alnilam	ε Ori	1.69	2,000			
32	Alnitak	ζ Ori	1.77	1,300			
48	Alphard	α Hya	2.00	180			
66	Alphecca	α CrB	2.23	75			
55	Alpheratz	α And	2.06	97			
45	Alsephina	δ Vel	1.96	80			
12	Altair	α Aql	0.76	17			
87	Aludra	η CMa	2.45	2,000			
81	Ankaa	α Phe	2.38	77			
15	Antares	α Sco	0.96	550			

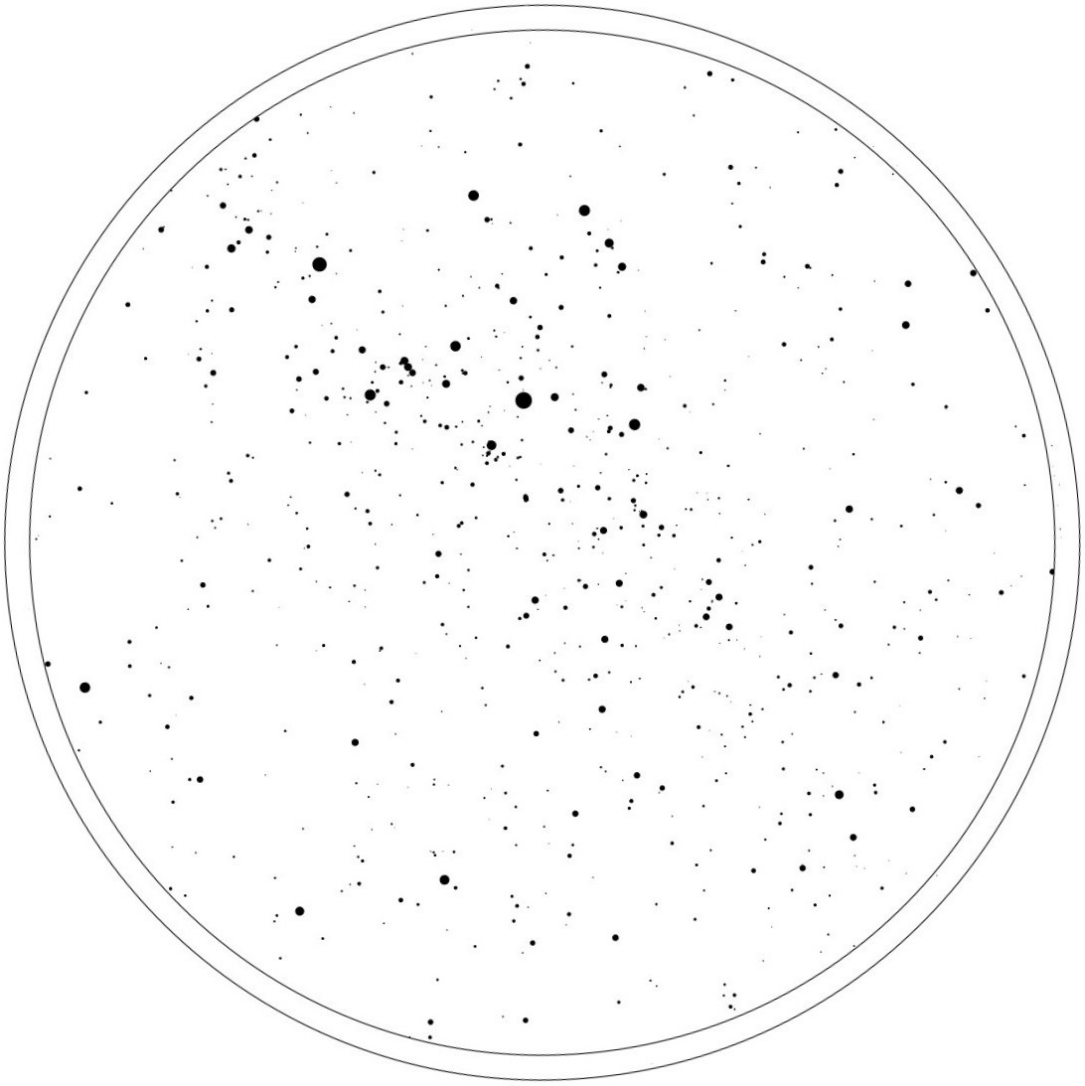
3	Arcturus	α Boo	-0.05	37			
64	Aspidiske	ι Car	2.21	690			
42	Atria	α TrA	1.91	390			
39	Avior	ε Car	1.86	600			
26	Bellatrix	γ Ori	1.64	250			
10	Betelgeuse	α Ori	0.50	550			
4	Canopus	α Car	0.00	310			
6	Capella	α Aur	0.08	43			
73	Caph	β Cas	2.28	54			
24	Castor	α Gem	1.62	51			
19	Deneb	α Cyg	1.25	2,600			
60	Denebola	β Leo	2.11	36			
50	Diphda	β Cet	2.02	96			
77	Dschubba	δ Sco	2.31	400			
34	Dubhe	α UMa	1.79	120			
27	Elnath	β Tau	1.65	130			
69	Eltanin	γ Dra	2.23	150			
83	Enif	ε Peg	2.40	670			
18	Fomalhaut	α PsA	1.16	25			
25	Gacrux	γ Cru	1.64	89			
11	Hadar	β Cen	0.61	390			
49	Hamal	α Ari	2.00	66			
74	Izar	ε Boo	2.29	202			
38	Kaus Australis	ε Sgr	1.85	140			
58	Kochab	β UMi	2.08	130			
78	Larawag	ε Sco	2.31	65			
90	Markab	α Peg	2.48	140			
88	Markeb	κ Vel	2.46	540			
41	Menkalinan	β Aur	1.90	80			
54	Menkent	θ Cen	2.06	59			
80	Merak	β UMa	2.37	79			
28	Miaplacidus	β Car	1.69	110			

20	Mimosa	β Cru	1.25	280			
67	Mintaka	δ Ori	2.23	900			
53	Mirach	β And	2.05	200			
35	Mirfak	α Per	1.80	510			
46	Mirzam	β CMa	1.98	500			
51	Mizar	ζ UMa	2.04	83			
63	Muhlifain	γ Cen	2.17	130			
71	Naos	ζ Pup	2.25	1,080			
52	Nunki	σ Sgr	2.05	230			
44	Peacock	α Pav	1.94	180			
86	Phecda	γ UMa	2.44	84			
47	Polaris	α UMi	1.98	430			
17	Pollux	β Gem	1.14	34			
8	Procyon	α CMi	0.34	11			
56	Rasalhague	α Oph	2.07	47			
21	Regulus	α Leo	1.39	79			
7	Rigel	β Ori	0.13	860			
85	Sabik	η Oph	2.43	88			
68	Sadr	γ Cyg	2.23	1,500			
59	Saiph	κ Ori	2.09	720			
37	Sargas	θ Sco	1.84	330			
84	Scheat	β Peg	2.42	200			
70	Schedar	α Cas	2.24	230			
23	Shaula	λ Sco	1.62	570			
1	Sirius	α CMa	-1.46	8.6			
16	Spica	α Vir	0.97	250			
65	Suhail	λ Vel	2.21	570			
62	Tiaki	β Gru	2.15	170			
5	Vega	α Lyr	0.03	25			
36	Wezen	δ CMa	1.82	1,800			

Problem 9.4 A sky chart from Nepal

Junior IOAA 2024, Kathmandu, Nepal Sky maps and charts OC1 45 points Solution p. 99

The map in the answer sheet represents the sky in azimuthal equidistant projection. The Zenith is at the center of the chart.



1. Estimate the latitude of the location from where this sky map is captured. [4 points]

2. Label the cardinal points (N, E, S, W) on the map and mark the three planets from the Solar System in the sky map with a small square around each of them. [7 points]

3. Draw on the map with continuous lines the celestial equator (EQ), the ecliptic (EC), and the local meridian (ME) and label them. [4.5 points]

4. Trace 10 constellations in the sky map and label them as C1, C2, ..., C10 and write the corresponding Latin name of the constellations or IAU code. [10 points]

5. Identify and circle the 4 brightest stars in the given sky map. Label the star from S1, the brightest, and continue with the fainter ones till number S4 for the faintest. Write the Bayer designation or the official IAU name of the four identified stars. [6 points]

6. Mark (with a small circle) positions of the following objects and label them. [10.5 points]

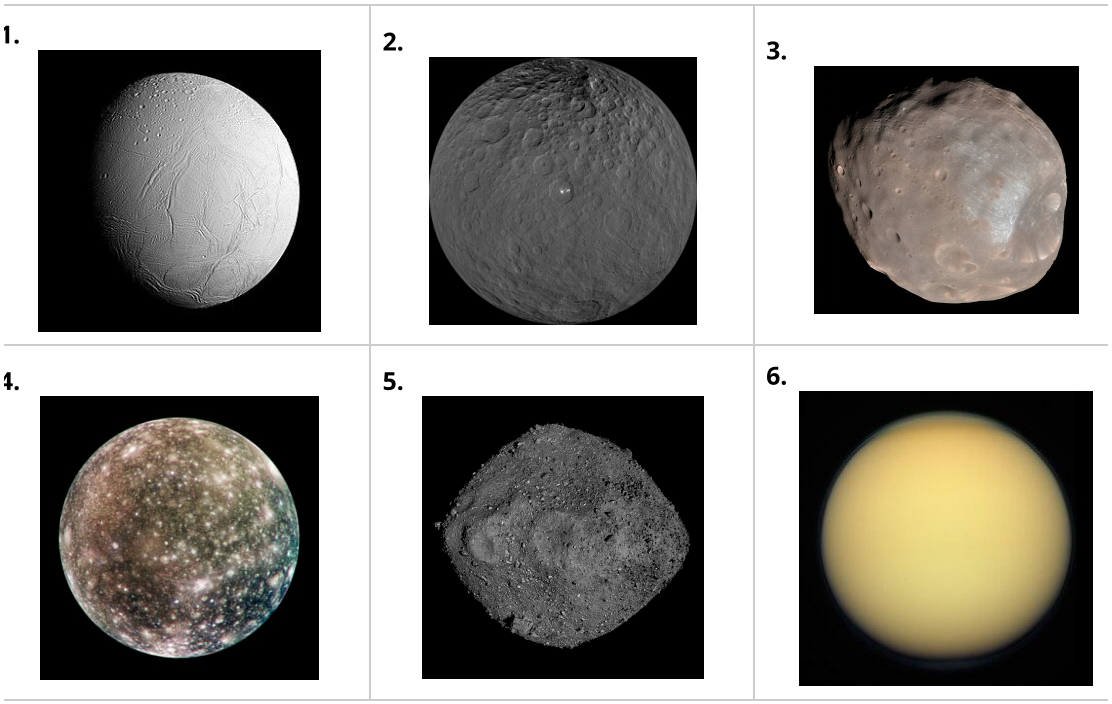
   M42, M81, M41, M31, γ Ori, α Per, α Cyg.

7. Estimate the equatorial coordinates of the star Aldebaran (α Tau). [3 points]

Problem 9.5 Classifying Solar System objects

Junior IOAA 2024, Kathmandu, Nepal Sky maps and charts OC2 9 points Solution p. 103

Fill in the numbers of the pictures as per their category.

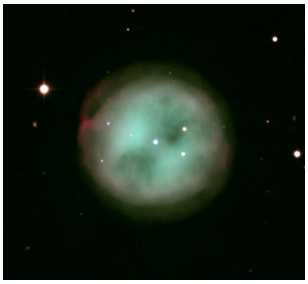
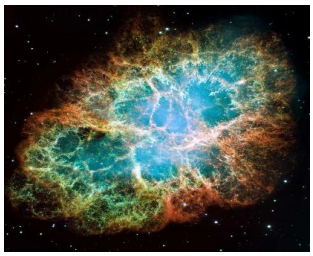


Asteroids _____ Dwarf Planets _____ Satellites _____

Problem 9.6 Identifying deep-sky objects

Junior IOAA 2024, Kathmandu, Nepal Sky maps and charts OC3 18 points Solution p. 104

Complete the table regarding the objects below.

O1		O4	
O2		O5	
O3		O6	

Object	Messier Number (e.g. M32)	Constellation (e.g. Ori or Orion)
O1		
O2		
O3		
O4		
O5		
O6		

Problem 9.7 Know Your Sky

Junior IOAA 2025, Piatra Neamț, Romania Sky maps and charts OM1 40 points Solution p. 104

Use the sky map Map-OM1 to answer the questions below. The map has a stereographic projection, and planets and the Moon are present on the map.

(OM1.1) Mark all the stars listed below with a square (□) drawn around each star on Map-OM1. Label every marked star with its corresponding object number. [10]

1	α Ari	Hamal	6	β CMi	Gomeisa
2	δ Cyg	Fawaris	7	α Lep	Arneb
3	β Dra	Rastaban	8	δ UMa	Megrez
4	α Cet	Menkar	9	β Tau	Elnath
5	δ Eri	Rana	10	α And	Alpheratz

(OM1.2) Use the provided plus signs (+) on the map and label them corresponding to the following eight Messier objects, M15, M31, M34, M35, M39, M42, M44, and M45. [8]

(OM1.3) The map is centered at the zenith. Draw the meridian and then draw with star-line patterns the constellations along or near the meridian and note the IAU abbreviations on the sheet. [10]

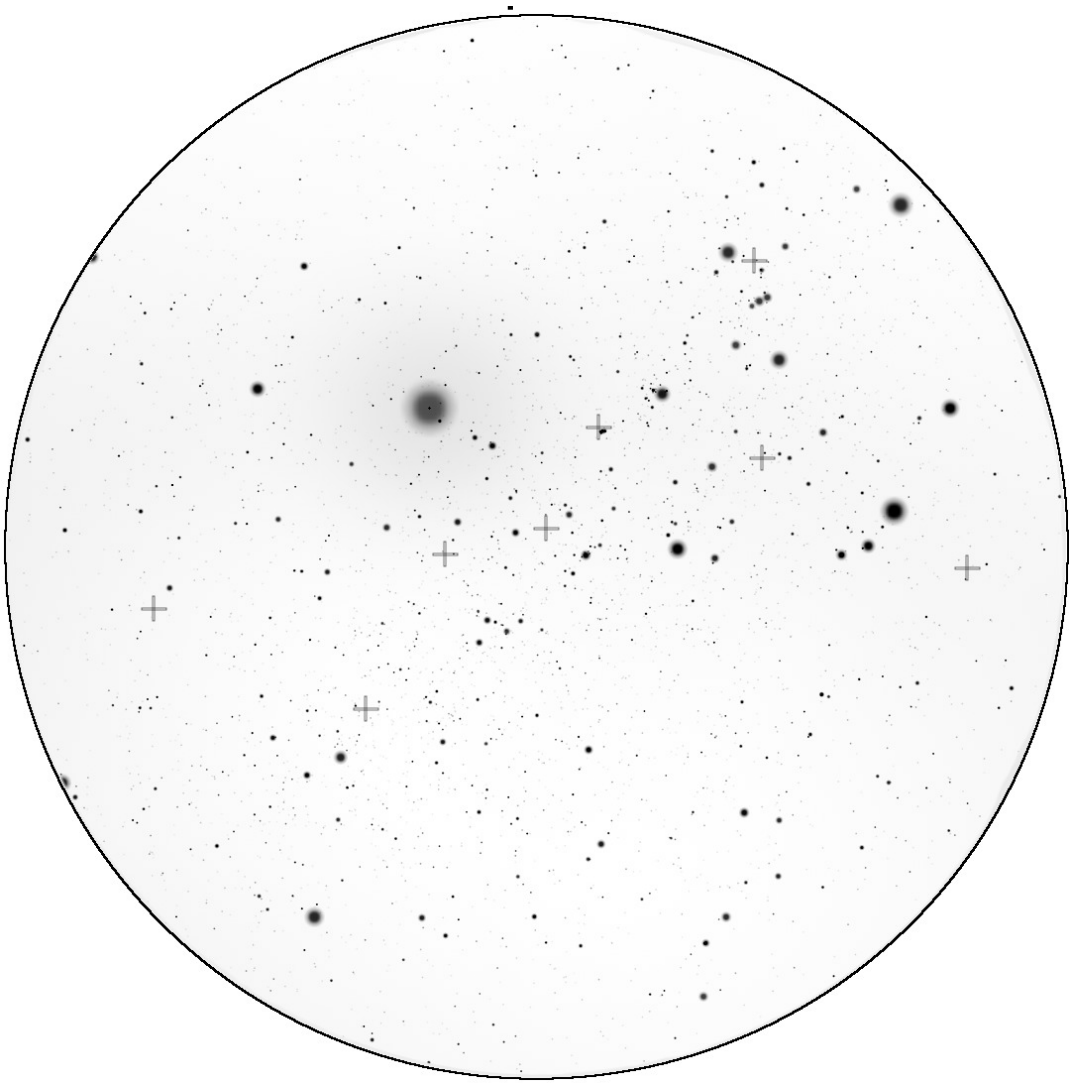
(OM1.4) Draw the celestial equator on Map-OM1. Mark it as "EQ". [2]

(OM1.5) Mark the cardinal points as N, S, W, and E on Map-OM1. [2]

(OM1.6) Draw the ecliptic on Map-OM1 and label it "EC". [2]

(OM1.7) Draw the galactic equator on Map-OM1 and label it "GE". Mark either the galactic center (GC) or anticenter (GA), whichever is visible on the map. [4]

(OM1.8) Draw the circumpolar circle on Map-OM1 and label it "CC". [2]



Problem 9.8 Chart and Time

Junior IOAA 2025, Piatra Neamț, Romania Sky maps and charts OM2 10 points Solution p. 105

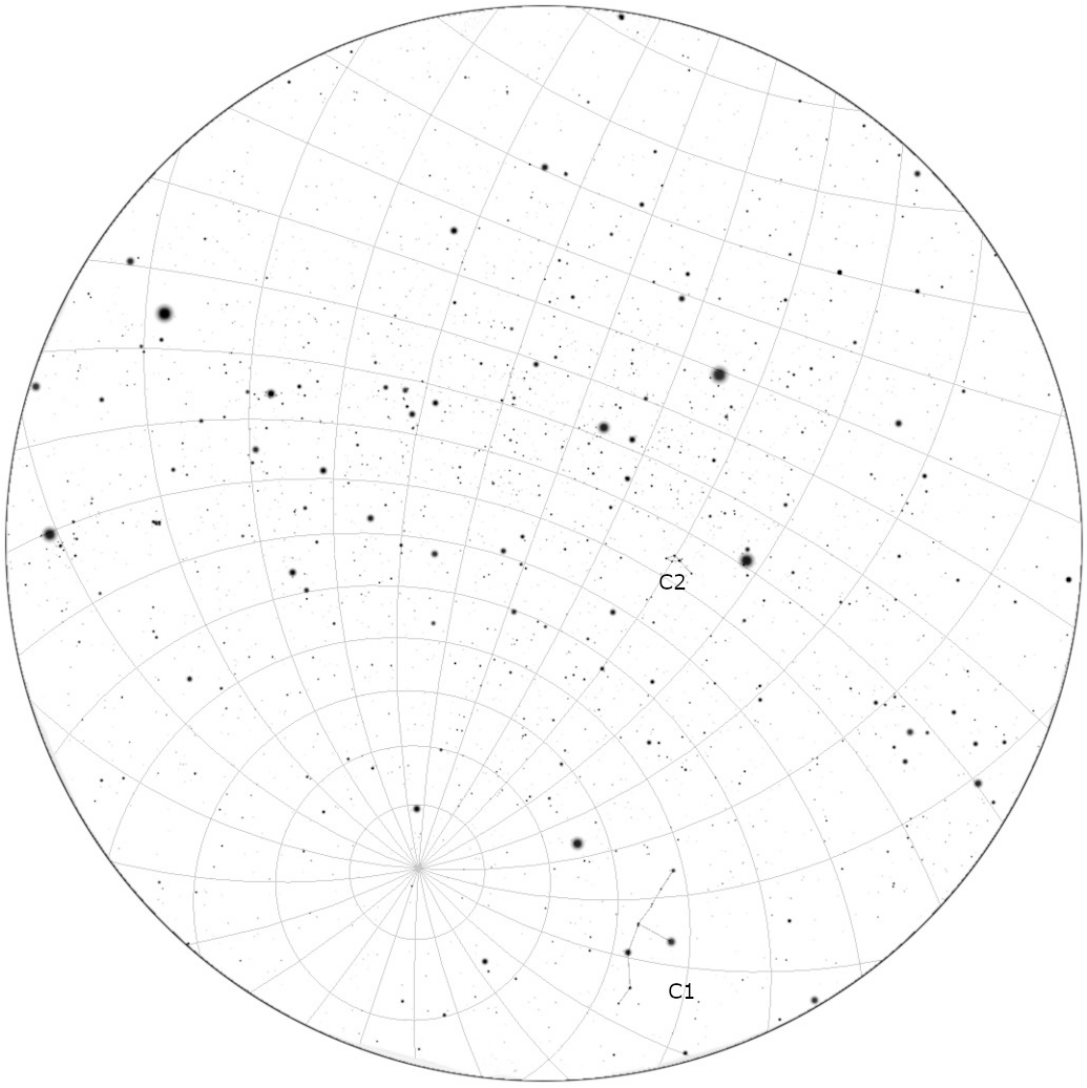
Use the OM2–Chart and Time page below. The map is centered on the zenith and shows two constellations, C1 and C2, together with the horizon and a coordinate grid.

(OM2.1) Identify the C1 and C2 constellations and write their Latin names or IAU abbreviations. [2]

(OM2.2) Draw an arrow each indicating the α stars of the C1 and C2 constellations. [2]

(OM2.3) One of the equinox points, either the vernal equinox (VE) or autumnal equinox (AE), is present on the map. Mark it with an X and label it VE or AE. [2]

(OM2.4) Identify which coordinate grid is present on the map. Mark one of ecliptic, azimuthal, galactic, or equatorial. [4]



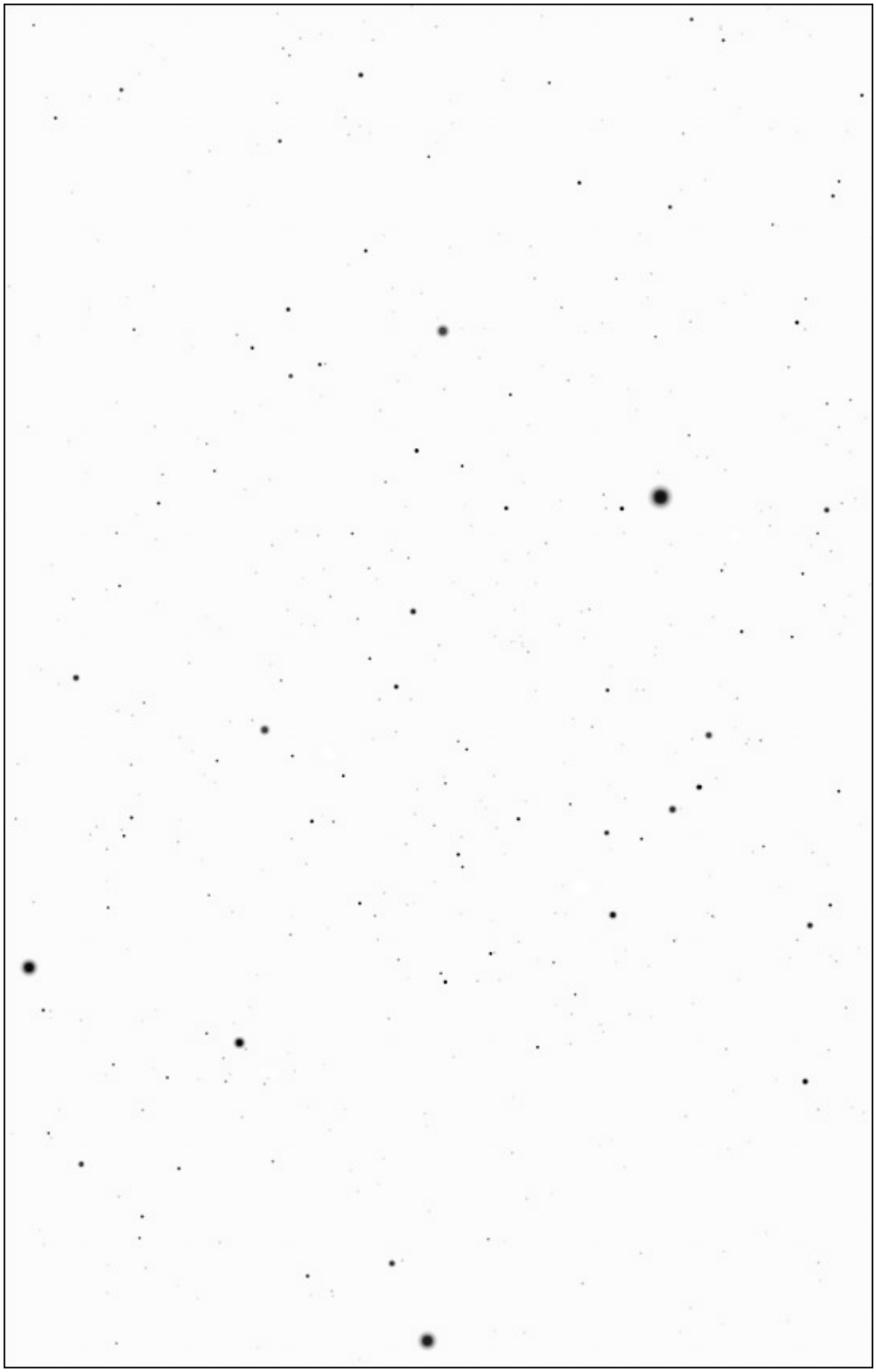
Problem 9.9 Missing Stars

Junior IOAA 2025, Piatra Neamț, Romania Sky maps and charts OM3 10 points Solution p. 105

Use the OM3–Missing Stars map. Four stars with visual magnitudes ≤ 2.5 are missing.

(OM3.1) Find which stars are missing and mark them on the map using a plus sign (+), labeling them S1, S2, S3, S4. [4]

(OM3.2) For each missing star, write its IAU star name or Bayer designation, and whether it is a double or multiple star. Write Y if it is, or N if it is not. [6]



Telescope

Problem 9.10 Sun observations

Junior IOAA 2023, Volos, Greece Telescope Q15 45 points Solution p. 106

You have 20 minutes in total. Avoid any direct viewing of the Sun. The telescope is already equipped with a solar filter and the finder scope has been removed or covered. Write answers on the separate answer sheet. Work on the question paper is not evaluated.

(a) Carefully point the telescope towards the Sun and centre it in the field of view. Ask the examiner to confirm successful targeting and record it on the answer sheet. If you cannot target the Sun, the examiner will do so and all points for this question are lost. [5 points]

(b) Within the disk on the answer sheet, draw all sunspots visible through the telescope. [10 points]

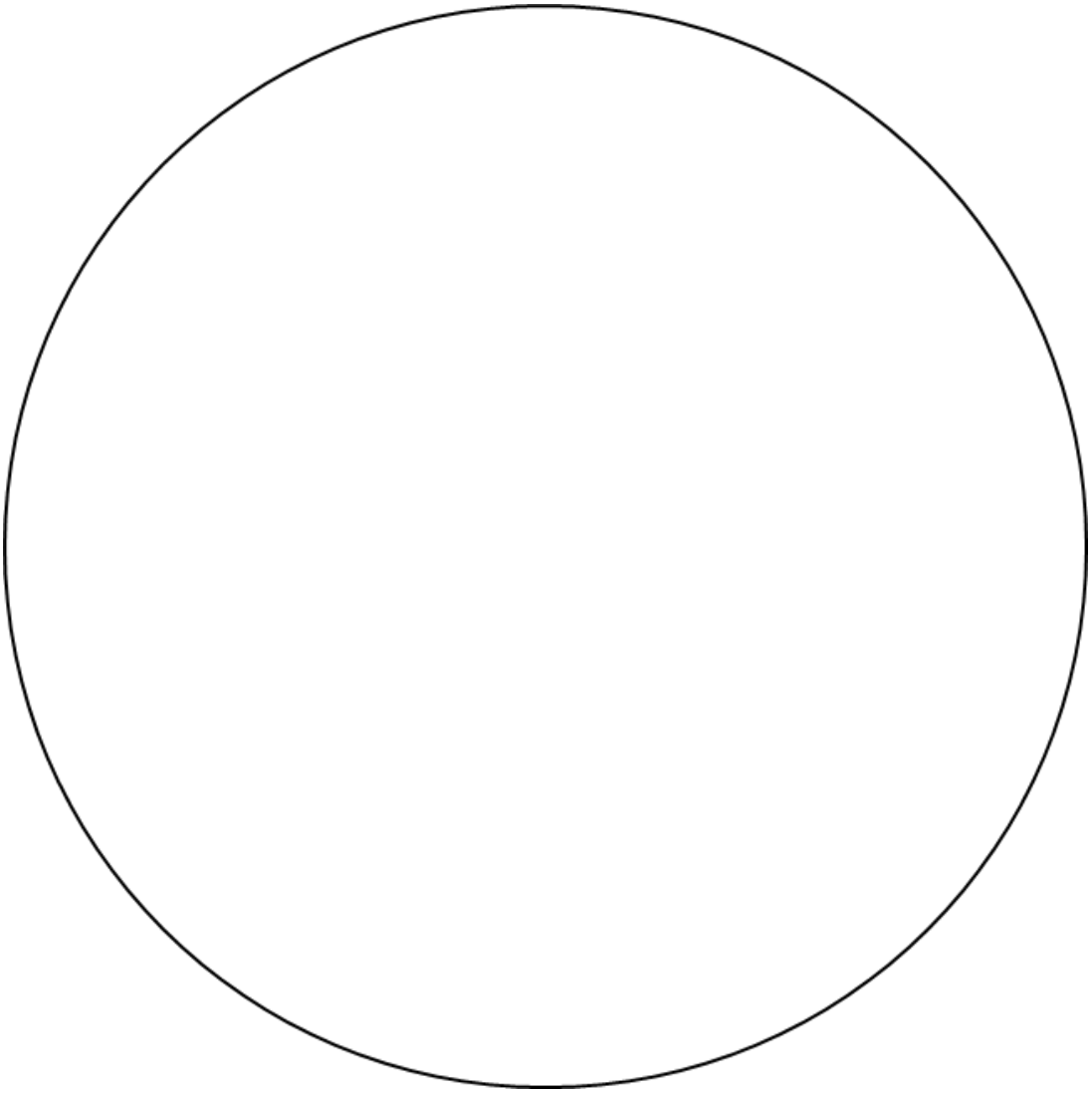
(c) Measure the Sun's declination using the telescope. [5 points]

(d) Estimate the Sun's right ascension in hours and minutes. [10 points]

(e) What will be the Sun's approximate declination in 3 months? [5 points]

(f) Call the examiner and demonstrate how to "scan" the celestial equator with the telescope, passing through all equatorial constellations. Give the examiner your answer sheet to record the result. [10 points]

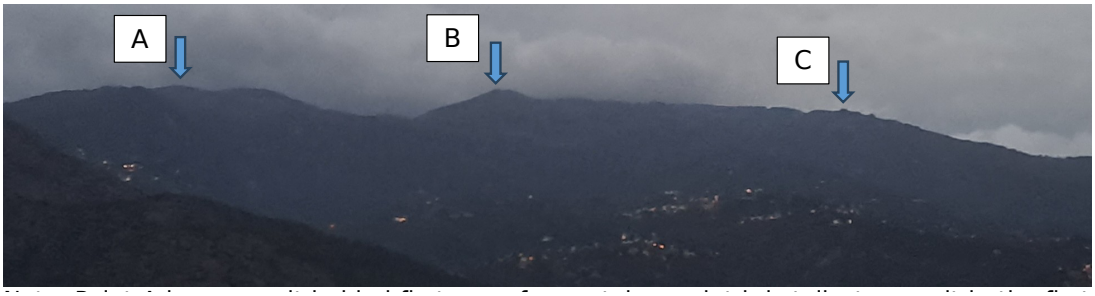
The solar disk supplied for part (b) follows.



Problem 9.11 Cloudy sky observations

Junior IOAA 2023, Volos, Greece Telescope Q16 45 points Solution p. 107

The source also provides this cloudy-weather observation task. You have 30 minutes. The telescope and finder tube are already aligned. Do not change the alignment. Both right-ascension and declination dials are calibrated. Write answers on the separate answer sheet. Answers/calculations on the question paper will not be evaluated. The north–south line is parallel to the balcony railing, with north on your left.



Point A is a summit behind the first row of mountains. Point B is the tallest summit in the first row, and point C is a square building at the right end of the mountain.

(A) Point the telescope to the three arrowed points and use its dials to determine their equatorial coordinates. [15 points]

(B) Find the angular distance between points A and C. [10 points]

(C) Call the examiner and demonstrate how to "scan" the celestial equator with the telescope, passing through all equatorial constellations. Give the examiner your answer sheet to record the result. [10 points]

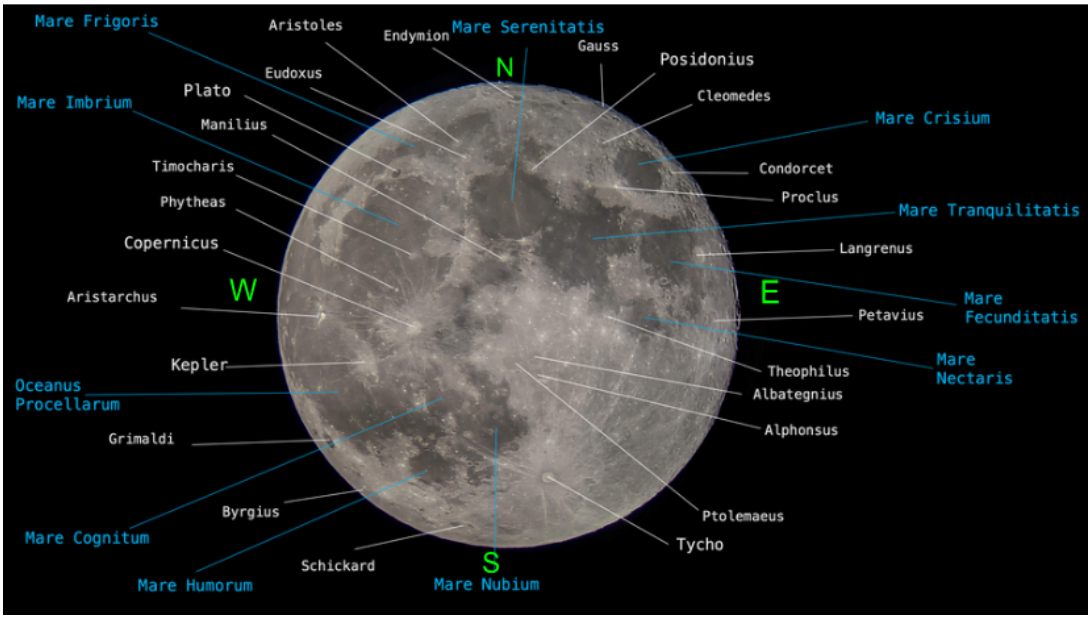
(D) Point the telescope towards equatorial coordinates $13^{\mathrm{h}}20^{\mathrm{m}}$ and 2° S. Ask the examiner to confirm successful targeting. [10 points]

Problem 9.12 The Moon

Junior IOAA 2024, Kathmandu, Nepal Telescope OT1 32 points Solution p. 107

You have 2 minutes 30 seconds to point and focus the telescope to the Moon. If you are not able to point it within 2 minutes 30 seconds, ask the supervisor to point it for you. [6 points] A zoomed-in view of the Moon will be shown for 90 seconds and a zoomed-out view of the Moon will be shown for the next 60 seconds. On your answer sheet,

1. Draw the Moon, including the lunar terminator. [4 points]
2. Label the cardinal points on the Moon drawn. [4 points]
3. With the help of the moon chart provided below, identify three visible craters and three maria and mark their positions on your drawing. [12 points]



Based on your observations, complete a table giving the age of the Moon (number of days since new moon), the hemisphere from where the Moon is being watched, and the official IAU name of the constellation in which the Moon is located. [6 points] *The live telescope view supplied during the examination is represented by the official solution image.*

Problem 9.13 Jupiter and its moons

Junior IOAA 2024, Kathmandu, Nepal Telescope OT2 16 points Solution p. 108

An animation of a planet and its moons will be displayed on the screen. The animation corresponds to a single location over a 3-day period.

1. Identify the planet shown in the images by ticking the correct option. Mercury, Venus, Mars, Jupiter, Saturn, Uranus, or Neptune.
2. Draw initial and final positions of the moons on your answer sheet and label the moons as A, B, C, and D. For example, if one moon is labeled as A in the first drawing, the same should be labeled as A in the second drawing.
3. Arrange the labels (A, B, C, D) of the observed moons according to their size from largest to smallest.

The examination animation is absent from the supplied archive. The official key includes its initial and final views.

Problem 9.14 Telescope Adjustment Knobs

Junior IOAA 2025, Piatra Neamț, Romania Telescope OT1 4 points Solution p. 109

Telescope-round instructions. There are five minutes to read the instructions before the round starts and 20 minutes with the telescope. Supervisors may assist in locking or unlocking the telescope axes on request. All telescopes have a 25 mm eyepiece. The scales are calibrated so that the hour angle axis represents right ascension. Handle the telescope carefully.

On your telescope, the adjustment knobs are marked with 1 and 2. Mark with an X on your answer sheet the answers to the following questions.

(a) Which coordinate does knob No. 1 refer to?

(b) Which coordinate does knob No. 2 refer to?

For each knob, choose altitude, right ascension, declination, or azimuth. *[The telescope and wall signs for this practical were supplied at the examination.]*

Problem 9.15 Coordinate Identification

Junior IOAA 2025, Piatra Neamț, Romania Telescope OT2 9 points Solution p. 109

On the wall opposite to your position, you will find a sign numbered 1. Point the telescope towards the specified sign and you will see a representation. For the star marked with the symbol $*$, answer the following, marking an X on the answer sheet.

(a) What is the name of the coordinate marked with letter A?

(b) What is the name of the coordinate marked with letter B?

(c) What is the name of the coordinate marked with letter C?

Choose among azimuth, local hour angle, sidereal hour angle, zenith angle, altitude, declination, and meridian angle. *[The original wall sign was supplied during the practical examination.]*

Problem 9.16 Constellation through the Eyepiece

Junior IOAA 2025, Piatra Neamț, Romania Telescope OT3 20 points Solution p. 109

On the wall opposite to your position, you will find a sign named O. Point the telescope towards the specified sign, where you should see a constellation. Answer the following.

(a) Draw the constellation seen through the eyepiece on your answer sheet.

(b) Specify the IAU name or abbreviation of the constellation.

(c) Mark the Alpha star with a circle on your drawing.

(d) Write the IAU name of the Alpha star.

(e) Between stars Gamma (γ) and Beta (β) of this constellation, there is a Messier object. Specify its IAU name or Messier catalogue number.

(f) Specify the type of the object. Choose galaxy, supernova, star cluster, planetary nebula, dwarf planet, or reflection nebula.

(g) On your drawing, mark this Messier object's position with a triangle (Δ).

[The original constellation sign was supplied during the practical examination.]

Problem 9.17 Objects and Telescope Coordinates

Junior IOAA 2025, Piatra Neamț, Romania Telescope OT4 27 points Solution p. 110

On the wall opposite to you, you will find signs named R, N and Y. Point the telescope towards each specific sign and, for each sign, answer the following.

(a) Write the IAU name or Messier Catalogue Number, if applicable, of the celestial object found on the sign.

(b) Give the right ascension read from the telescope for the sign.

(c) Give the declination read from the telescope for the sign.

[The original object signs were supplied during the practical examination.]

Night observation

Problem 9.18 Observations with the Naked Eye

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Night observation Q14 Solution p. 111

Working time is 30 minutes, including five minutes to become familiar with the tasks. The available equipment is a Newtonian telescope with EQ5 mount, 200 mm primary mirror diameter and 1000 mm focal length, plus 25 mm and 10 mm eyepieces.

A. Observations with the naked eye.

1. Estimate the sidereal time at the start of the test, within ± 10 minutes.
2. Find, within $\pm 3°$, the angular distances between α Per (Mirfak) and β And (Mirach), and between β Cas (Caph) and γ Cyg (Sadar).

3. Specify two constellations at the meridian, their type of culmination, and their alpha stars.
4. Estimate Jupiter's right ascension within ±10 minutes.
5. For β Ari (Sheratan), estimate the apparent magnitude within ±0.4, declination within ±3°, and right ascension within ±10 minutes.

Problem 9.19 Observations Using the Astronomical Telescope

Junior IOAA 2022, Câmpulung Moldovenesc, Romania Night observation Q15 Solution p. 111

Working time is 30 minutes, including five minutes to become familiar with the tasks. The available equipment is a Newtonian telescope with EQ5 mount, 200 mm primary mirror diameter and 1000 mm focal length, plus 25 mm and 10 mm eyepieces.

B. Observations using the astronomical telescope.

1. Point the telescope at β Cyg and specify the star's type.
2. Point the telescope at Messier M45. State its constellation and object type.
3. Point the telescope at Jupiter with the 25 mm eyepiece. Draw what you observe and identify the Galilean satellites visible with the 10 mm eyepiece.

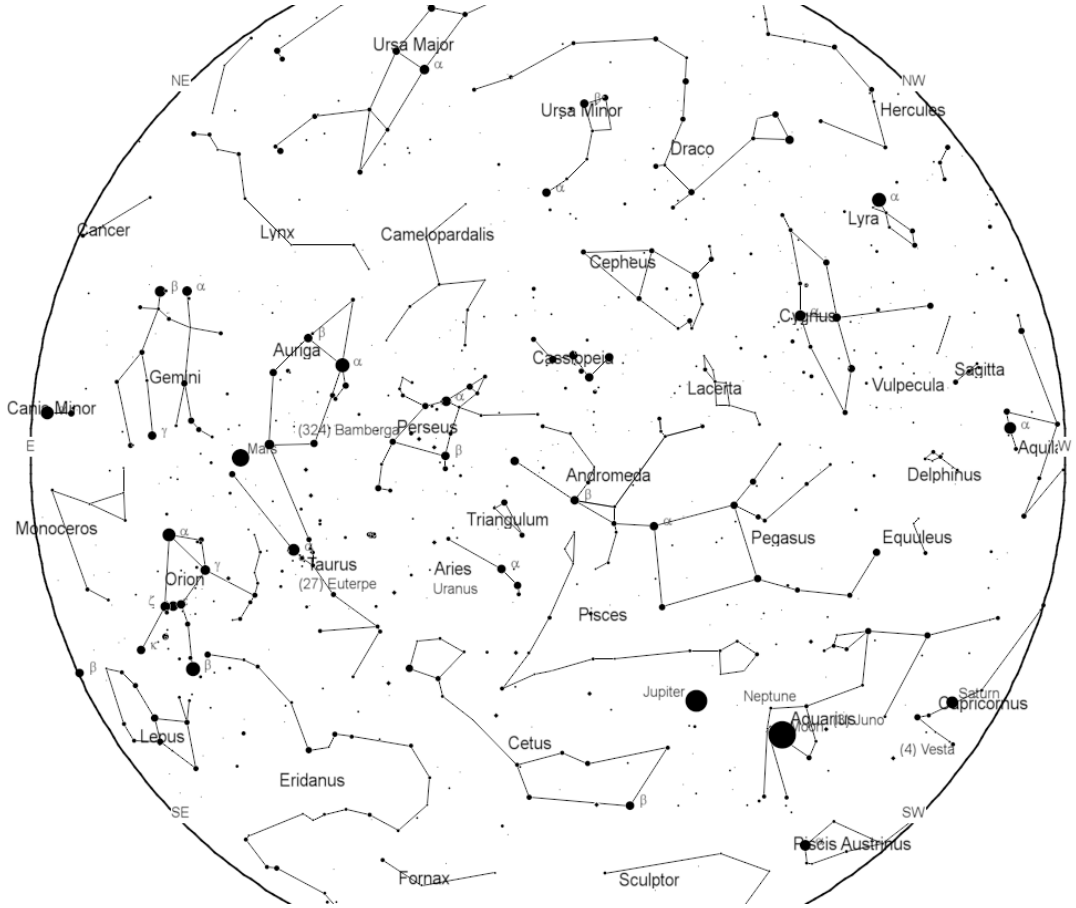
Solutions

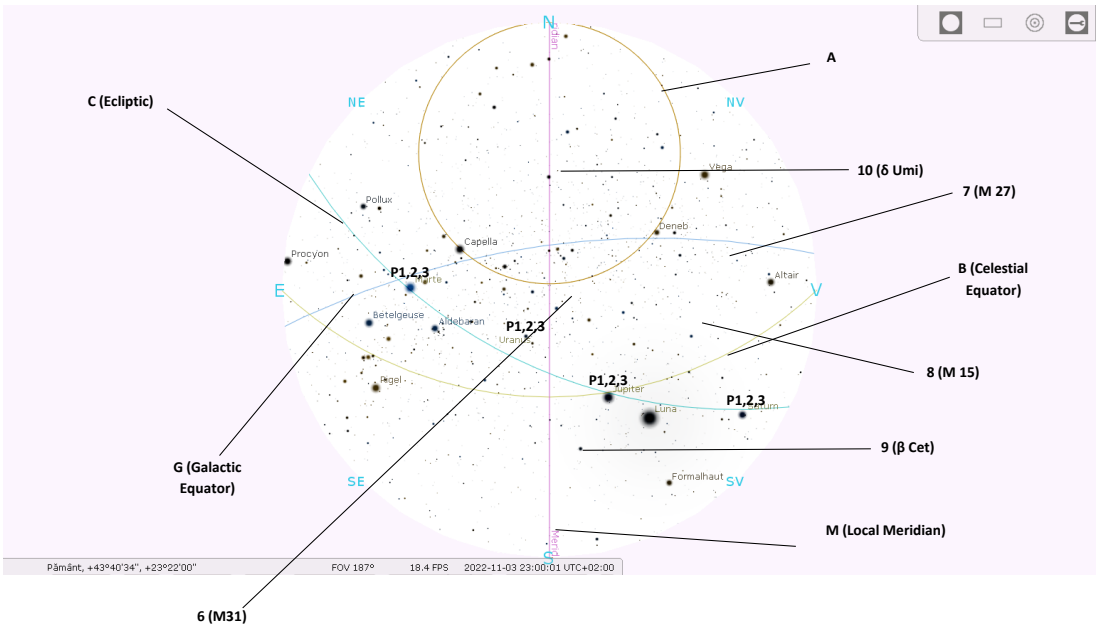
Solution 9.1 Blind Sky Map

Problem p. 73

[The separately issued blind sky map and the referenced IAU-code Table 1 are absent from the supplied question PDF. Two annotated charts survive in the official solution and are reproduced there.]

1. $\varphi = 43°40'$.
2. See the annotated maps below.
3. See the maps. Any three visible planets are accepted in any order.
4. See the maps.
5. See the maps.
6. See the maps.
7. $s = 1^{\mathrm{h}}34^{\mathrm{m}}$.
8. $\alpha = 19^{\mathrm{h}}51^{\mathrm{m}}53^{\mathrm{s}}$ or $H = 05^{\mathrm{h}}41^{\mathrm{m}}59^{\mathrm{s}}$, and $\delta = +8°56'$.





Solution 9.2 Navigating the night sky I

Problem p. 74

[No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.]

Solution 9.3 Navigating the night sky II

Problem p. 78

[No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.]

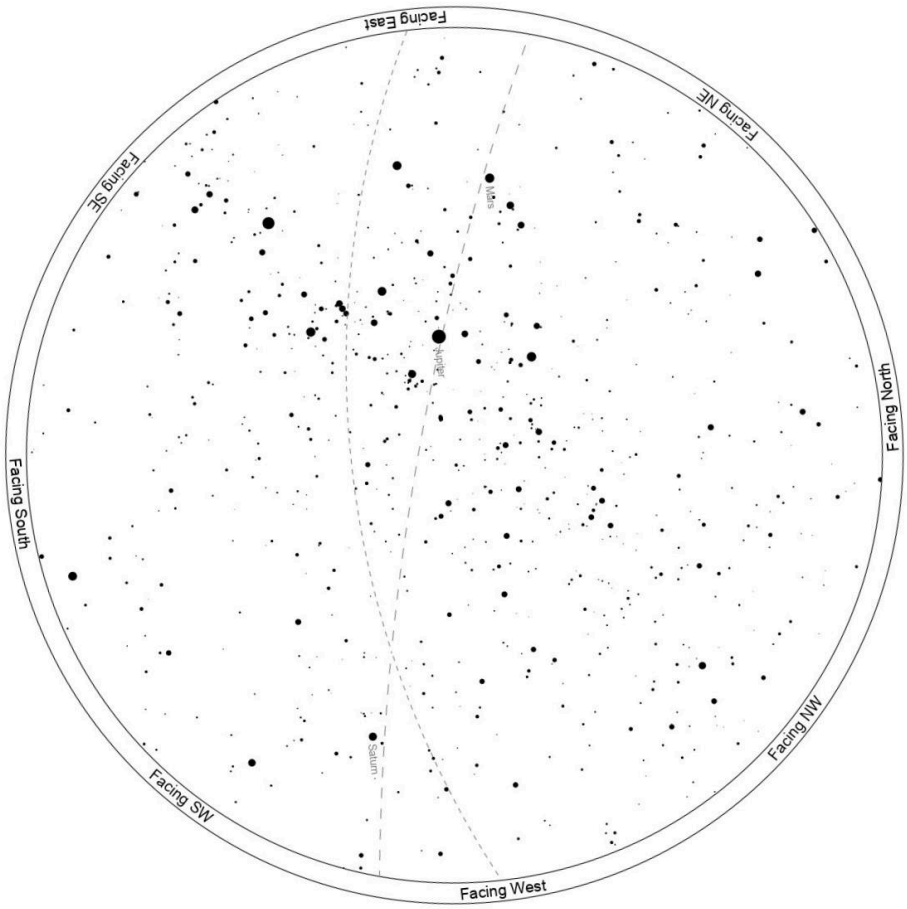
Solution 9.4 A sky chart from Nepal

Problem p. 84

[The combined question paper and answer key use a 45-point OC1 within a 72-point observation round. The separate observation PDF instead gives OC1 43.5 points. The official key calls 2.5 points full credit for part 7 although the question allocates 3 points.]

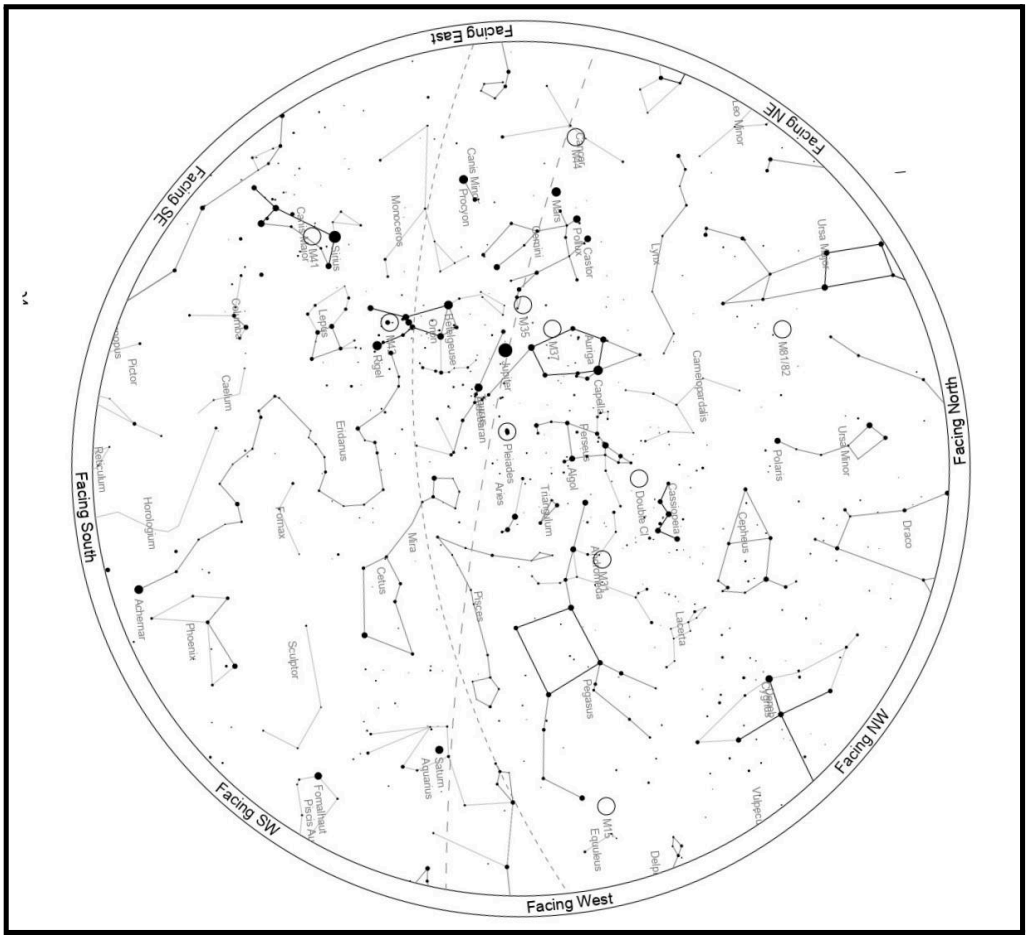
1. The latitude is 27°7′N. Answers between 27° and 28° gain all 4 points. Each degree of inaccuracy loses 0.5 point. For example, 26° or 29° receives 3.5 points.

2. The cardinal points and three planets are indicated on the map. Each 10° of inaccuracy loses 0.5 point for cardinal points. The planets have to be circled correctly for full points. Naming them is unnecessary.

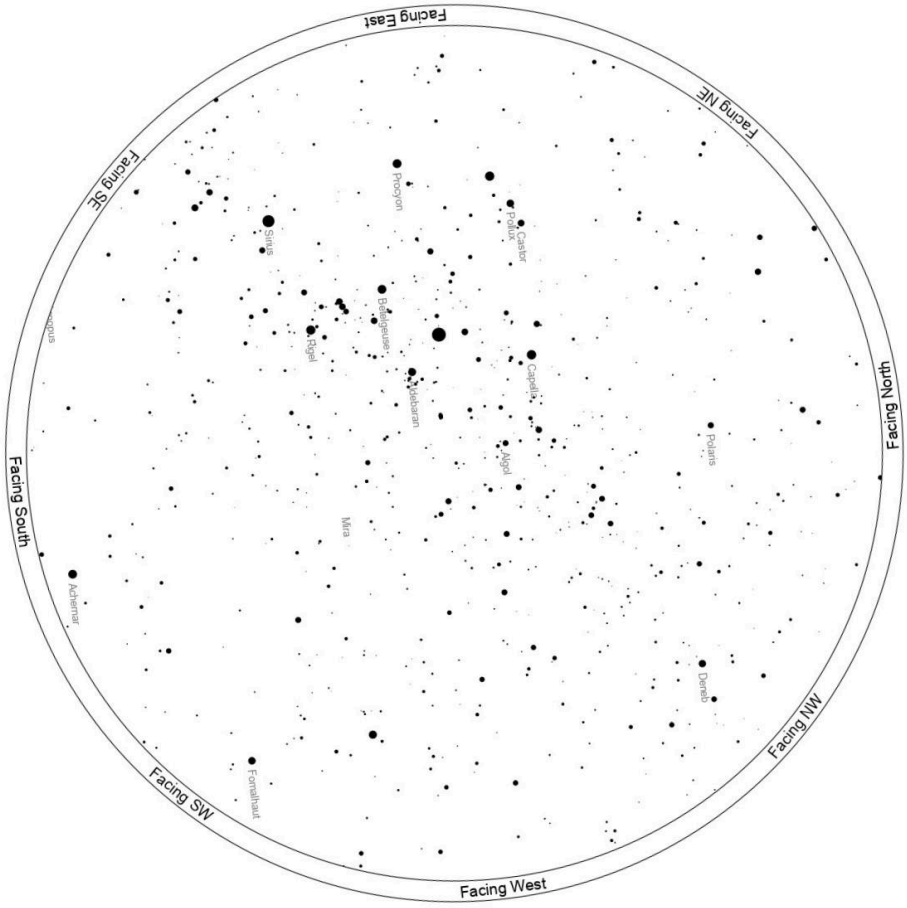


3. Use the same map, except that the local meridian runs straight from North to South. Each 10° of inaccuracy loses 0.5 point.

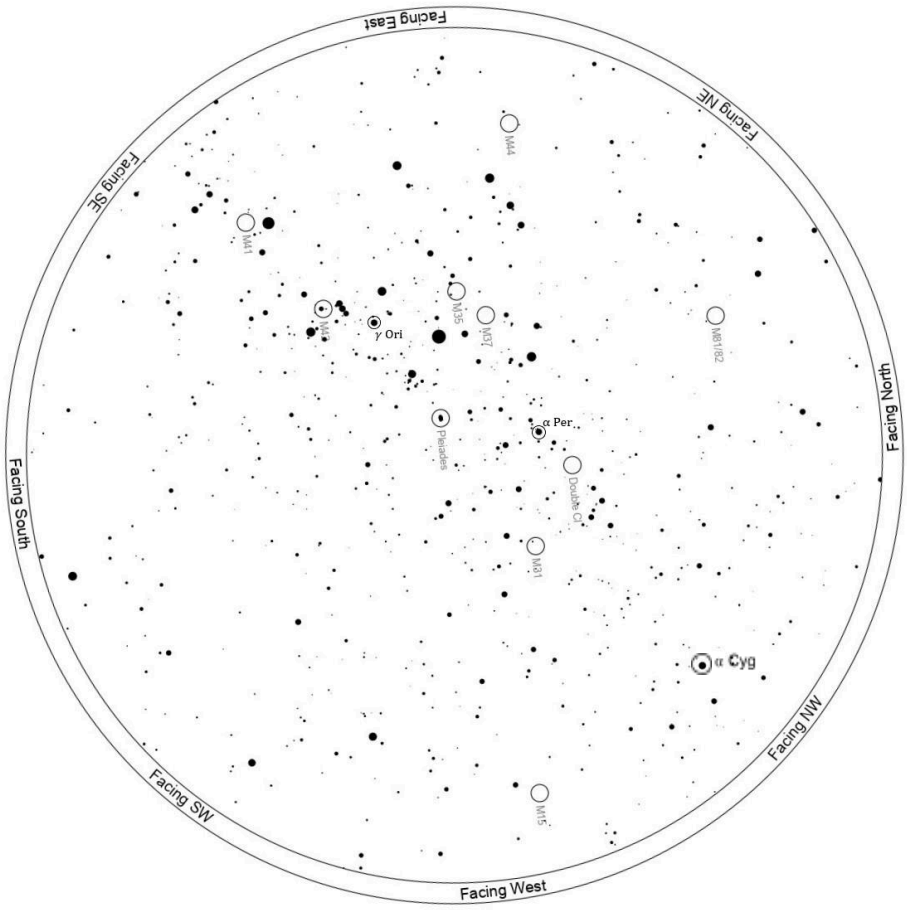
4. Any constellations shown in the map below are acceptable. A tracing including at least the brightest stars earns full points. Missing one or more prominent stars loses 0.5 point.



5. In descending brightness order, the four stars are α Canis Majoris (Sirius), α Aurigae (Capella), β Orionis (Rigel), and α Canis Minoris (Procyon). For each star, 0.5 point is earned for the correct Bayer name or abbreviation, 0.5 for the right order, and 0.5 for the correct map marking.



6. The labeled map shows M42, M81, M41, M31, γ Orionis, α Persei and α Cygni. Labels within the indicated circles earn full credit. A label in the correct constellation but at the wrong place earns 0.5 point.



7. Aldebaran has right ascension $4^{\mathrm{h}}36^{\mathrm{m}}$ and declination $16°31′$. The official tolerance ranges are

RA range	Declination range	Points
3.5–5.5 h	15.5°–17.5°	2.5
2.5–6.5 h	14.5°–18.5°	2
1.5–7.5 h	13.5°–19.5°	1.5
0.5–8.5 h	12.5°–20.5°	1
−0.5–9.5 h	11.5°–21.5°	0.5

Anything beyond these ranges earns no credit.

Solution 9.5 Classifying Solar System objects

Problem p. 86

[The combined question paper and official key use six pictures and 9 points. The separate observation PDF gives nine pictures and 13.5 points. This edition follows the matching six-picture version.]

Asteroids	Dwarf Planets	Satellites
5	2	1, 3, 4, 6

Each correctly categorized picture earns 1.5 points.

Solution 9.6 Identifying deep-sky objects

Problem p. 86

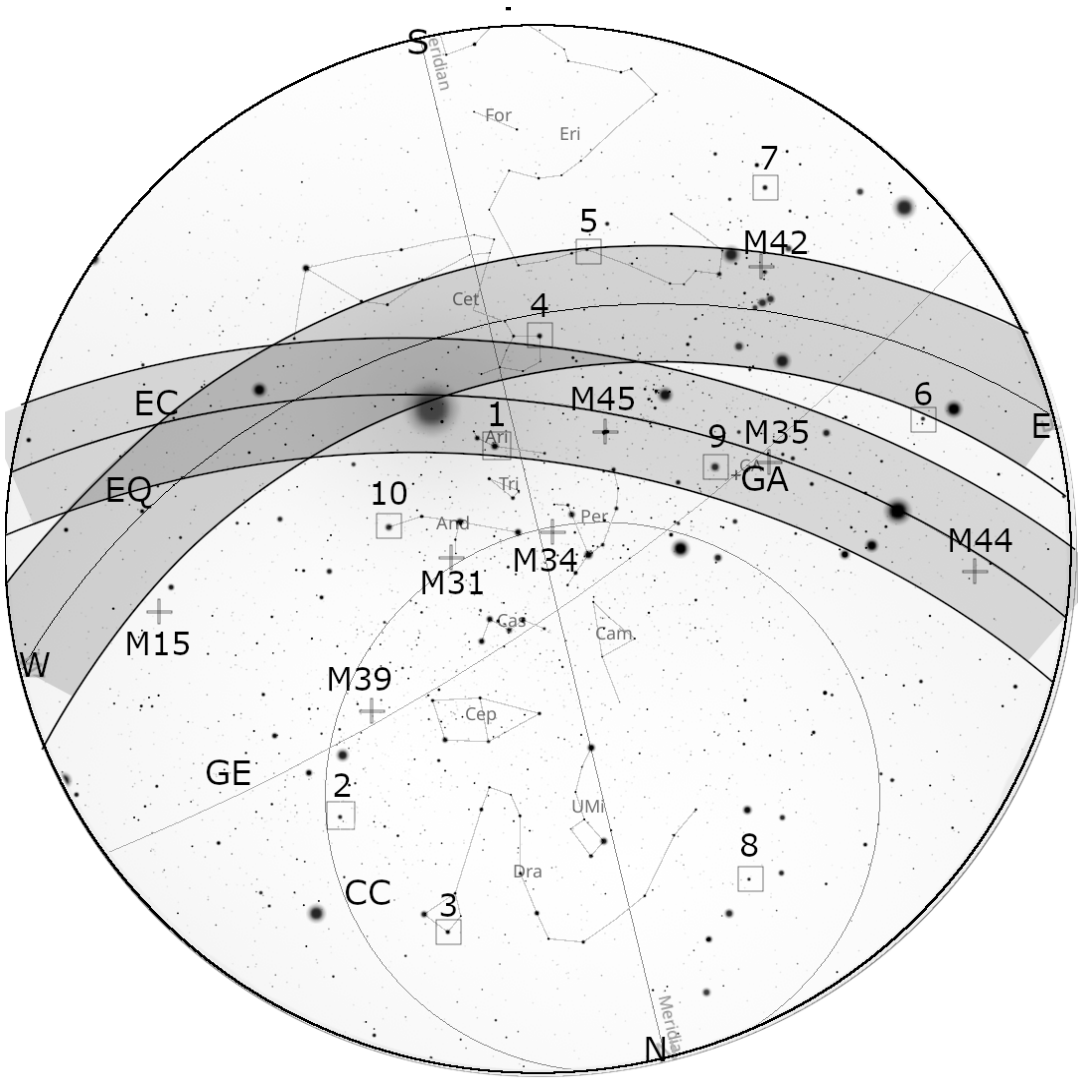
Object	Messier number	Constellation
O1	M97	Ursa Major
O2	M101	Ursa Major
O3	M51	Canes Venatici
O4	M45	Taurus
O5	M1	Taurus
O6	M42 or M43	Orion

Each correct Messier number or constellation earns 1.5 points. Both M42 and M43 receive full marks for O6.

Solution 9.7 Know Your Sky

Problem p. 87

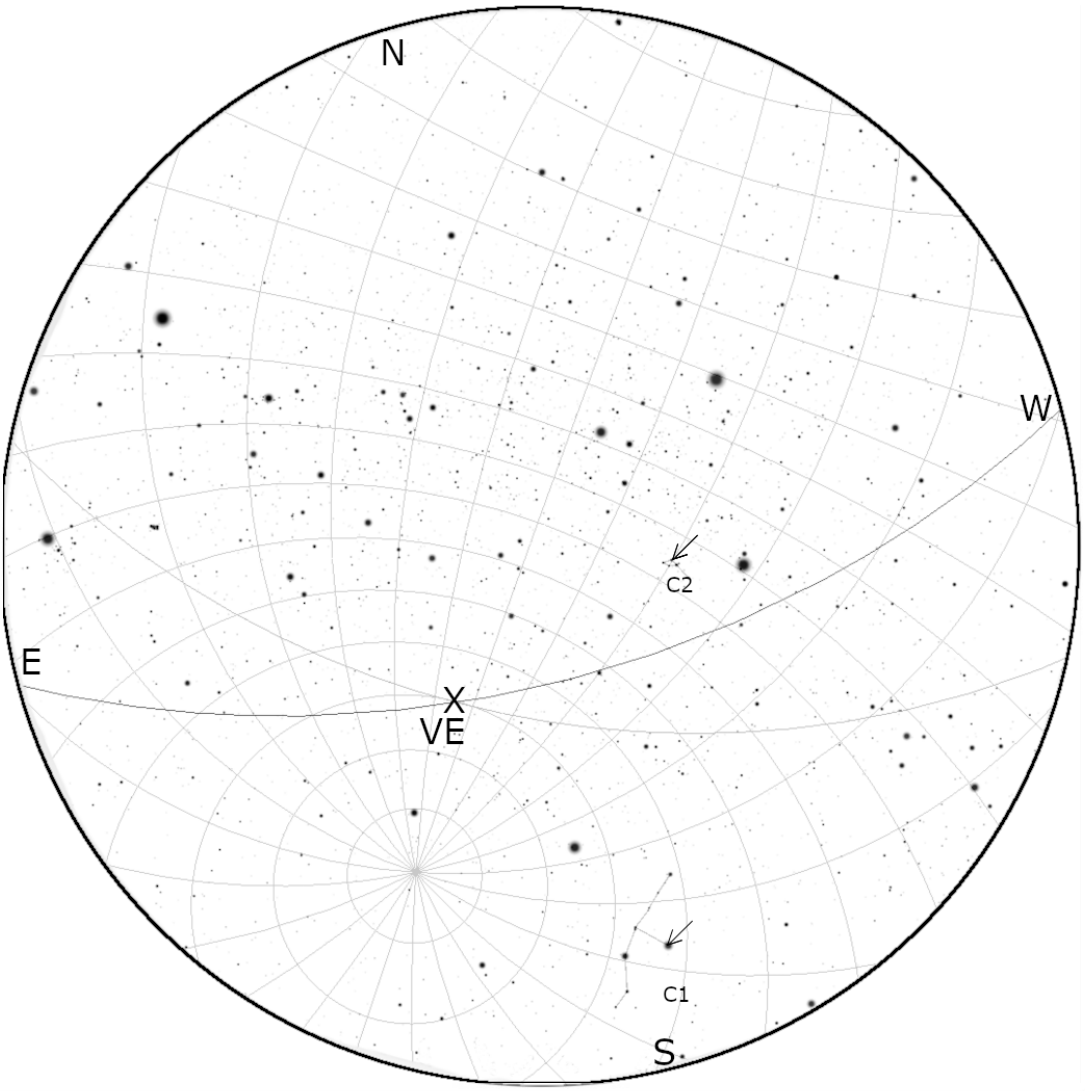
The official answer is the marked Map-OM1 below. It shows the ten numbered stars, eight Messier objects, meridian and nearby constellation patterns, celestial equator, cardinal points, ecliptic, galactic plane and visible galactic center or anticenter, and circumpolar circle.



Solution 9.8 Chart and Time

Problem p. 89

C1 is Grus (Gru) and C2 is Delphinus (Del). The coordinate grid is equatorial. The official marked map shows the α stars and the equinox point.



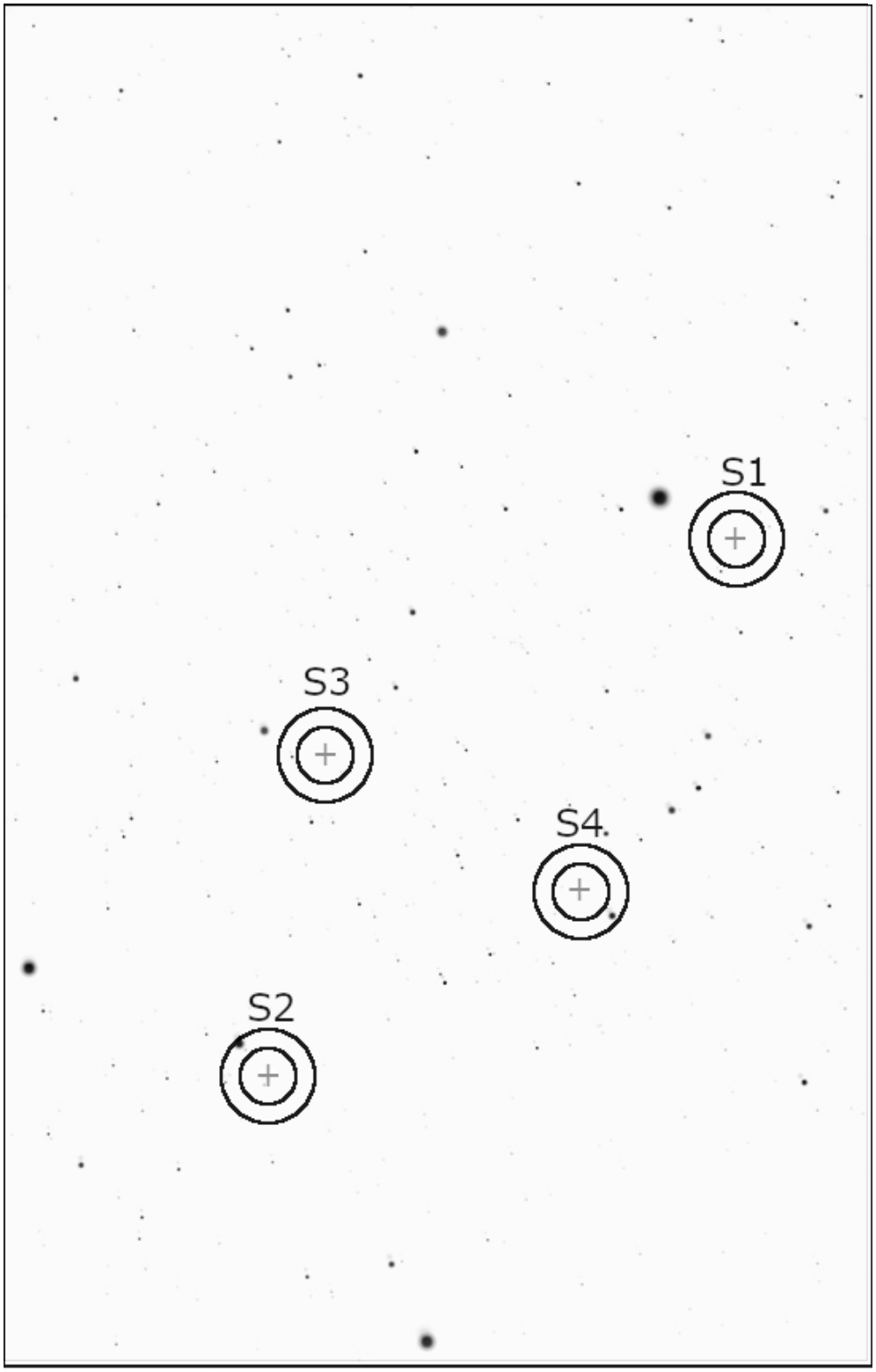
Solution 9.9 Missing Stars

Problem p. 90

The official key identifies the missing stars as follows.

Star	Name	Double/multiple?
1	Izar	Y, double star
2	Castor	Y, double star
3	Algieba	Y, double star
4	Merak	N

Their positions are shown on the official marked map.



Solution 9.10 Sun observations

Problem p. 92

[No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.]

Solution 9.11 Cloudy sky observations

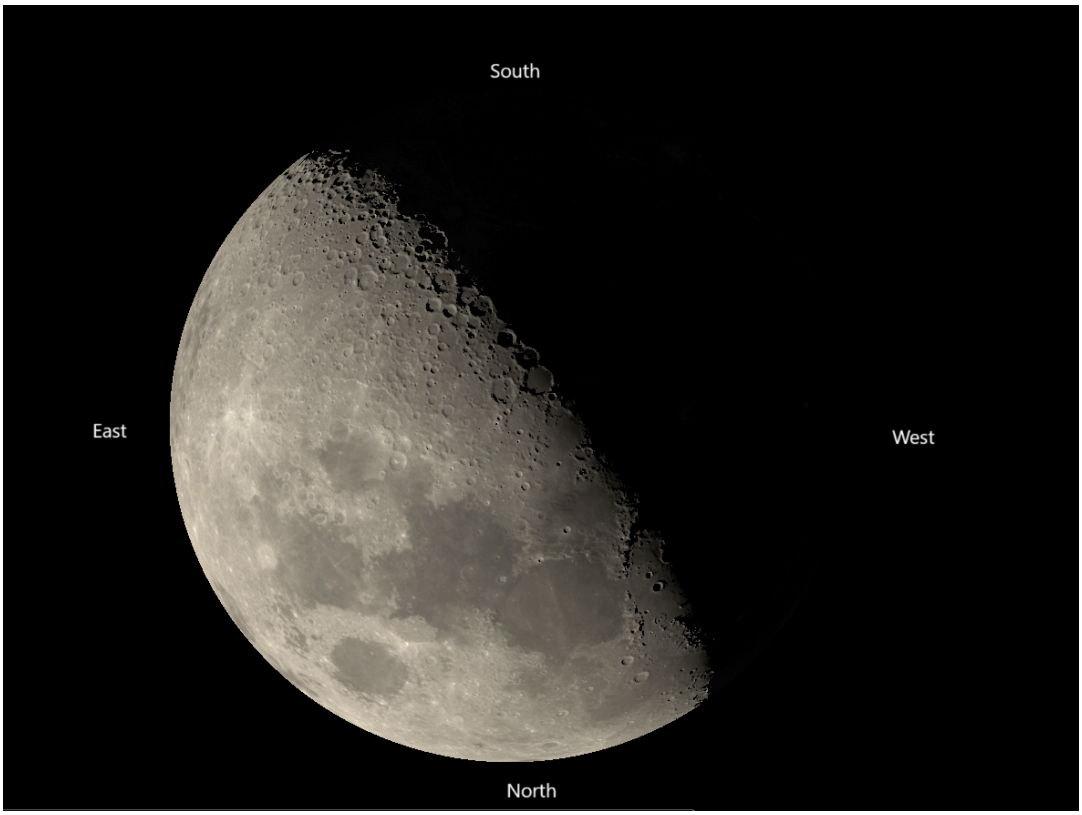
Problem p. 93

[No official solution is included in the supplied 2023 observation papers. The surviving file contains the question and blank answer sheets.]

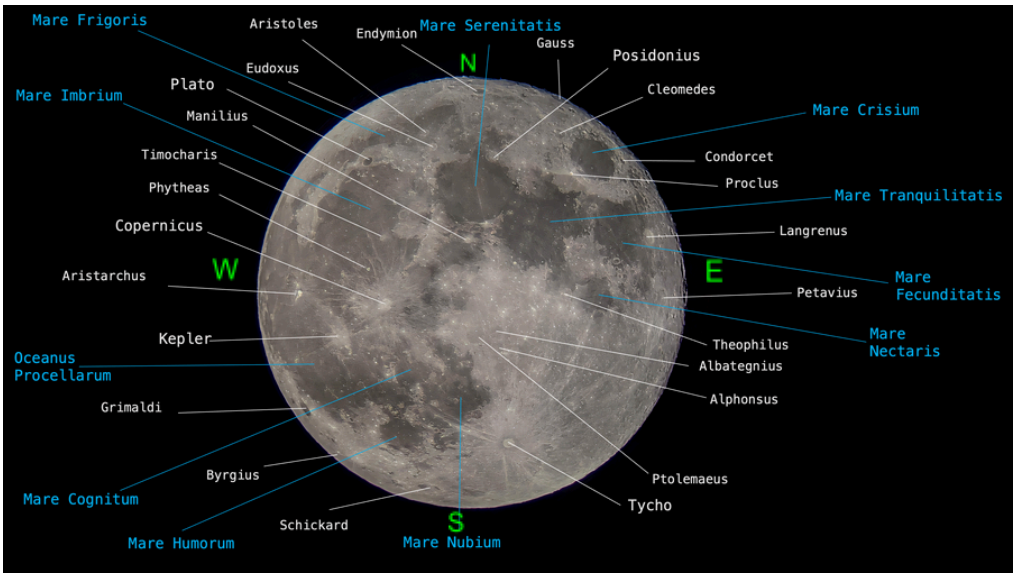
Solution 9.12 The Moon

Problem p. 94

The supervisor awards 6 points for pointing.



1. Drawing the Moon earns 2 points and drawing the lunar terminator properly earns 2 points.
2. The cardinal points earn 4 points.
3. Each of three maria earns 2 points and each of three craters earns 2 points. The labeled reference image is shown below.



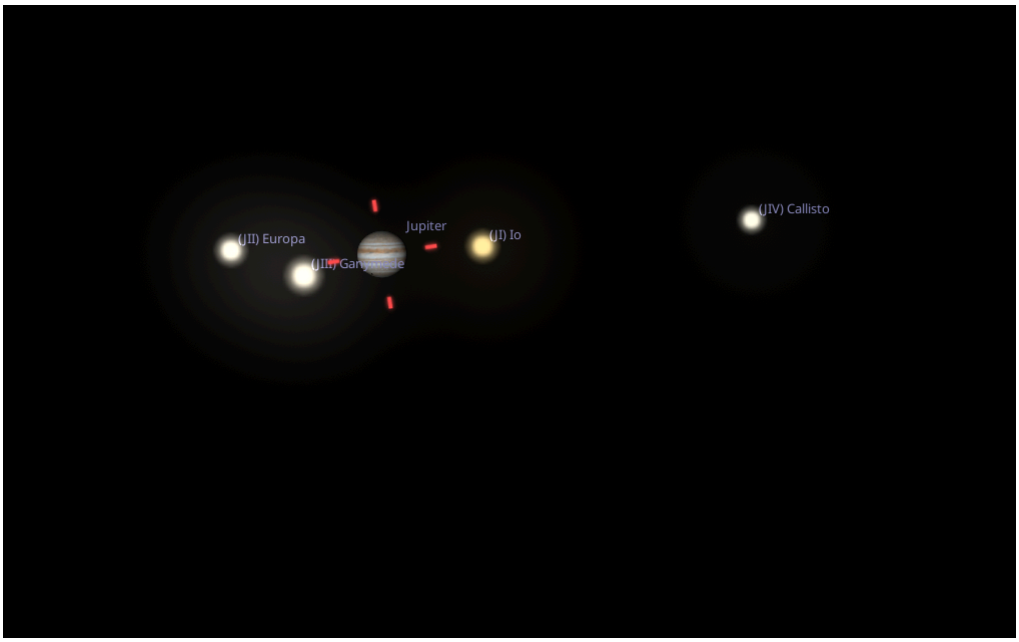
4. The Moon's age is 8.3 days. Values of 8–8.5 days gain full marks, and 7.5–9 days gain half marks. (2 points) The observed hemisphere is Southern. (2 points) The Moon is in Taurus. (2 points)

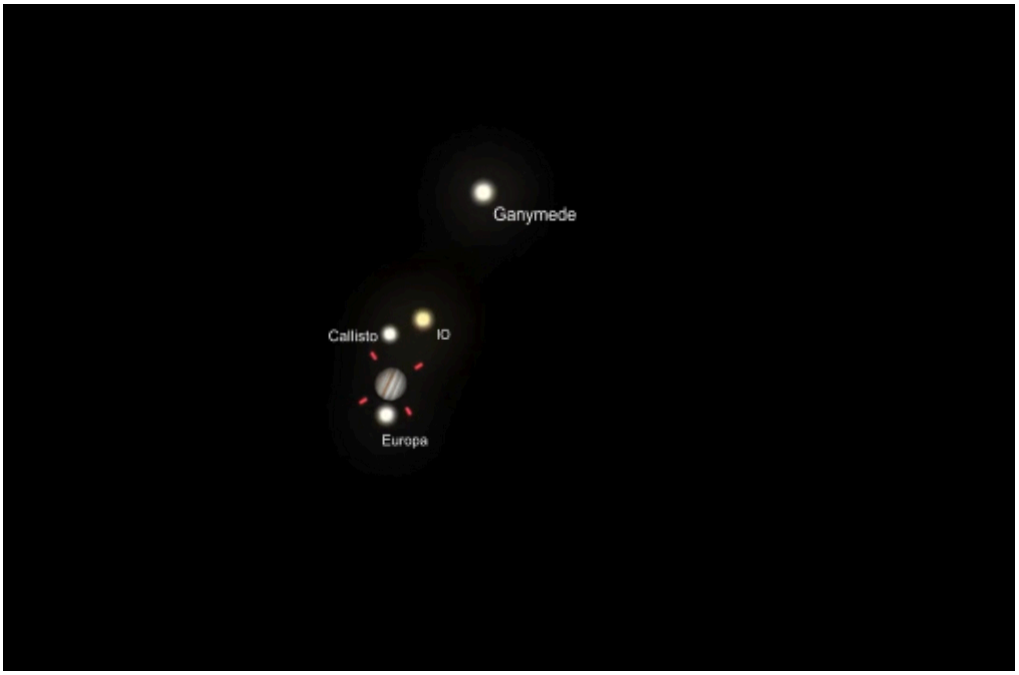
Solution 9.13 Jupiter and its moons

Problem p. 95

1. The planet is Jupiter. (2 points)

2. The initial and final positions earn 4 points, and correct consistent labeling earns 4 points.





Students use A, B, C, D labels rather than names. The same moon must carry the same label in both images.

3. In descending size order, the moons are Ganymede, Callisto, Io, and Europa. Each correct position in the order earns 1.5 points.

Solution 9.14 Telescope Adjustment Knobs

Problem p. 95

(a) Right ascension. (2 points)

(b) Declination. (2 points)

Solution 9.15 Coordinate Identification

Problem p. 95

(a) Altitude. (3 points)

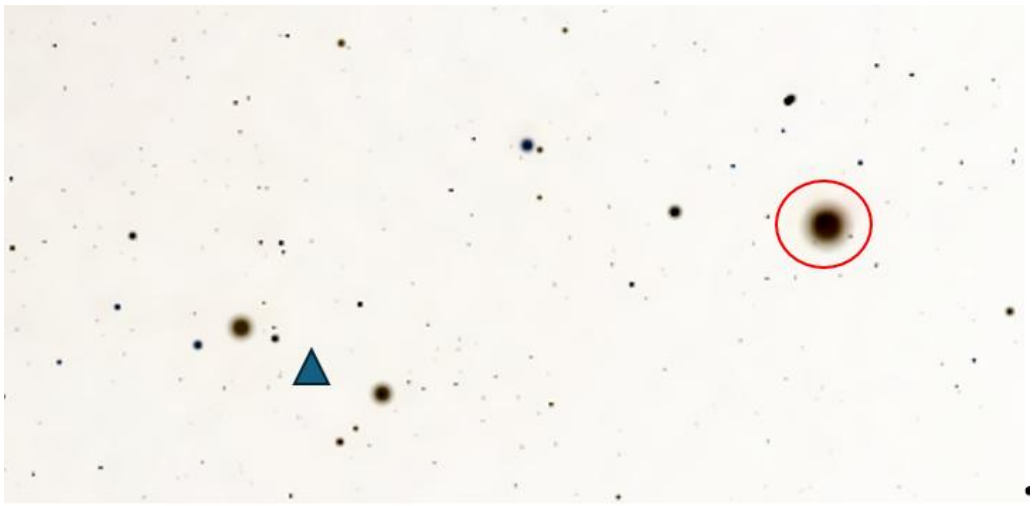
(b) Zenith angle. (3 points)

(c) Azimuth. (3 points)

Solution 9.16 Constellation through the Eyepiece

Problem p. 96

(a), (c), (g) The official drawing is below. Award 5 points for the constellation drawing, 2 points for marking the star, and 2 points for marking M57.



(b) Lyra or Lyr. (2 points)

(d) Vega. (3 points)

(e) Ring Nebula or M57. (3 points)

(f) Planetary nebula. (3 points)

Solution 9.17 Objects and Telescope Coordinates

Problem p. 96

The signs identify R as the Orion Nebula (M42), N as the Andromeda Galaxy (M31), and Y as the double star Albireo. Award 3 points for each correct answer. The official key gives the following coordinates for the 18 telescopes. Right ascensions within $\pm 10^{\mathrm{m}}$ and declinations within $\pm 2^{\circ}$ are accepted.

Telescope	Sign	Right ascension	Declination
1	R	$22^{\mathrm{h}}10^{\mathrm{m}}$	32°
1	N	$22^{\mathrm{h}}30^{\mathrm{m}}$	26°
1	Y	$23^{\mathrm{h}}00^{\mathrm{m}}$	22°
2	R	$21^{\mathrm{h}}30^{\mathrm{m}}$	40°
2	N	$22^{\mathrm{h}}00^{\mathrm{m}}$	32°
2	Y	$22^{\mathrm{h}}30^{\mathrm{m}}$	30°
3	R	$20^{\mathrm{h}}00^{\mathrm{m}}$	44°
3	N	$20^{\mathrm{h}}50^{\mathrm{m}}$	40°
3	Y	$21^{\mathrm{h}}30^{\mathrm{m}}$	38°
4	R	$19^{\mathrm{h}}40^{\mathrm{m}}$	48°
4	N	$20^{\mathrm{h}}20^{\mathrm{m}}$	44°
4	Y	$21^{\mathrm{h}}00^{\mathrm{m}}$	42°
5	R	$19^{\mathrm{h}}40^{\mathrm{m}}$	48°
5	N	$20^{\mathrm{h}}20^{\mathrm{m}}$	44°
5	Y	$21^{\mathrm{h}}20^{\mathrm{m}}$	42°
6	R	$22^{\mathrm{h}}10^{\mathrm{m}}$	42°
6	N	$22^{\mathrm{h}}30^{\mathrm{m}}$	35°
6	Y	$23^{\mathrm{h}}10^{\mathrm{m}}$	30°
7	R	$20^{\mathrm{h}}40^{\mathrm{m}}$	47°
7	N	$21^{\mathrm{h}}20^{\mathrm{m}}$	40°

Telescope	Sign	Right ascension	Declination
7	Y	$22^{h}10^{m}$	36°
8	R	$21^{h}30^{m}$	34°
8	N	$22^{h}00^{m}$	28°
8	Y	$22^{h}40^{m}$	24°
9	R	$22^{h}10^{m}$	24°
9	N	$22^{h}40^{m}$	18°
9	Y	$23^{h}20^{m}$	16°
10	R	$19^{h}50^{m}$	46°
10	N	$20^{h}30^{m}$	48°
10	Y	$19^{h}00^{m}$	48°
11	R	$22^{h}30^{m}$	30°
11	N	$23^{h}00^{m}$	28°
11	Y	$21^{h}55^{m}$	36°
12	R	$20^{h}00^{m}$	42°
12	N	$20^{h}30^{m}$	40°
12	Y	$19^{h}10^{m}$	44°
13	R	$21^{h}00^{m}$	32°
13	N	$21^{h}30^{m}$	32°
13	Y	$20^{h}20^{m}$	36°
14	R	$21^{h}30^{m}$	36°
14	N	$22^{h}00^{m}$	34°
14	Y	$21^{h}00^{m}$	42°
15	R	$21^{h}30^{m}$	40°
15	N	$22^{h}00^{m}$	38°
15	Y	$20^{h}50^{m}$	44°
16	R	$21^{h}20^{m}$	36°
16	N	$21^{h}50^{m}$	36°
16	Y	$20^{h}40^{m}$	40°
17	R	$20^{h}50^{m}$	32°
17	N	$21^{h}20^{m}$	32°
17	Y	$20^{h}10^{m}$	36°
18	R	$21^{h}00^{m}$	36°
18	N	$21^{h}30^{m}$	36°
18	Y	$20^{h}20^{m}$	40°

Solution 9.18 Observations with the Naked Eye

Problem p. 96

[No official solution or marking key for the 2022 naked-eye night-observation tasks is included in the supplied archive.]

Solution 9.19 Observations Using the Astronomical Telescope

Problem p. 97

[No official solution or marking key for the 2022 telescope night-observation tasks is included in the supplied archive.]

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